

q -Pochhammer Symbols and q -Binomial Coefficients

Algebra, Combinatorics, Analysis, Arithmetic, and Summation Theory

A self-contained, proof-oriented monograph

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Abstract

The finite and infinite q -Pochhammer symbols and the Gaussian, or q -binomial, coefficients form a small set of notational primitives from which a remarkable amount of mathematics can be built. They simultaneously encode weighted subsets, partitions inside rectangles, lattice paths, subspaces over finite fields, Schubert cells in Grassmannians, noncommutative binomial expansions, q -difference operators, basic hypergeometric summations, theta products, partition identities, and congruences at roots of unity.

This monograph develops those themes from first principles. It proves the algebraic laws of shifted factorials; the polynomial, reciprocal, and cyclotomic structure of Gaussian coefficients; finite and infinite q -binomial theorems; weighted generalizations through elementary and complete symmetric functions; combinatorial and finite-geometric models; unimodality by $\mathfrak{sl}_2$ representation theory; the q -Gauss, q -Chu–Vandermonde, and q -Pfaff–Saalschütz summations; Jackson calculus and the q -gamma and q -beta functions; Jacobi’s triple product, Euler’s pentagonal theorem, and the Rogers–Ramanujan identities through Bailey pairs; cyclotomic factorization, q -Lucas and q -Babbage congruences; negative upper indices and q -Newton interpolation; and a recent family of central q -binomial/Lambert-series evaluations. A separate part develops denominator-free Gaussian algebra over arbitrary commutative rings, geometric interpolation and Richardson filters, and the dyadic specialization $q = 1/2$ that governs exact Fabius–Rvachev and Thue–Morse formulas. It derives reciprocal-dyadic and arbitrary-dyadic values, Appell and Bell representations, composition and determinant formulas, the infinite sinc product, finite box splines, and a uniformly convergent Gaussian Toeplitz scheme. Every displayed theorem used in the main development is accompanied by a proof, and the final appendices collect formulas, limiting transitions, and computational recipes.

Preface and source map

The aim is not merely to list identities, but to expose the mechanisms that repeatedly generate them. Three such mechanisms dominate:

1. finite products split into residue classes, blocks, or initial and tail factors;
2. coefficient extraction turns product factorizations into convolution identities;
3. functional equations determine analytic q -series from one initial value.

Once these mechanisms are visible, many formulas that initially appear unrelated become instances of the same principle.

The supplied archive was used throughout. In particular, the weighted subset framework of Konvalina motivates [Chapter 4](#); Kupershmidt’s treatment of the q -Newton binomial informs [Chapters 11](#) and [17](#); Straub’s seminar on q -binomial congruences is reflected in [Chapter 16](#); the formalized q -series project of Lau, Lee, and Ono motivates the algebraic-versus-analytic distinctions in [Chapter 24](#) and supplies a modern Bailey-pair route to the Rogers–Ramanujan identities; and the central q -binomial paper of Chern, Dilcher, and Jiu is developed, with a complete proof, in [Chapter 22](#). The archive’s reference chapters and encyclopedia snapshots were checked against the standard sources listed in the bibliography. Additional material was drawn from the classical books and articles of Andrews, Askey, Bailey, Gasper, Rahman, Rota, Goldman, Reiner, Stanton, White, and others, as well as the NIST Digital Library of Mathematical Functions.

The recursive $\text{\TeX}$ corpus under `Analysis/FabiusFunction/docs` in Vladimir Reshetnikov’s [ProveIt](#) repository was surveyed as a second source corpus. The present source map is pinned to the following state:

repository commit	a289ff83097f1cc7f896c49b7e62d10188a8d62c,
documentation tree	02fbf025858f75f2e4323e8c605098f3c307f530.

The q -relevant current and archival lineages include the primary Fabius–Rvachev exposition, the mathematical glossary, the dedicated *Dyadic q -Calculus* article, the dyadic- q connections report, and the associated research-frontier studies. Their half-base identities and the more recent denominator-free geometric-filter layer are synthesized in [Chapters 18](#) to [21](#). The Lean development is used as a theorem inventory and verification map, not as a substitute for proof: the mathematical arguments needed here are given in full. See [\[37\]](#).

Two regimes for q

Throughout the finite algebraic chapters, q may be an indeterminate and identities are understood in a polynomial or rational-function ring. In analytic chapters, unless stated otherwise,

$$q \in \mathbb{C}, \quad 0 < |q| < 1.$$

For the Jackson integral and positivity statements we usually assume $0 < q < 1$. In the dyadic Fabius part, the base is fixed at $q = 1/2$; auxiliary translations are denoted by θ or τ so that they cannot be confused with the base. A capital Q is used for the cardinality of a finite field, so Q is a prime power and normally $Q > 1$.

Conventions

Empty products equal 1, empty sums equal 0, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \quad \text{when } k < 0 \text{ or } k > n$$

for nonnegative integral n . Multiple parameters are compressed as

$$(a_1, \dots, a_r; q)_n = \prod_{j=1}^r (a_j; q)_n.$$

The notation $(a; q)_\infty$ is analytic when $|q| < 1$; when formal power series are intended, that will be said explicitly.

Contents

Abstract	i
Preface and source map	ii
 I The finite algebra	 1
1 The algebra of q-shifted factorials	2
1.1 Definitions and first examples	2
1.2 Concatenation, dissection, and reversal	2
1.3 Negative indices	3
1.4 Partial fractions and logarithmic derivatives	4
1.5 Infinite products and Lambert series	5
1.5.1 Complex orders	5
1.6 The dilogarithmic limit	7
1.7 Classical shifted factorials as a limit	8
 2 q-integers and Gaussian coefficients	 9
2.1 Definitions	9
2.2 The two Pascal recurrences	9
2.3 Symmetry, reciprocity, and degree	10
2.4 Multinomial coefficients and flags	11
2.5 Adjacent ratios and efficient recurrences	11
2.6 Coefficient moments	11
 3 Finite q-binomial theorems and inversion	 13
3.1 Euler's finite product expansions	13
3.2 Reciprocal finite products	14
3.3 q -Vandermonde convolution	14
3.4 q -binomial inversion	15
3.5 A q -Cauchy product identity	15
 4 Weighted binomial coefficients and symmetric functions	 17
4.1 Elementary and complete weighted coefficients	17
4.2 Classical, Stirling, and Gaussian specializations	18
4.3 Weighted inversion	19
4.4 The subset–subspace analogy	19

5	Partitions, words, paths, and inversion statistics	20
5.1	Binary words	20
5.2	Partitions inside a rectangle	20
5.3	Lattice paths and area	21
5.4	Durfee squares and the central convolution	21
5.5	Words with several letters	22
5.6	Rogers–Szegő polynomials	22
6	Finite fields, Grassmannians, and rank enumeration	23
6.1	Counting subspaces	23
6.2	Flags and q -multinomial coefficients	24
6.3	Matrices of fixed rank	24
6.4	Schubert cells and a topological meaning	24
6.5	Growth over finite fields	25
7	Symmetry and unimodality	26
7.1	A representation-theoretic lemma	26
7.2	Exterior powers and Gaussian coefficients	27
II	Infinite series and basic hypergeometric summation	28
8	The infinite q-binomial theorem	29
8.1	A functional-equation proof	29
8.2	Formal and analytic interpretations	30
8.3	Limits in the number of factors	30
9	Basic hypergeometric series	32
9.1	Definition and convergence	32
9.2	Heine’s transformation	32
9.3	The q -Gauss sum	33
9.4	A second finite Cauchy identity	33
9.5	The q -Pfaff–Saalschütz sum	34
9.6	The two q -Chu–Vandermonde sums	35
9.7	A hierarchy of mechanisms	35
10	Bilateral series and Ramanujan’s ${}_1\psi_1$ sum	36
10.1	Definition	36
11	q-difference calculus and the quantum binomial	38
11.1	The q -derivative	38
11.2	q -Taylor and Newton interpolation	39
11.3	The quantum plane	40
11.4	Jackson’s q -integral	41
12	The q-gamma and q-beta functions	42
12.1	Definition and functional equation	42
12.2	A q -Bohr–Møllerup theorem	43
12.3	Classical limit	43

12.4	The Jackson q -beta integral	44
12.5	Multiplication and reflection-type formulas	45
III	Theta products and partition identities	46
13	Jacobi's triple product	47
13.1	A bilateral theta series	47
13.2	Euler's pentagonal number theorem	48
13.3	Jacobi's cubic identity	48
13.4	A theta form of the q -gamma reflection product	49
14	Partitions and product-sum identities	50
14.1	Unrestricted, distinct, and odd parts	50
14.2	Bounded partitions	50
14.3	Finite approximations and stabilization	51
15	Bailey pairs and the Rogers–Ramanujan identities	52
15.1	Bailey pairs	52
15.2	Bailey inversion	52
15.3	The limiting Bailey lemma	53
15.4	The Rogers–Ramanujan identities	54
IV	Cyclotomic arithmetic and extended indices	56
16	Cyclotomic factorization and congruences	57
16.1	Cyclotomic valuations and carries	57
16.2	The q -Lucas congruence	58
16.3	Cyclic sieving for subsets	59
16.4	A q -Babbage congruence modulo a cyclotomic square	60
17	Negative upper indices and geometric Newton interpolation	61
17.1	Generalized Gaussian coefficients	61
17.2	An arbitrary upper parameter	62
17.3	Two complex arguments and generalized binomial series	63
17.4	Newton interpolation on the grid $1, q, q^2, \dots$	64
V	Geometric grids and the dyadic half-base theory	66
18	Denominator-free Gaussian algebra and geometric filters	67
18.1	Recursive Gaussian polynomials over an arbitrary ring	67
18.2	Complete homogeneous functions on a geometric alphabet	68
18.3	A universal moment-residual theorem	69
18.4	Explicit geometric interpolation and Richardson weights	70

19	The Fabius distribution, moments, and dyadic Appell calculus	71
19.1	The random series and its self-similarity	71
19.2	Moments, the entire product, and inverse-dyadic values	72
19.3	Appell polynomials and a box-spline formula	73
19.4	Binary digits and a sparse global expansion	74
20	Thue–Morse cancellation and exact half-base Gaussian formulas	76
20.1	Mersenne arithmetic and binary subspaces	76
20.2	The dyadic nodal annihilator	76
20.3	Thue–Morse blocks and their forced zero	77
20.4	A coefficient-extraction principle	78
20.5	Finite Gaussian formulas for the Appell polynomials	78
20.6	Arbitrary dyadic numerators in one shot	79
20.7	The arbitrary-argument nested q -series	79
21	Bernoulli–Bell formulas, sinc products, and a q-Toeplitz limit	81
21.1	A composition formula for the reciprocal dyadic values	81
21.2	A lower-Hessenberg determinant	82
21.3	Bernoulli cumulants and complete Bell polynomials	83
21.4	The Rvachev density and its infinite sinc product	84
21.5	Finite centered uniform splines	84
21.6	The half-base Gaussian Toeplitz filter	85
21.7	Synthesis of the half-base theory	87
VI	Central sums, limits, and effective methods	88
22	Central q-binomial sums and Lambert series	89
22.1	The central-factor reduction	89
22.2	A q -beta evaluation	90
22.3	Deleting a singular term	91
22.4	Separate finite and infinite components	92
22.5	The classical finite part	93
23	Asymptotic regimes and Bernoulli expansions	96
23.1	A complete expansion as $q \rightarrow 1$	96
23.2	Fixed q and growing parameters	97
23.3	The regime $q > 1$	98
23.4	Extreme specializations	99
24	Computation, certification, and formal proof architecture	100
24.1	Dynamic programming for Gaussian polynomials	100
24.2	A factorization algorithm	101
24.3	Telescoping certificates for finite sums	101
24.4	Denominator clearing and specialization	102
24.5	Formal series versus analytic series	102
A	Formula atlas	104

B	Limit and specialization dictionary	105
B.1	Elementary limits	105
B.2	Classical shadows of the main summations	105
B.3	Roots of unity	106
C	Proof-dependency guide	107
	Index	111

Part I

The finite algebra

Chapter 1

The algebra of q -shifted factorials

1.1 Definitions and first examples

Definition 1.1. For $n \in \mathbb{N}_0$ the q -shifted factorial, or q -Pochhammer symbol, is

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_0 = 1.$$

For parameters $a_1, \dots, a_r$ we write

$$(a_1, \dots, a_r; q)_n = \prod_{i=1}^r (a_i; q)_n.$$

When $0 < |q| < 1$, the infinite symbol is

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The word “shifted” refers to the geometric progression $a, aq, aq^2, \dots$. The first few values are

$$(a; q)_1 = 1 - a, \quad (a; q)_2 = (1 - a)(1 - aq), \quad (a; q)_3 = (1 - a)(1 - aq)(1 - aq^2).$$

Important specializations include

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad (-q; q)_n = \prod_{j=1}^n (1 + q^j), \quad (a; q^r)_n = \prod_{j=0}^{n-1} (1 - aq^{rj}).$$

1.2 Concatenation, dissection, and reversal

Proposition 1.2 (Concatenation law). For $m, n \in \mathbb{N}_0$,

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

Consequently, whenever the denominator is nonzero,

$$\frac{(a; q)_{m+n}}{(a; q)_m} = (aq^m; q)_n.$$

Proof. Split the defining product at $j = m$:

$$\prod_{j=0}^{m+n-1} (1 - aq^j) = \left(\prod_{j=0}^{m-1} (1 - aq^j) \right) \left(\prod_{j=m}^{m+n-1} (1 - aq^j) \right).$$

In the second product put $j = m + r$, where $0 \leq r < n$. □

Proposition 1.3 (Dissection into residue classes). *For integers $r, n \geq 1$,*

$$(a; q)_{rn} = \prod_{s=0}^{r-1} (aq^s; q^r)_n.$$

If $|q| < 1$, then also

$$(a; q)_{\infty} = \prod_{s=0}^{r-1} (aq^s; q^r)_{\infty}.$$

Proof. Every integer j with $0 \leq j < rn$ has a unique representation $j = s + rt$ with $0 \leq s < r$ and $0 \leq t < n$. Reordering the finite product gives the first identity. For the infinite identity, the factors are absolutely summably close to 1 because $\sum_{j \geq 0} |aq^j| < \infty$, so the product may be regrouped into the r residue classes. □

Proposition 1.4 (Base reversal). *For $n \geq 0$ and $a \neq 0$,*

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1}; q^{-1})_n.$$

Equivalently,

$$(a; q^{-1})_n = (-a)^n q^{-\binom{n}{2}} (a^{-1}; q)_n.$$

Proof. Factor $-aq^j$ from each term:

$$1 - aq^j = -aq^j (1 - a^{-1}q^{-j}).$$

Multiplying for $0 \leq j < n$ gives the factor $(-a)^n q^{0+1+\dots+(n-1)} = (-a)^n q^{\binom{n}{2}}$. The remaining product is $(a^{-1}; q^{-1})_n$. □

1.3 Negative indices

The concatenation law suggests the only extension to negative integers that preserves $(a; q)_{m+n} = (a; q)_m (aq^m; q)_n$.

Definition 1.5. For $n \geq 1$, define

$$(a; q)_{-n} = \frac{1}{(aq^{-n}; q)_n},$$

provided no factor in the denominator vanishes.

Proposition 1.6 (Closed form for a negative index). *For $n \geq 1$ and $a \neq 0$,*

$$(a; q)_{-n} = \frac{(-1)^n a^{-n} q^{\binom{n+1}{2}}}{(q/a; q)_n}.$$

Moreover, for all integers m, n for which both sides are defined,

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n.$$

Proof. By definition,

$$(aq^{-n}; q)_n = \prod_{j=0}^{n-1} (1 - aq^{-n+j}).$$

Factor $-aq^{-n+j}$ from each factor. Their product is

$$(-a)^n q^{-n^2 + \binom{n}{2}} = (-a)^n q^{-\binom{n+1}{2}},$$

and the remaining product is

$$\prod_{j=0}^{n-1} (1 - a^{-1} q^{n-j}) = (q/a; q)_n.$$

Taking the reciprocal proves the formula. The extended concatenation identity follows by cancelling the overlapping finite factors; the cases in which one of $m, n, m+n$ is negative reduce directly to the definition. $\square$

Corollary 1.7 (A terminating factor). *If $0 \leq k \leq N$, then*

$$(q^{-N}; q)_k = (-1)^k q^{-Nk + \binom{k}{2}} \frac{(q; q)_N}{(q; q)_{N-k}}.$$

In particular, $(q^{-N}; q)_k = 0$ for $k > N$.

Proof. Apply base reversal to $(q^{-N}; q)_k$:

$$(q^{-N}; q)_k = (-1)^k q^{-Nk + \binom{k}{2}} (q^{N-k+1}; q)_k.$$

The last factor is $(q; q)_N / (q; q)_{N-k}$. For $k > N$, the defining product contains the factor $1 - q^0$. $\square$

1.4 Partial fractions and logarithmic derivatives

Theorem 1.8 (Finite partial-fraction decomposition). *Assume $q^i \neq q^j$ for $0 \leq i < j \leq n$. Then*

$$\frac{1}{(z; q)_{n+1}} = \sum_{j=0}^n \frac{(-1)^j q^{\binom{j+1}{2}}}{(q; q)_j (q; q)_{n-j}} \frac{1}{1 - zq^j}.$$

Proof. The left side has simple poles at $z = q^{-j}$, $0 \leq j \leq n$. The coefficient of $(1 - zq^j)^{-1}$ is

$$A_j = \left. \frac{1 - zq^j}{(z; q)_{n+1}} \right|_{z=q^{-j}} = \frac{1}{\prod_{r \neq j} (1 - q^{r-j})}.$$

For $r < j$, put $s = j - r$ and obtain

$$\prod_{r=0}^{j-1} (1 - q^{r-j}) = \prod_{s=1}^j (1 - q^{-s}) = (-1)^j q^{-\binom{j+1}{2}} (q; q)_j.$$

The factors with $r > j$ multiply to $(q; q)_{n-j}$. Thus A_j is the displayed coefficient. Subtracting the proposed right side from the left gives a rational function with no poles and tending to 0 at infinity, hence it is identically zero. $\square$

Proposition 1.9 (Logarithmic derivatives). *Away from its zeros,*

$$\frac{\partial}{\partial a} \log(a; q)_n = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}.$$

If $a = a(x)$ is differentiable, then

$$\frac{d}{dx} \log(a(x); q)_n = -a'(x) \sum_{j=0}^{n-1} \frac{q^j}{1 - a(x)q^j}.$$

Proof. Differentiate the logarithm of the finite product term by term. □

1.5 Infinite products and Lambert series

Theorem 1.10 (Convergence and zero set). *Let $0 < |q| < 1$. The product $(a; q)_\infty$ converges locally uniformly as a function of $a \in \mathbb{C}$ and therefore defines an entire function. Its zeros are precisely*

$$a = q^{-j}, \quad j = 0, 1, 2, \dots,$$

and all are simple.

Proof. On a compact set $K \subset \mathbb{C}$ let $M = \max_{a \in K} |a|$. Then

$$\sum_{j=0}^{\infty} \sup_{a \in K} |aq^j| \leq \frac{M}{1 - |q|} < \infty.$$

The standard convergence criterion for infinite products shows that $\prod_j (1 - aq^j)$ converges uniformly on K ; equivalently, one may apply uniform convergence to the tails after choosing an analytic branch of the logarithm near 1. Uniform limits of holomorphic functions are holomorphic, so the limit is entire.

If $a = q^{-j}$, one factor vanishes. Conversely, if no factor vanishes, then the tail product is nonzero because the sum of the absolute deviations from 1 converges, while the finite initial product is nonzero. At $a = q^{-j}$ exactly one factor vanishes and its derivative is $-q^j \neq 0$, whereas all other factors are nonzero; the zero is simple. □

1.5.1 Complex orders

Finite and negative integral indices extend naturally to a complex order once an infinite product is available. Fix $0 < |q| < 1$ and a branch of the logarithm at q , denoted by $\log q$, and write $q^\alpha = \exp(\alpha \log q)$.

Definition 1.11. For $\alpha \in \mathbb{C}$, define

$$(a; q)_\alpha := \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \tag{1.1}$$

whenever the denominator is nonzero.

Proposition 1.12 (Complex concatenation). *Whenever the displayed quantities are defined,*

$$(a; q)_{\alpha+\beta} = (a; q)_\alpha (aq^\alpha; q)_\beta. \tag{1.2}$$

In particular,

$$(a; q)_{\alpha+1} = (1 - aq^\alpha)(a; q)_\alpha. \tag{1.3}$$

For integral α , (1.1) agrees with the finite and negative-index definitions already given.

Proof. By definition,

$$(a; q)_\alpha (aq^\alpha; q)_\beta = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty} \frac{(aq^\alpha; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty} = \frac{(a; q)_\infty}{(aq^{\alpha+\beta}; q)_\infty}.$$

This is (1.2). Taking $\beta = 1$ gives (1.3). If $\alpha = n \geq 0$, the tail factorization $(a; q)_\infty = (a; q)_n (aq^n; q)_\infty$ recovers the finite product. If $\alpha = -n < 0$, the same relation, applied after shifting by $-n$, gives $(a; q)_{-n} = 1/(aq^{-n}; q)_n$. $\square$

Proposition 1.13 (Derivative with respect to the order). *On every domain avoiding the poles of (1.1),*

$$\frac{\partial}{\partial \alpha} \log(a; q)_\alpha = (\log q) \sum_{j=0}^{\infty} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}. \quad (1.4)$$

The series is locally uniformly convergent.

Proof. Only the denominator in (1.1) depends on α . Differentiate its locally uniformly convergent logarithm term by term:

$$-\frac{\partial}{\partial \alpha} \sum_{j \geq 0} \log(1 - aq^{\alpha+j}) = (\log q) \sum_{j \geq 0} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}.$$

On a compact set avoiding the poles, the denominators of the tail are bounded away from zero and the numerators form a geometric majorant, proving local uniform convergence. $\square$

Theorem 1.14 (Lambert-series expansion). *If $|q| < 1$ and $|a| < 1$, then*

$$\log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^m}{m(1 - q^m)}.$$

Consequently,

$$\frac{\partial}{\partial a} \log(a; q)_\infty = - \sum_{m=1}^{\infty} \frac{a^{m-1}}{1 - q^m} = - \sum_{j=0}^{\infty} \frac{q^j}{1 - aq^j}.$$

All series converge locally uniformly in $|a| < 1$.

Proof. For $|a| < 1$, use $\log(1 - u) = - \sum_{m \geq 1} u^m/m$ and absolute convergence:

$$\sum_{j \geq 0} \sum_{m \geq 1} \frac{|a|^m |q|^j}{m} = \sum_{m \geq 1} \frac{|a|^m}{m(1 - |q|^m)} < \infty.$$

Thus the order of summation may be exchanged:

$$\sum_{j \geq 0} \log(1 - aq^j) = - \sum_{m \geq 1} \frac{a^m}{m} \sum_{j \geq 0} q^{jm}.$$

The geometric sum gives the first formula. Local uniform convergence permits termwise differentiation. Expanding $(1 - aq^j)^{-1}$ geometrically gives the second form of the derivative. $\square$

Corollary 1.15 (Divisor sums). *As a formal power series,*

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

where $\sigma_1(n) = \sum_{d|n} d$.

Proof. Logarithmic differentiation gives

$$-q \frac{d}{dq} \log(q; q)_\infty = \sum_{j \geq 1} \frac{jq^j}{1 - q^j} = \sum_{j \geq 1} \sum_{r \geq 1} jq^{jr}.$$

The coefficient of q^n is the sum of all j dividing n . □

1.6 The dilogarithmic limit

The singular regime $q \rightarrow 1^-$ turns an infinite product into an exponential of order $(1 - q)^{-1}$. The dilogarithm is the natural leading term.

Theorem 1.16 (Asymptotic expansion near $q = 1$). *Fix $0 < \rho < 1$ and an integer $N \geq 1$. Uniformly for $|a| \leq \rho$, as $t \rightarrow 0^+$,*

$$\begin{aligned} \log(a; e^{-t})_\infty &= -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) \\ &\quad - \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} t^{2r-1} \text{Li}_{2-2r}(a) + O_{\rho, N}(t^{2N-1}), \end{aligned}$$

where B_{2r} are Bernoulli numbers and $\text{Li}_s(a) = \sum_{m \geq 1} a^m / m^s$ for $|a| < 1$.

Proof. By Theorem 1.14,

$$\log(a; e^{-t})_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - e^{-mt})}.$$

The Bernoulli expansion at the origin is

$$\frac{1}{1 - e^{-x}} = \frac{1}{x} + \frac{1}{2} + \sum_{r=1}^{N-1} \frac{B_{2r}}{(2r)!} x^{2r-1} + R_N(x),$$

with $R_N(x) = O_N(x^{2N-1})$ for $0 \leq x \leq 1$. Substitution of the displayed polynomial terms gives respectively

$$-\frac{1}{t} \sum_{m \geq 1} \frac{a^m}{m^2}, \quad -\frac{1}{2} \sum_{m \geq 1} \frac{a^m}{m} = \frac{1}{2} \log(1 - a),$$

and the polylogarithmic terms in the statement.

It remains to bound the total remainder. Split the m -sum at $m \leq t^{-1}$. In this range,

$$\sum_{m \leq t^{-1}} \frac{|a|^m}{m} |R_N(mt)| \leq C_N t^{2N-1} \sum_{m \geq 1} \rho^m m^{2N-2} = O_{\rho, N}(t^{2N-1}).$$

For $m > t^{-1}$, subtracting finitely many asymptotic terms leaves a quantity bounded by a polynomial in mt plus $(1 - e^{-mt})^{-1}$; the factor ρ^m makes the tail exponentially small in $1/t$, hence smaller than every power of t . This proves the uniform remainder estimate. □

1.7 Classical shifted factorials as a limit

Proposition 1.17 (Finite classical limit). *For fixed $x \in \mathbb{C}$ and $n \in \mathbb{N}_0$,*

$$\lim_{q \rightarrow 1} \frac{(q^x; q)_n}{(1 - q)^n} = x(x + 1) \cdots (x + n - 1) = (x)_n,$$

where the limit is taken through values for which a consistent branch of q^x is chosen.

Proof. Write

$$\frac{(q^x; q)_n}{(1 - q)^n} = \prod_{j=0}^{n-1} \frac{1 - q^{x+j}}{1 - q}.$$

Each factor tends to $x + j$, since $q^u = 1 + u(q - 1) + o(q - 1)$ for fixed u . □

Chapter 2

q -integers and Gaussian coefficients

2.1 Definitions

Definition 2.1. For $n \in \mathbb{N}_0$ define the q -integer and q -factorial by

$$[n]_q = \frac{1 - q^n}{1 - q} = 1 + q + \cdots + q^{n-1}, \quad [n]_q! = \prod_{j=1}^n [j]_q.$$

For $0 \leq k \leq n$, the *Gaussian coefficient* is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Outside $0 \leq k \leq n$ it is defined to be 0.

The notation is initially a rational expression, but the recurrences below show that it is a polynomial in q with nonnegative integral coefficients.

Proposition 2.2 (Product forms). For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k} = \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j}.$$

Proof. Cancel $(q; q)_{n-k}$ from $(q; q)_n$ using [Theorem 1.2](#). □

2.2 The two Pascal recurrences

Theorem 2.3 (q -Pascal recurrences). For $1 \leq k \leq n-1$,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q, \tag{2.1}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k \end{bmatrix}_q. \tag{2.2}$$

Both identities remain valid at the boundary under the zero convention.

Proof. Put the two terms on the right of (2.1) over the common denominator $(q; q)_k(q; q)_{n-k}$. Their numerator is

$$(q; q)_{n-1}((1 - q^k) + q^k(1 - q^{n-k})) = (q; q)_{n-1}(1 - q^n) = (q; q)_n.$$

The proof of (2.2) is the same, using $q^{n-k}(1 - q^k) + (1 - q^{n-k}) = 1 - q^n$. $\square$

Corollary 2.4 (Polynomiality and positivity). *For all $0 \leq k \leq n$, the Gaussian coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$ belongs to $\mathbb{N}_0[q]$.*

Proof. Induct on n . The boundary values are 1. Either recurrence in Theorem 2.3 expresses an interior coefficient as a sum of two polynomials with nonnegative integral coefficients. $\square$

2.3 Symmetry, reciprocity, and degree

Theorem 2.5 (Elementary structure). *For $0 \leq k \leq n$,*

- (i) $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$;
- (ii) $\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q$;
- (iii) $\begin{bmatrix} n \\ k \end{bmatrix}_q$ has degree $k(n-k)$, constant term 1, and leading coefficient 1;
- (iv) its coefficient sequence is palindromic.

Proof. Symmetry is immediate from the factorial definition. For reciprocity, use

$$(q^{-1}; q^{-1})_m = (-1)^m q^{-\binom{m+1}{2}} (q; q)_m$$

for $m = n, k, n-k$. The signs cancel and the exponent is

$$-\binom{n+1}{2} + \binom{k+1}{2} + \binom{n-k+1}{2} = -k(n-k).$$

Since $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is a polynomial with constant term 1, reciprocity says

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}.$$

Hence its degree is at most $k(n-k)$ and its leading coefficient equals its constant coefficient. The product form shows that the order at infinity is exactly $k(n-k)$, so the degree is exact. The same reciprocal identity equates coefficients at complementary degrees and proves palindromicity. $\square$

Corollary 2.6 (Classical limit). *For fixed n, k ,*

$$\lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q = \binom{n}{k}.$$

Proof. Use the q -factorial form and $[j]_q \rightarrow j$. $\square$

2.4 Multinomial coefficients and flags

Definition 2.7. If $n_1, \dots, n_r \geq 0$ and $n_1 + \dots + n_r = n$, define

$$\left[\begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \frac{(q; q)_n}{(q; q)_{n_1} \cdots (q; q)_{n_r}}.$$

Proposition 2.8 (Factorization into binomials). *Let $N_j = n_j + n_{j+1} + \dots + n_r$. Then*

$$\left[\begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \prod_{j=1}^{r-1} \left[\begin{matrix} N_j \\ n_j \end{matrix} \right]_q.$$

In particular, every q -multinomial coefficient is a polynomial with nonnegative integral coefficients.

Proof. Write out the factorial ratios. Consecutive numerator and denominator factors telescope:

$$\prod_{j=1}^{r-1} \frac{(q; q)_{N_j}}{(q; q)_{n_j} (q; q)_{N_{j+1}}} = \frac{(q; q)_n}{\prod_{j=1}^r (q; q)_{n_j}}.$$

Positivity follows from [Theorem 2.4](#). □

2.5 Adjacent ratios and efficient recurrences

Proposition 2.9. *For $0 \leq k < n$,*

$$\frac{\left[\begin{matrix} n \\ k+1 \end{matrix} \right]_q}{\left[\begin{matrix} n \\ k \end{matrix} \right]_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}.$$

For $0 \leq k \leq n-1$,

$$\frac{\left[\begin{matrix} n \\ k \end{matrix} \right]_q}{\left[\begin{matrix} n-1 \\ k \end{matrix} \right]_q} = \frac{1 - q^n}{1 - q^{n-k}}.$$

Proof. Cancel common factors in the product or factorial definitions. □

These ratios are useful numerically, but near roots of unity they hide cancellations. The polynomial recurrences of [Theorem 2.3](#) are then preferable.

2.6 Coefficient moments

Write

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{j=0}^D c_j q^j, \quad D = k(n-k),$$

and choose an exponent X with probability $\Pr(X = j) = c_j / \left[\begin{matrix} n \\ k \end{matrix} \right]_q$. The combinatorial meaning of X will be established in [Chapter 5](#); its moments can already be found algebraically.

Theorem 2.10 (Mean and variance). *For the preceding distribution,*

$$\mathbb{E}X = \frac{k(n-k)}{2}, \quad \text{Var}(X) = \frac{k(n-k)(n+1)}{12}.$$

Equivalently,

$$\left. \frac{d}{dq} \left[\begin{matrix} n \\ k \end{matrix} \right]_q \right|_{q=1} = \frac{k(n-k)}{2} \left[\begin{matrix} n \\ k \end{matrix} \right].$$

Proof. The palindromic identity $c_j = c_{D-j}$ immediately gives $\mathbb{E}X = D/2$ and the derivative formula.

For the variance, use the moment-generating function

$$M(t) = \frac{[n]_q e^t}{\binom{n}{k}_q} = \prod_{i=1}^k \frac{e^{(n-k+i)t} - 1}{(n-k+i)t} \frac{it}{e^{it} - 1}.$$

The local expansion

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \frac{u^2}{24} + O(u^3)$$

gives

$$\log M(t) = \frac{t}{2} \sum_{i=1}^k (n-k) + \frac{t^2}{24} \sum_{i=1}^k ((n-k+i)^2 - i^2) + O(t^3).$$

The first sum is $k(n-k)$. The second equals

$$k(n-k)^2 + 2(n-k) \frac{k(k+1)}{2} = k(n-k)(n+1).$$

Since the first and second derivatives of $\log M$ at 0 are the mean and variance, respectively, the result follows. $\square$

Chapter 3

Finite q -binomial theorems and inversion

3.1 Euler's finite product expansions

Theorem 3.1 (Finite q -binomial theorem). *For $n \geq 0$,*

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \quad (3.1)$$

$$(-z; q)_n = \sum_{k=0}^n q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (3.2)$$

Proof. It is enough to prove the first identity; replace z by $-z$ for the second. The case $n = 0$ is clear. Suppose the formula holds for n . Then

$$(z; q)_{n+1} = (1 - zq^n)(z; q)_n.$$

The coefficient of z^k on the right is

$$(-1)^k q^{\binom{k}{2}} \left(\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q \right).$$

By the second q -Pascal recurrence, the parenthesis is $\begin{bmatrix} n+1 \\ k \end{bmatrix}_q$. This completes the induction. $\square$

Corollary 3.2 (Alternating sums). *For $n > 0$,*

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0.$$

More generally,

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2} + mk} \begin{bmatrix} n \\ k \end{bmatrix}_q = (q^m; q)_n.$$

Proof. Set $z = 1$ and $z = q^m$, respectively, in (3.1). $\square$

Corollary 3.3 (Weighted subset sum). *For $0 \leq k \leq n$,*

$$\sum_{0 \leq i_1 < \dots < i_k \leq n-1} q^{i_1 + \dots + i_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Equivalently,

$$\sum_{1 \leq j_1 < \dots < j_k \leq n} q^{j_1 + \dots + j_k} = q^{\binom{k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Proof. Expand $\prod_{i=0}^{n-1} (1 + zq^i)$ by choosing either 1 or zq^i from each factor, and compare the coefficient of z^k with (3.2). $\square$

3.2 Reciprocal finite products

Theorem 3.4 (Reciprocal finite q -binomial theorem). *For $n \geq 0$ and $|z| < 1$,*

$$\frac{1}{(z; q)_{n+1}} = \sum_{k=0}^{\infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k.$$

The identity also holds formally in $\mathbb{Z}[q][[z]]$.

Proof. Let $H_n(z) = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$. The second q -Pascal recurrence gives

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q - q^n \begin{bmatrix} n+k-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

Therefore

$$(1 - zq^n)H_n(z) = H_{n-1}(z),$$

where $H_0(z) = \sum_{k \geq 0} z^k = (1 - z)^{-1}$. Iterating yields

$$H_n(z) = \frac{1}{(1 - z)(1 - zq) \cdots (1 - zq^n)}.$$

All operations are valid formally; analytically they are justified for $|z| < 1$. $\square$

3.3 q -Vandermonde convolution

Theorem 3.5 (q -Vandermonde convolution). *For $m, n, r \in \mathbb{N}_0$,*

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_{j \in \mathbb{Z}} q^{(m-j)(r-j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q, \quad (3.3)$$

$$= \sum_{j \in \mathbb{Z}} q^{j(n-r+j)} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ r-j \end{bmatrix}_q. \quad (3.4)$$

Only finitely many terms are nonzero.

Proof. Factor

$$(-z; q)_{m+n} = (-z; q)_m (-zq^m; q)_n$$

and use (3.2). The coefficient of z^r on the left is $q^{\binom{r}{2}} \begin{bmatrix} m+n \\ r \end{bmatrix}_q$. The term with j chosen from the first factor and $r-j$ from the second has exponent

$$\binom{j}{2} + m(r-j) + \binom{r-j}{2}.$$

Subtracting $\binom{r}{2}$ gives $(m-j)(r-j)$, proving (3.3). Splitting the product in the reverse order gives (3.4). $\square$

Corollary 3.6 (Central form). *For $m, n \geq 0$,*

$$\begin{bmatrix} m+n \\ m \end{bmatrix}_q = \sum_j q^{j^2} \begin{bmatrix} m \\ j \end{bmatrix}_q \begin{bmatrix} n \\ j \end{bmatrix}_q.$$

Proof. Set $r = m$ in (3.3) and put $\ell = m - j$. Then the exponent becomes ℓ^2 , while symmetry gives $\begin{bmatrix} m \\ m-\ell \end{bmatrix}_q = \begin{bmatrix} m \\ \ell \end{bmatrix}_q$ and the second factor is $\begin{bmatrix} n \\ \ell \end{bmatrix}_q$. $\square$

3.4 q -binomial inversion

Theorem 3.7 (q -binomial inversion). *For sequences (a_n) and (b_n) in a commutative ring containing q , the relations*

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k$$

are equivalent to

$$a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k.$$

Proof. Substitute the first relation into the proposed inverse. The coefficient of a_j becomes

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

The factorial definition gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q.$$

Put $r = n - k$. The coefficient is

$$\begin{bmatrix} n \\ j \end{bmatrix}_q \sum_{r=0}^{n-j} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n-j \\ r \end{bmatrix}_q.$$

By [Theorem 3.2](#), this equals 0 when $n > j$ and 1 when $n = j$. Thus the composition is the identity. The same argument in the opposite order proves equivalence. $\square$

Corollary 3.8 (Matrix orthogonality). *For $0 \leq j \leq n$,*

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \delta_{n,j}.$$

Proof. This is the kernel computation in the preceding proof. $\square$

3.5 A q -Cauchy product identity

Theorem 3.9 (q -Cauchy identity). *For u, v in a commutative ring,*

$$(uv; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (u; q)_k v^k (v; q)_{n-k}.$$

Proof. Induct on n . The assertion is clear at $n = 0$. Let the right side be $C_n(u, v)$. Use

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q + \begin{bmatrix} n \\ k \end{bmatrix}_q$$

and split C_{n+1} accordingly. In the first part shift $k \mapsto k+1$, use $(u; q)_{k+1} = (u; q)_k (1 - uq^k)$, and combine with the second part after using $(v; q)_{n+1-k} = (v; q)_{n-k} (1 - vq^{n-k})$. The bracket that remains is

$$(1 - vq^{n-k}) + vq^{n-k} (1 - uq^k) = 1 - uvq^n,$$

independent of k . Therefore

$$C_{n+1}(u, v) = (1 - uvq^n)C_n(u, v) = (uv; q)_{n+1}.$$

□

Remark 3.10. Taking $u = 0$ recovers the tautology $(0; q)_k = 1$ inside a weighted expansion; taking special limits of u or v produces several finite convolution formulas used later in the proof of the q -Pfaff–Saalschütz summation.

Chapter 4

Weighted binomial coefficients and symmetric functions

The ordinary and Gaussian binomial coefficients are instances of a broader construction in which each available object carries a weight. This point of view, developed systematically by Konvalina, places subset selection, multiset selection, Stirling numbers, and Gaussian coefficients in one framework.

4.1 Elementary and complete weighted coefficients

Definition 4.1. Let $\mathbf{w} = (w_1, \dots, w_n)$ be elements of a commutative ring. Define

$$C_k^n(\mathbf{w}) = \sum_{1 \leq i_1 < \dots < i_k \leq n} w_{i_1} \cdots w_{i_k},$$

$$S_k^n(\mathbf{w}) = \sum_{1 \leq i_1 \leq \dots \leq i_k \leq n} w_{i_1} \cdots w_{i_k}.$$

We set $C_0^n = S_0^n = 1$. Thus C_k^n is the elementary symmetric function $e_k(w_1, \dots, w_n)$ and S_k^n is the complete homogeneous symmetric function $h_k(w_1, \dots, w_n)$.

Theorem 4.2 (Weighted Pascal recurrences). *For $n, k \geq 1$,*

$$C_k^n(\mathbf{w}) = C_k^{n-1}(\mathbf{w}) + w_n C_{k-1}^{n-1}(\mathbf{w}), \quad (4.1)$$

$$S_k^n(\mathbf{w}) = S_k^{n-1}(\mathbf{w}) + w_n S_{k-1}^{n-1}(\mathbf{w}). \quad (4.2)$$

Proof. A k -element subset of $\{1, \dots, n\}$ either omits n , contributing C_k^{n-1} , or contains n , in which case removal of n leaves a $(k-1)$ -element subset and the weight acquires a factor w_n . This proves (4.1).

For multisets, separate those with no occurrence of n from those with at least one occurrence. Removing one occurrence of n from the latter leaves an arbitrary multiset of size $k-1$ drawn from all n types. This proves (4.2). $\square$

Theorem 4.3 (Generating products). *One has*

$$\prod_{i=1}^n (1 + w_i z) = \sum_{k=0}^n C_k^n(\mathbf{w}) z^k, \quad (4.3)$$

$$\prod_{i=1}^n \frac{1}{1 - w_i z} = \sum_{k=0}^{\infty} S_k^n(\mathbf{w}) z^k. \quad (4.4)$$

The second identity is formal in z , or analytic when $|w_i z| < 1$ for all i .

Proof. In the first product, choosing $w_i z$ from exactly the factors indexed by a subset I produces $z^{|I|} \prod_{i \in I} w_i$. Summing over subsets gives (4.3). For the second, expand each factor geometrically:

$$\frac{1}{1 - w_i z} = \sum_{r_i \geq 0} w_i^{r_i} z^{r_i}.$$

The coefficient of z^k sums over weak compositions $r_1 + \cdots + r_n = k$, which are equivalent to multisets of size k . $\square$

Corollary 4.4 (Elementary–complete orthogonality). *For $m \geq 0$,*

$$\sum_{j=0}^m (-1)^j C_j^n(\mathbf{w}) S_{m-j}^n(\mathbf{w}) = \delta_{m,0}.$$

Proof. Multiply $\prod_i (1 - w_i z)$ by its reciprocal and compare the coefficient of z^m , using Theorem 4.3 with z replaced by $-z$ in the elementary generating function. $\square$

4.2 Classical, Stirling, and Gaussian specializations

Theorem 4.5 (A specialization table). *The following identities hold.*

(i) *If $w_i = 1$, then*

$$C_k^n = \binom{n}{k}, \quad S_k^n = \binom{n+k-1}{k}.$$

(ii) *If $w_i = q^{i-1}$, then*

$$C_k^n = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q, \quad S_k^n = \begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q.$$

(iii) *For the unsigned Stirling numbers of the first kind $c(n, k)$ and Stirling numbers of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$,*

$$c(n+1, k+1) = C_{n-k}^n(1, 2, \dots, n),$$

$$\left\{ \begin{smallmatrix} n+k \\ n \end{smallmatrix} \right\} = S_k^n(1, 2, \dots, n).$$

Proof. For $w_i = 1$, C_k^n counts k -subsets. The multiset count is the stars-and-bars number $\binom{n+k-1}{k}$.

For $w_i = q^{i-1}$, the elementary identity is Theorem 3.3. For the complete identity, Theorem 4.3 gives

$$\prod_{i=0}^{n-1} \frac{1}{1 - zq^i} = \frac{1}{(z; q)_n}.$$

The coefficient of z^k is $\begin{bmatrix} n+k-1 \\ k \end{bmatrix}_q$ by Theorem 3.4.

Finally,

$$x(x+1) \cdots (x+n) = \sum_{j=0}^{n+1} c(n+1, j) x^j.$$

The coefficient of x^{k+1} is $e_{n-k}(1, 2, \dots, n)$. The standard ordinary generating function

$$\prod_{i=1}^n \frac{1}{1 - iz} = \sum_{k \geq 0} \left\{ \begin{smallmatrix} n+k \\ n \end{smallmatrix} \right\} z^k$$

follows from the recurrence for Stirling numbers of the second kind, or by induction on n ; comparison with (4.4) proves the last formula. $\square$

4.3 Weighted inversion

Theorem 4.6 (Symmetric-function inversion). *Let (a_m) and (b_m) be sequences. The transformations*

$$b_m = \sum_{j=0}^m S_{m-j}^n(\mathbf{w}) a_j$$

and

$$a_m = \sum_{j=0}^m (-1)^{m-j} C_{m-j}^n(\mathbf{w}) b_j$$

are inverse to each other.

Proof. The generating functions $A(z) = \sum a_m z^m$ and $B(z) = \sum b_m z^m$ satisfy

$$B(z) = A(z) \prod_{i=1}^n (1 - w_i z)^{-1}.$$

Multiplication by $\prod_i (1 - w_i z)$ gives the proposed inverse coefficient formula. Equivalently, use the orthogonality in [Theorem 4.4](#). $\square$

4.4 The subset–subspace analogy

The choice $w_i = q^{i-1}$ is more than a convenient specialization. It replaces the Boolean lattice of subsets by the lattice of subspaces over a finite field. Ordinary binomial coefficients count subsets by cardinality; Gaussian coefficients count subspaces by dimension. The two-term weighted recurrence mirrors the decomposition of a subspace according to its relation with a fixed hyperplane. This analogy is made exact in [Chapter 6](#).

Chapter 5

Partitions, words, paths, and inversion statistics

5.1 Binary words

Let $\mathcal{W}_{n,k}$ be the set of binary words of length n containing k ones. Define

$$\text{inv}(w) = \# \{ (i, j) : i < j, w_i = 1, w_j = 0 \}.$$

Theorem 5.1 (Inversion generating function). *For $0 \leq k \leq n$,*

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \sum_{w \in \mathcal{W}_{n,k}} q^{\text{inv}(w)}.$$

Proof. Let $F_{n,k}(q)$ denote the right side. If the last letter is 1, deletion of it gives a word in $\mathcal{W}_{n-1,k-1}$ and creates no new inversion. If the last letter is 0, deletion gives a word in $\mathcal{W}_{n-1,k}$ and the final zero forms an inversion with each of the k ones, contributing a factor q^k . Hence

$$F_{n,k} = F_{n-1,k-1} + q^k F_{n-1,k}.$$

The boundary values are 1, so $F_{n,k}$ and $\left[\begin{matrix} n \\ k \end{matrix} \right]_q$ obey the same recurrence and initial conditions. $\square$

Corollary 5.2. *The random variable in [Theorem 2.10](#) is the inversion number of a uniformly random word in $\mathcal{W}_{n,k}$.*

Proof. The coefficient of q^j in [Theorem 5.1](#) counts words with j inversions, while $|\mathcal{W}_{n,k}| = \binom{n}{k}$. $\square$

5.2 Partitions inside a rectangle

A partition $\lambda = (\lambda_1, \dots, \lambda_k)$ is said to fit in a $k \times m$ rectangle if

$$m \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0.$$

Its size is $|\lambda| = \lambda_1 + \dots + \lambda_k$.

Theorem 5.3 (Rectangular partition generating function). *For $m, k \geq 0$,*

$$\left[\begin{matrix} m+k \\ k \end{matrix} \right]_q = \sum_{\lambda \subseteq k \times m} q^{|\lambda|}.$$

Proof. Take a word $w \in \mathcal{W}_{m+k,k}$ and let the positions of its ones be $1 \leq i_1 < \cdots < i_k \leq m+k$. Define

$$\lambda_j = m + j - i_j \quad (1 \leq j \leq k).$$

Then $0 \leq \lambda_j \leq m$ and

$$\lambda_j - \lambda_{j+1} = i_{j+1} - i_j - 1 \geq 0.$$

Thus λ fits in the rectangle. Conversely, $i_j = m + j - \lambda_j$ recovers a strictly increasing sequence of positions, so this is a bijection. The number λ_j is exactly the number of zeros to the right of the j th one, hence $|\lambda| = \text{inv}(w)$. Apply [Theorem 5.1](#). $\square$

Corollary 5.4 (Conjugation and complement). *Conjugating Ferrers diagrams proves $\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \left[\begin{smallmatrix} m+k \\ m \end{smallmatrix} \right]_q$. Taking the complement in the rectangle proves*

$$\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_{q^{-1}} = q^{-mk} \left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q.$$

Proof. Conjugation exchanges a $k \times m$ rectangle with an $m \times k$ rectangle without changing size. Complementation sends size r to $mk - r$, so the generating polynomial is reciprocal of degree mk . $\square$

5.3 Lattice paths and area

Encode a binary word by a path from $(0,0)$ to (m,k) , reading 0 as an east step and 1 as a north step. With a suitable choice of side, the number of unit squares between the path and the boundary is the inversion number.

Corollary 5.5 (Area generating function). *The polynomial $\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$ is the generating function of monotone lattice paths from $(0,0)$ to (m,k) by the area under the path:*

$$\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q = \sum_P q^{\text{area}(P)}.$$

Proof. Under the word–path encoding, every north step contributes the number of later east steps, which is precisely the number of squares in its row between the path and the chosen boundary. Summing over north steps gives $\text{inv}(w)$. $\square$

5.4 Durfee squares and the central convolution

The largest square in the northwest corner of a partition diagram is its Durfee square.

Theorem 5.6 (Durfee-square decomposition). *For $m, n \geq 0$,*

$$\left[\begin{smallmatrix} m+n \\ m \end{smallmatrix} \right]_q = \sum_{j=0}^{\min(m,n)} q^{j^2} \left[\begin{smallmatrix} m \\ j \end{smallmatrix} \right]_q \left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]_q.$$

Proof. This is [Theorem 3.6](#); we give the partition proof. A partition in an $m \times n$ rectangle with Durfee square of side j consists of the $j \times j$ square, a partition to its right fitting in a $j \times (n-j)$ rectangle, and a partition below it fitting in an $(m-j) \times j$ rectangle. By [Theorem 5.3](#), their generating functions are

$$q^{j^2} \left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]_q \left[\begin{smallmatrix} m \\ j \end{smallmatrix} \right]_q.$$

The Durfee size is unique, so summing over j counts every partition exactly once. $\square$

5.5 Words with several letters

Let $\mathcal{W}(n_1, \dots, n_r)$ be the words containing n_i copies of the letter i , ordered by $1 < \dots < r$. Let $\text{inv}(w)$ count pairs $i < j$ of positions with $w_i > w_j$.

Theorem 5.7 (Multiset inversion enumerator). *If $n = n_1 + \dots + n_r$, then*

$$\left[\begin{matrix} n \\ n_1, \dots, n_r \end{matrix} \right]_q = \sum_{w \in \mathcal{W}(n_1, \dots, n_r)} q^{\text{inv}(w)}.$$

Proof. Choose the positions of the largest letter r . If their binary indicator has inversion number s , then s counts precisely the inversions in which the larger letter is r . The remaining positions contain a word on $1, \dots, r-1$, and its inversions are independent. Therefore the generating function factors as

$$\left[\begin{matrix} n \\ n_r \end{matrix} \right]_q \left[\begin{matrix} n - n_r \\ n_1, \dots, n_{r-1} \end{matrix} \right]_q.$$

Iterating and using [Theorem 2.8](#) proves the formula. $\square$

5.6 Rogers–Szegő polynomials

Definition 5.8. The Rogers–Szegő polynomial is

$$H_n(z; q) = \sum_{k=0}^n \left[\begin{matrix} n \\ k \end{matrix} \right]_q z^k.$$

Proposition 5.9 (Basic recurrence). *One has*

$$H_{n+1}(z; q) = zH_n(z; q) + H_n(qz; q), \quad H_0(z; q) = 1.$$

Moreover, if $|q| < 1$, then

$$\sum_{n=0}^{\infty} \frac{H_n(z; q)}{(q; q)_n} t^n = \frac{1}{(t; q)_{\infty} (zt; q)_{\infty}}$$

for $|t| < 1$ and $|zt| < 1$.

Proof. Insert the first q -Pascal recurrence into H_{n+1} :

$$\sum_k \left(\left[\begin{matrix} n \\ k-1 \end{matrix} \right]_q + q^k \left[\begin{matrix} n \\ k \end{matrix} \right]_q \right) z^k = zH_n(z; q) + H_n(qz; q).$$

For the generating function, use

$$\frac{\left[\begin{matrix} n \\ k \end{matrix} \right]_q}{(q; q)_n} = \frac{1}{(q; q)_k (q; q)_{n-k}}.$$

Writing $n = k + j$ and separating the double sum gives

$$\left(\sum_{j \geq 0} \frac{t^j}{(q; q)_j} \right) \left(\sum_{k \geq 0} \frac{(zt)^k}{(q; q)_k} \right).$$

Each factor equals the reciprocal infinite product by Euler's identity, proved later in [Theorem 8.2](#); alternatively, the equality follows already by taking $n \rightarrow \infty$ in [Theorem 3.4](#). This gives the stated product. $\square$

Chapter 6

Finite fields, Grassmannians, and rank enumeration

6.1 Counting subspaces

Theorem 6.1 (Gaussian coefficients count subspaces). *Let Q be a prime power. The number of k -dimensional linear subspaces of $\mathbb{F}_Q^n$ is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

Proof. Count ordered linearly independent k -tuples in $\mathbb{F}_Q^n$. The first vector may be any nonzero vector, the second any vector outside the span of the first, and so on, giving

$$(Q^n - 1)(Q^n - Q) \cdots (Q^n - Q^{k-1}).$$

Each k -dimensional subspace has

$$(Q^k - 1)(Q^k - Q) \cdots (Q^k - Q^{k-1})$$

ordered bases. Dividing and cancelling Q^i in corresponding factors yields

$$\prod_{i=0}^{k-1} \frac{Q^{n-i} - 1}{Q^{k-i} - 1} = \prod_{j=1}^k \frac{1 - Q^{n-k+j}}{1 - Q^j} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

□

Theorem 6.2 (Geometric Pascal recurrence). *Let H be a fixed hyperplane in $\mathbb{F}_Q^n$. Then*

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_Q + Q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q.$$

Proof. A k -subspace U is either contained in H or not. The contained subspaces contribute $\begin{bmatrix} n-1 \\ k \end{bmatrix}_Q$. If $U \not\subseteq H$, then $W = U \cap H$ has dimension $k-1$. Fixing W , the quotient $\mathbb{F}_Q^n/W$ has dimension $n-k+1$, and H/W is a hyperplane. The subspace U/W is a line not contained in H/W . The total number of lines in $\mathbb{F}_Q^{n-k+1}$ minus those in its hyperplane is

$$\frac{Q^{n-k+1} - 1}{Q - 1} - \frac{Q^{n-k} - 1}{Q - 1} = Q^{n-k}.$$

There are $\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_Q$ choices for W , proving the formula. □

6.2 Flags and q -multinomial coefficients

Theorem 6.3 (Partial flags). *Let $d_i = n_1 + \cdots + n_i$. The number of flags*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_r = \mathbb{F}_Q^n, \quad \dim(V_i/V_{i-1}) = n_i,$$

is

$$\begin{bmatrix} n \\ n_1, \dots, n_r \end{bmatrix}_Q.$$

Proof. Choose V_1 in $\begin{bmatrix} n \\ n_1 \end{bmatrix}_Q$ ways. In the quotient $\mathbb{F}_Q^n/V_1$, choose V_2/V_1 in $\begin{bmatrix} n-n_1 \\ n_2 \end{bmatrix}_Q$ ways, and continue. The product is the factorization in [Theorem 2.8](#). $\square$

6.3 Matrices of fixed rank

Theorem 6.4 (Rank enumeration). *The number of $m \times n$ matrices of rank r over $\mathbb{F}_Q$ is*

$$N_{m,n,r}(Q) = \begin{bmatrix} m \\ r \end{bmatrix}_Q \begin{bmatrix} n \\ r \end{bmatrix}_Q |\mathrm{GL}_r(\mathbb{F}_Q)|,$$

where

$$|\mathrm{GL}_r(\mathbb{F}_Q)| = \prod_{i=0}^{r-1} (Q^r - Q^i).$$

Equivalently,

$$N_{m,n,r}(Q) = \frac{\prod_{i=0}^{r-1} (Q^m - Q^i)(Q^n - Q^i)}{\prod_{i=0}^{r-1} (Q^r - Q^i)}.$$

Proof. Every rank- r matrix A can be factored as $A = UV$, where U is an $m \times r$ matrix of full column rank and V is an $r \times n$ matrix of full row rank. There are $\prod_{i=0}^{r-1} (Q^m - Q^i)$ choices for U and $\prod_{i=0}^{r-1} (Q^n - Q^i)$ choices for V . Two pairs (U, V) and (U', V') give the same A exactly when

$$U' = UG, \quad V' = G^{-1}V$$

for a unique $G \in \mathrm{GL}_r(\mathbb{F}_Q)$. Thus every matrix is counted $|\mathrm{GL}_r|$ times. Substituting the subspace-count formulas gives the first expression. $\square$

6.4 Schubert cells and a topological meaning

Let $\mathrm{Gr}(k, n)$ be the complex Grassmannian of k -planes in $\mathbb{C}^n$. Its Schubert cell decomposition is indexed by partitions inside a $k \times (n - k)$ rectangle.

Theorem 6.5 (Poincaré polynomial of the Grassmannian). *The integral cohomology of $\mathrm{Gr}(k, n)$ is concentrated in even degrees, and its Poincaré polynomial is*

$$P_{\mathrm{Gr}(k,n)}(t) = \sum_{j \geq 0} \mathrm{rank} H^{2j}(\mathrm{Gr}(k, n); \mathbb{Z}) t^{2j} = \begin{bmatrix} n \\ k \end{bmatrix}_{t^2}.$$

Proof. Fix the standard flag in $\mathbb{C}^n$. Every k -plane has a unique row-reduced echelon matrix. Its pivot positions determine a binary word with k ones, hence a partition λ inside the $k \times (n - k)$ rectangle. Once the pivots are fixed, the entries allowed to vary freely occupy exactly the boxes of λ ; the corresponding stratum is therefore an affine complex space $\mathbb{C}^{|\lambda|}$. Ordering the strata by inclusion of diagrams yields a CW decomposition with one real cell of dimension $2|\lambda|$ for each λ . Since all cells have even dimension, every cellular boundary map is zero. Thus the rank of H^{2j} is the number of such partitions of size j . Apply [Theorem 5.3](#) with $q = t^2$. $\square$

Remark 6.6 (The “same polynomial” principle). For a prime power Q , evaluating $\begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = Q$ counts rational points of a Grassmannian over $\mathbb{F}_Q$. Evaluating it at $q = t^2$ records the even Betti numbers of the complex Grassmannian. This is an elementary instance of the broad principle that polynomial point counts over finite fields often shadow topological invariants over $\mathbb{C}$.

6.5 Growth over finite fields

Proposition 6.7 (Dimension-dominant bounds). *For $Q > 1$ and $0 \leq k \leq n$,*

$$Q^{k(n-k)} \leq \begin{bmatrix} n \\ k \end{bmatrix}_Q \leq \frac{Q^{k(n-k)}}{(Q^{-1}; Q^{-1})_\infty}.$$

Proof. Rewrite the product as

$$\begin{bmatrix} n \\ k \end{bmatrix}_Q = Q^{k(n-k)} \prod_{j=1}^k \frac{1 - Q^{-(n-k+j)}}{1 - Q^{-j}}.$$

Each ratio is at least 1 because $n - k + j \geq j$, proving the lower bound. For the upper bound, discard the numerator factors and note

$$\prod_{j=1}^k (1 - Q^{-j}) \geq \prod_{j=1}^{\infty} (1 - Q^{-j}) = (Q^{-1}; Q^{-1})_\infty.$$

$\square$

Chapter 7

Symmetry and unimodality

Palindromicity follows immediately from reciprocity. Unimodality is subtler: the coefficients rise toward the middle and then fall. A concise conceptual proof comes from the representation theory of $\mathfrak{sl}_2$.

7.1 A representation-theoretic lemma

Let $\mathfrak{sl}_2(\mathbb{C})$ have generators E, F, H with

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

Lemma 7.1 (Structure of finite-dimensional $\mathfrak{sl}_2$ -modules). *Every finite-dimensional complex $\mathfrak{sl}_2$ -module is a direct sum of irreducibles L_d indexed by $d \in \mathbb{N}_0$. The module L_d has dimension $d + 1$ and H -weights*

$$d, d - 2, \dots, -d,$$

each with multiplicity one.

Proof. We include the standard argument. Restrict a representation to the compact real form $\mathfrak{su}_2$. A finite-dimensional Lie-algebra representation integrates to the simply connected compact group $SU(2)$ by exponentiating the representing matrices; the Lie relations ensure consistency of the local exponentials, and simple connectedness removes monodromy. Starting from any Hermitian inner product, average it over $SU(2)$ with normalized Haar measure. The result is invariant. Hence the orthogonal complement of every invariant subspace is invariant, proving complete reducibility.

It remains to classify an irreducible module V . In the invariant Hermitian structure, a suitable scalar multiple of H is self-adjoint, so H is diagonalizable. Choose an eigenvector v whose eigenvalue d has maximal real part. Since E raises an H -weight by 2, maximality gives $Ev = 0$. The commutation relations imply by induction

$$HF^j v = (d - 2j)F^j v, \quad EF^j v = j(d - j + 1)F^{j-1} v.$$

Because V is finite-dimensional, let m be maximal with $F^m v \neq 0$. Applying E to $F^{m+1} v = 0$ gives

$$(m + 1)(d - m)F^m v = 0,$$

so $d = m \in \mathbb{N}_0$. The vectors $v, Fv, \dots, F^d v$ have distinct weights and are linearly independent. Their span is invariant under E, F, H and is nonzero; irreducibility makes it all of V . The displayed weights and multiplicities follow. $\square$

7.2 Exterior powers and Gaussian coefficients

Let L_{n-1} have a weight basis $v_0, \dots, v_{n-1}$ with $Hv_i = (n-1-2i)v_i$.

Proposition 7.2 (Exterior-power character). *For $0 \leq k \leq n$,*

$$\text{ch} \left(\bigwedge^k L_{n-1} \right) (x) = x^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{x^{-2}}.$$

Proof. A basis of the exterior power consists of $v_{i_1} \wedge \dots \wedge v_{i_k}$ with $0 \leq i_1 < \dots < i_k \leq n-1$. Its weight is

$$k(n-1) - 2(i_1 + \dots + i_k).$$

Therefore

$$\text{ch} = x^{k(n-1)} \sum_{i_1 < \dots < i_k} (x^{-2})^{i_1 + \dots + i_k}.$$

Use [Theorem 3.3](#) with $q = x^{-2}$. The prefactor becomes

$$x^{k(n-1)} x^{-2 \binom{k}{2}} = x^{k(n-k)}.$$

□

Theorem 7.3 (Unimodality of Gaussian coefficients). *If*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = c_0 + c_1 q + \dots + c_D q^D, \quad D = k(n-k),$$

then

$$c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$$

and $c_j = c_{D-j}$. Thus the coefficient sequence is symmetric and unimodal.

Proof. Symmetry was proved in [Theorem 2.5](#). By [Theorem 7.1](#), decompose

$$\bigwedge^k L_{n-1} \cong \bigoplus_{d \geq 0} m_d L_d.$$

All weights in this exterior power have the same parity as D . The character of one summand L_d , placed inside the ambient sequence of weights

$$D, D-2, \dots, -D,$$

contributes a consecutive block of ones centered at weight 0: it is zero until weight d is reached, then one at every step down to $-d$, then zero again. As one moves from weight D toward 0, such a block can turn on but cannot turn off. A nonnegative sum of these centered blocks therefore has weakly increasing multiplicities toward the middle.

By [Theorem 7.2](#), the multiplicity of weight $D-2j$ is exactly c_j . Hence $c_0 \leq c_1 \leq \dots \leq c_{\lfloor D/2 \rfloor}$. □

Remark 7.4. The proof explains more than the inequality: the successive differences $c_j - c_{j-1}$ record multiplicities of irreducible summands with prescribed highest weight. Stronger strict-unimodality results require additional arguments and have a small family of exceptional cases; ordinary unimodality is exactly what the $\mathfrak{sl}_2$ decomposition supplies for free.

Part II

Infinite series and basic hypergeometric summation

Chapter 8

The infinite q -binomial theorem

8.1 A functional-equation proof

Theorem 8.1 (Infinite q -binomial theorem). *Let $|q| < 1$ and $|z| < 1$. Then*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

The convergence is locally uniform in (a, z) on compact subsets of $\mathbb{C} \times \{|z| < 1\}$.

Proof. Set

$$F(z) = \sum_{n=0}^{\infty} c_n z^n, \quad c_n = \frac{(a; q)_n}{(q; q)_n}.$$

Since $c_n/c_{n-1} = (1 - aq^{n-1})/(1 - q^n) \rightarrow 1$, the radius of convergence is 1 unless the series terminates; in every case it converges for $|z| < 1$. The coefficient recurrence is equivalent to

$$(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}.$$

Comparing coefficients shows that

$$(1 - z)F(z) = (1 - az)F(qz), \quad F(0) = 1.$$

Now put $G(z) = (az; q)_{\infty}/(z; q)_{\infty}$. Using $(z; q)_{\infty} = (1 - z)(qz; q)_{\infty}$ gives the same functional equation and $G(0) = 1$.

To prove uniqueness, iterate the functional equation:

$$F(z) = \prod_{j=0}^{N-1} \frac{1 - azq^j}{1 - zq^j} F(q^N z).$$

As $N \rightarrow \infty$, $q^N z \rightarrow 0$, so $F(q^N z) \rightarrow 1$. The finite product tends to $G(z)$. Hence $F = G$. □

Corollary 8.2 (Euler's two infinite identities). *For $|q| < 1$,*

$$(z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q; q)_n} z^n, \quad z \in \mathbb{C}, \quad (8.1)$$

$$\frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n}, \quad |z| < 1. \quad (8.2)$$

Proof. Set $a = 0$ in [Theorem 8.1](#) to obtain (8.2). For (8.1), start with the finite identity

$$(z; q)_N = \sum_{n=0}^N (-1)^n q^{\binom{n}{2}} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n.$$

For fixed n , $\begin{bmatrix} N \\ n \end{bmatrix}_q \rightarrow 1/(q; q)_n$ as $N \rightarrow \infty$. The limiting series converges for every z because its successive-term ratio has magnitude asymptotic to $|z||q|^n$. To justify passage to the limit uniformly on $|z| \leq R$, note that

$$\left| \begin{bmatrix} N \\ n \end{bmatrix}_q \right| \leq \frac{\prod_{j \geq 1} (1 + |q|^j)}{\inf_{r \geq 0} |(q; q)_r|},$$

where the denominator is positive because $(q; q)_r \rightarrow (q; q)_\infty \neq 0$. Thus the summands, extended by zero for $n > N$, are dominated by a constant multiple of $|q|^{\binom{n}{2}} R^n$. Dominated convergence, together with local uniform convergence of the products, now permits passage to the limit. $\square$

Corollary 8.3 (A ratio expansion). *For $|z| < 1$,*

$$\frac{(a_1 z; q)_\infty \cdots (a_r z; q)_\infty}{(b_1 z; q)_\infty \cdots (b_s z; q)_\infty}$$

can be expanded by multiplying r copies of (8.1) and s copies of (8.2). In particular, every coefficient is a finite convolution of q -Pochhammer ratios.

Proof. Each numerator factor has an entire series expansion and each reciprocal denominator factor converges for $|b_j z| < 1$. Cauchy multiplication is valid in their common disk of absolute convergence. $\square$

8.2 Formal and analytic interpretations

Proposition 8.4 (Formal version). *If q is an indeterminate, then*

$$\frac{(az; q)_\infty}{(z; q)_\infty} = \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} z^n$$

holds in the q -adically complete ring $\mathbb{Q}(a)[[q]][[z]]$. More generally, it remains valid after any specialization into a complete topological ring for which the displayed products and series converge.

Proof. The products are well defined q -adically: apart from their initial factors, the j th factors differ from 1 by a multiple of q^j . For a formal power series $F(z) = \sum c_n z^n$, the equation

$$(1 - z)F(z) = (1 - az)F(qz), \quad c_0 = 1,$$

determines c_n recursively by $(1 - q^n)c_n = (1 - aq^{n-1})c_{n-1}$, because $1 - q^n$ is a unit in $\mathbb{Q}(a)[[q]]$. Both sides satisfy this equation, so their coefficients agree. The same uniqueness argument works in every stated complete specialization. $\square$

8.3 Limits in the number of factors

Proposition 8.5. *For fixed k and $|q| < 1$,*

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}.$$

More generally,

$$\lim_{n \rightarrow \infty} \begin{bmatrix} n+k \\ k \end{bmatrix}_q = \frac{1}{(q; q)_k}.$$

Proof. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k}.$$

Every numerator factor tends to 1. □

Remark 8.6. The two Euler identities are therefore the stable limits of the finite product and reciprocal-product formulas. This stability is one reason finite q -binomial identities so often serve as polynomial approximations to infinite q -series identities.

Chapter 9

Basic hypergeometric series

9.1 Definition and convergence

Definition 9.1. The basic hypergeometric series is

$${}_r\phi_s \left[\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right] = \sum_{n=0}^{\infty} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left((-1)^n q^{\binom{n}{2}} \right)^{1+s-r} z^n,$$

provided no denominator factor vanishes. If one numerator parameter is q^{-N} , the series terminates at $n = N$.

Proposition 9.2 (Elementary convergence classification). *Assume the series does not terminate and the denominator parameters avoid poles.*

- (i) *If $r \leq s$, the series converges for every z .*
- (ii) *If $r = s + 1$, its radius of convergence is 1.*
- (iii) *If $r > s + 1$, the terms generally fail to tend to zero for $z \neq 0$; such series are normally interpreted only when terminating or formally.*

Proof. Let T_n be the n th summand. Since $(1 - aq^n) \rightarrow 1$ for each fixed parameter,

$$\frac{T_{n+1}}{T_n} = z \frac{\prod_{i=1}^r (1 - a_i q^n)}{(1 - q^{n+1}) \prod_{j=1}^s (1 - b_j q^n)} (-1)^{1+s-r} q^{n(1+s-r)}.$$

The ratio tends to 0 when $r \leq s$, to z when $r = s + 1$, and has unbounded magnitude when $r > s + 1$ and $z \neq 0$. Apply the ratio test. $\square$

The infinite q -binomial theorem is the evaluation

$${}_1\phi_0 \left[\begin{matrix} a \\ - \end{matrix}; q, z \right] = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

9.2 Heine's transformation

Theorem 9.3 (Heine transformation). *Initially for $|z| < 1$ and $|b| < 1$,*

$${}_2\phi_1 \left[\begin{matrix} a, b \\ c \end{matrix}; q, z \right] = \frac{(b, az; q)_{\infty}}{(c, z; q)_{\infty}} {}_2\phi_1 \left[\begin{matrix} c/b, z \\ az \end{matrix}; q, b \right].$$

The identity extends wherever both sides are defined by analytic continuation.

Proof. Use the tail-factor identity to write

$$\frac{(b; q)_n}{(c; q)_n} = \frac{(b; q)_\infty (cq^n; q)_\infty}{(c; q)_\infty (bq^n; q)_\infty}.$$

By the infinite q -binomial theorem,

$$\frac{(cq^n; q)_\infty}{(bq^n; q)_\infty} = \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k q^{nk}.$$

Insert this into the ${}_2\phi_1$ series and interchange the absolutely convergent sums:

$$\frac{(b; q)_\infty}{(c; q)_\infty} \sum_{k \geq 0} \frac{(c/b; q)_k}{(q; q)_k} b^k \sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} (zq^k)^n.$$

The inner sum is

$$\frac{(azq^k; q)_\infty}{(zq^k; q)_\infty} = \frac{(az; q)_\infty}{(z; q)_\infty} \frac{(z; q)_k}{(az; q)_k}.$$

Collecting factors yields the claimed transformation. Both sides are meromorphic functions of the parameters, so equality extends from the initial open domain by the identity theorem. $\square$

9.3 The q -Gauss sum

Theorem 9.4 (q -Gauss summation). *If $|c/(ab)| < 1$, then*

$${}_2\phi_1 \left[\begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right] = \frac{(c/a, c/b; q)_\infty}{(c, c/(ab); q)_\infty}.$$

Proof. Set $z = c/(ab)$ in [Theorem 9.3](#). Then $az = c/b$, so one numerator parameter of the transformed series cancels its denominator parameter:

$${}_2\phi_1 \left[\begin{matrix} c/b, c/(ab) \\ c/b \end{matrix}; q, b \right] = {}_1\phi_0 \left[\begin{matrix} c/(ab) \\ - \end{matrix}; q, b \right] = \frac{(c/a; q)_\infty}{(b; q)_\infty}.$$

Multiplication by the prefactor in Heine's transformation gives the result. The initially stronger restrictions are removed by analytic continuation within the convergence domain. $\square$

9.4 A second finite Cauchy identity

Lemma 9.5 (Cauchy convolution II). *For a, b, c in a commutative ring,*

$$(ac; q)_n (bc; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k c^k (c; q)_{n-k} (abcq^k; q)_{n-k}.$$

Proof. Apply the q -Cauchy identity [Theorem 3.9](#) with $u = bq^k$, $v = ac$, and length $n - k$:

$$(abcq^k; q)_{n-k} = \sum_{r=0}^{n-k} \begin{bmatrix} n-k \\ r \end{bmatrix}_q (bq^k; q)_r (ac)^r (ac; q)_{n-k-r}.$$

Insert this in the right side of the proposed identity. Use

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n-k \\ r \end{bmatrix}_q = \begin{bmatrix} n \\ k+r \end{bmatrix}_q \begin{bmatrix} k+r \\ k \end{bmatrix}_q, \quad (b; q)_k (bq^k; q)_r = (b; q)_{k+r}.$$

Put $m = k + r$ and use

$$(c; q)_{n-k} = (c; q)_{n-m} (cq^{n-m}; q)_{m-k}.$$

The double sum becomes

$$\sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} (ac; q)_{n-m} \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q (a; q)_k a^{m-k} (cq^{n-m}; q)_{m-k}.$$

In the inner sum replace $j = m - k$ and apply [Theorem 3.9](#) with $u = cq^{n-m}$ and $v = a$; it equals $(acq^{n-m}; q)_m$. Hence

$$(ac; q)_{n-m} (acq^{n-m}; q)_m = (ac; q)_n$$

can be factored out. The remaining sum is

$$(ac; q)_n \sum_{m=0}^n \begin{bmatrix} n \\ m \end{bmatrix}_q (b; q)_m c^m (c; q)_{n-m} = (ac; q)_n (bc; q)_n$$

by one final application of the q -Cauchy identity. □

9.5 The q -Pfaff–Saalschütz sum

Theorem 9.6 (q -Pfaff–Saalschütz summation). For $n \in \mathbb{N}_0$,

$${}_3\phi_2 \left[\begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q \right] = \frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n}.$$

Proof. In [Theorem 9.5](#), replace its parameters by

$$a \mapsto a, \quad b \mapsto b, \quad c \mapsto \frac{c}{ab}.$$

This gives

$$\frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a, b; q)_k \left(\frac{c}{ab} \right)^k \frac{(c/(ab); q)_{n-k} (cq^k; q)_{n-k}}{(c, c/(ab); q)_n}.$$

We show that the k th summand is the k th term of the stated ${}_3\phi_2$. The tail identities give

$$\frac{(cq^k; q)_{n-k}}{(c; q)_n} = \frac{1}{(c; q)_k}.$$

Also, reversal of the two finite products gives

$$(q^{-n}; q)_k = (-1)^k q^{-nk + \binom{k}{2}} (q^{n-k+1}; q)_k,$$

$$(abq^{1-n}/c; q)_k = \left(-\frac{ab}{c} \right)^k q^{k(1-n) + \binom{k}{2}} (cq^{n-k}/(ab); q)_k.$$

Therefore

$$\begin{aligned}
 & \frac{(q^{-n}; q)_k q^k}{(q; q)_k (abq^{1-n}/c; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k}} \\
 &= \left(\frac{c}{ab}\right)^k \frac{(q^{n-k+1}; q)_k}{(q; q)_k} \frac{(c/(ab); q)_n}{(c/(ab); q)_{n-k} (cq^{n-k}/(ab); q)_k} \\
 &= \left[\begin{matrix} n \\ k \end{matrix} \right]_q \left(\frac{c}{ab}\right)^k.
 \end{aligned}$$

Substitution yields exactly

$$\sum_{k=0}^n \frac{(a, b, q^{-n}; q)_k}{(q, c, abq^{1-n}/c; q)_k} q^k.$$

□

9.6 The two q -Chu–Vandermonde sums

Corollary 9.7 (q -Chu–Vandermonde). For $n \in \mathbb{N}_0$,

$${}_2\phi_1 \left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, \frac{cq^n}{a} \right] = \frac{(c/a; q)_n}{(c; q)_n}, \quad (9.1)$$

$${}_2\phi_1 \left[\begin{matrix} q^{-n}, a \\ c \end{matrix}; q, q \right] = a^n \frac{(c/a; q)_n}{(c; q)_n}. \quad (9.2)$$

Proof. For (9.1), put $b = q^{-n}$ in the q -Gauss sum. The infinite products telescope:

$$\frac{(cq^n, c/a; q)_\infty}{(c, cq^n/a; q)_\infty} = \frac{(c/a; q)_n}{(c; q)_n}.$$

Because the series terminates, no convergence restriction remains.

For (9.2), let $b \rightarrow 0$ in Theorem 9.6, with its other numerator parameter equal to a . Then $(b; q)_k \rightarrow 1$ and $(abq^{1-n}/c; q)_k \rightarrow 1$, so the left side becomes the displayed ${}_2\phi_1$. On the right,

$$\frac{(c/b; q)_n}{(c/(ab); q)_n} \rightarrow a^n$$

by comparing the leading factors of the two degree- n polynomials in $1/b$. □

9.7 A hierarchy of mechanisms

The proofs in this chapter form a hierarchy:

$$\text{finite } q\text{-Pascal} \implies q\text{-Cauchy} \implies q\text{-Pfaff–Saalschütz},$$

while

$$\text{infinite } q\text{-binomial} \implies \text{Heine} \implies q\text{-Gauss} \implies \text{first } q\text{-Chu–Vandermonde}.$$

The finite chain is valid in a commutative ring after denominators are cleared. The infinite chain requires convergence or a suitable complete formal topology. This distinction becomes important in computer formalization; see Chapter 24.

Chapter 10

Bilateral series and Ramanujan's ${}_1\psi_1$ sum

10.1 Definition

For integral n , negative q -shifted factorials are understood as in [Theorem 1.6](#). Define

$${}_1\psi_1 \left[\begin{matrix} a \\ b \end{matrix}; q, z \right] = \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(b; q)_n} z^n.$$

The positive tail behaves like a geometric series in z , while the negative tail behaves like a geometric series in $b/(az)$.

Proposition 10.1 (Convergence annulus). *If denominator factors do not vanish, the ${}_1\psi_1$ series converges absolutely in*

$$|b/a| < |z| < 1.$$

Proof. For $n \rightarrow +\infty$, the ratio of consecutive terms tends to z . For the negative tail put $n = -m$ and use the negative-index formula. The ratio of successive terms in m tends to $b/(az)$. The ratio test gives the annulus. $\square$

Theorem 10.2 (Ramanujan's ${}_1\psi_1$ summation). *For $|b/a| < |z| < 1$,*

$${}_1\psi_1 \left[\begin{matrix} a \\ b \end{matrix}; q, z \right] = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}.$$

Proof. We first prove the formula for $b = q^{m+1}$, where $m \in \mathbb{N}_0$. In that case $1/(b; q)_n = 0$ for $n < -m$ in the limiting sense, so shift $n = k - m$:

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(q^{m+1}; q)_n} z^n &= z^{-m} \frac{(a; q)_{-m}}{(q^{m+1}; q)_{-m}} \sum_{k=0}^{\infty} \frac{(aq^{-m}; q)_k}{(q; q)_k} z^k \\ &= z^{-m} \frac{(q; q)_m}{(aq^{-m}; q)_m} \frac{(aq^{-m}z; q)_\infty}{(z; q)_\infty}. \end{aligned}$$

The second equality is the infinite q -binomial theorem. Reverse the finite products:

$$\begin{aligned} (aq^{-m}; q)_m &= (-a)^m q^{-m(m+1)/2} (q/a; q)_m, \\ (aq^{-m}z; q)_m &= (-az)^m q^{-m(m+1)/2} (q/(az); q)_m. \end{aligned}$$

Since $(aq^{-m}z; q)_\infty = (aq^{-m}z; q)_m (az; q)_\infty$, we obtain

$$\frac{(q; q)_m}{(q/a; q)_m} \frac{(q/(az); q)_m (az; q)_\infty}{(z; q)_\infty}.$$

On the other hand, substituting $b = q^{m+1}$ in the proposed product and cancelling tails gives exactly the same expression:

$$\frac{(q; q)_\infty}{(q^{m+1}; q)_\infty} \frac{(q^{m+1}/a; q)_\infty}{(q/a; q)_\infty} \frac{(q/(az); q)_\infty}{(q^{m+1}/(az); q)_\infty} \frac{(az; q)_\infty}{(z; q)_\infty}.$$

For fixed a, z with $|z| < 1$, both sides are analytic functions of b in a neighborhood of 0 contained in $|b| < |az|$. Absolute convergence is locally uniform there by [Theorem 10.1](#). For all sufficiently large m , the points $b = q^{m+1}$ lie in that neighborhood, and they accumulate at 0. The identity theorem therefore proves equality near 0. Both sides are meromorphic functions of b throughout the disk $|b| < |az|$: the series gives the meromorphic continuation termwise on compacta avoiding denominator zeros, and the product quotient is visibly meromorphic. Uniqueness of meromorphic continuation proves the identity everywhere in that disk where both sides are defined, which is precisely the asserted annulus for fixed a and z . $\square$

Remark 10.3. Ramanujan's sum contains the infinite q -binomial theorem and several theta-product identities as limiting or specialized cases. It is a model example of a bilateral identity: the second inequality in the convergence annulus is controlled by the positive tail, while the first is controlled by the negative tail.

Chapter 11

q -difference calculus and the quantum binomial

11.1 The q -derivative

Assume $q \neq 1$. For $x \neq 0$ define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1 - q)x}.$$

At $x = 0$ one uses the continuous extension when it exists.

Proposition 11.1 (Elementary rules). *For $n \geq 1$,*

$$D_q x^n = [n]_q x^{n-1}.$$

The product rules are

$$\begin{aligned} D_q(fg)(x) &= g(x)D_q f(x) + f(qx)D_q g(x), \\ &= f(x)D_q g(x) + g(qx)D_q f(x). \end{aligned}$$

Moreover,

$$D_q(f(q^r x)) = q^r (D_q f)(q^r x)$$

for integral r .

Proof. The monomial identity follows from $(x^n - q^n x^n)/((1 - q)x) = [n]_q x^{n-1}$. For the first product rule, insert and subtract $f(qx)g(x)$ in $f(x)g(x) - f(qx)g(qx)$. For the second, insert and subtract $f(x)g(qx)$. The scaling rule follows directly from the definition. $\square$

Theorem 11.2 (q -Leibniz rule). *For $n \in \mathbb{N}_0$,*

$$D_q^n(fg)(x) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}x)(D_q^{n-k} g)(x).$$

Proof. Induct on n . The assertion is the ordinary product at $n = 0$. Apply D_q to the k th term at level n . By the product and scaling rules, it contributes

$$q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^{k+1} f)(q^{n-k}x)(D_q^{n-k} g)(x)$$

to index $k + 1$, and

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k+1}x)(D_q^{n-k+1}g)(x)$$

to index k . The total coefficient of index k is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the second q -Pascal recurrence. □

11.2 q -Taylor and Newton interpolation

Define the q -falling power based at a by

$$(x-a)_q^m = \prod_{j=0}^{m-1} (x - q^j a), \quad (x-a)_q^0 = 1.$$

Lemma 11.3. For $m \geq 1$,

$$D_q(x-a)_q^m = [m]_q (x-a)_q^{m-1}.$$

Consequently,

$$D_q^j (x-a)_q^m = \frac{[m]_q!}{[m-j]_q!} (x-a)_q^{m-j}$$

for $0 \leq j \leq m$.

Proof. Let $P_m(x) = (x-a)_q^m$. Then

$$P_m(qx) = q^m (x - q^{-1}a) \prod_{j=0}^{m-2} (x - q^j a).$$

Subtracting from $P_m(x) = (x - q^{m-1}a) \prod_{j=0}^{m-2} (x - q^j a)$ gives

$$P_m(x) - P_m(qx) = (1 - q^m)x P_{m-1}(x).$$

Divide by $(1 - q)x$. Iteration proves the second formula. □

Theorem 11.4 (q -Taylor formula). Assume $[j]_q \neq 0$ for $1 \leq j \leq N$. If f is a polynomial of degree at most N , then

$$f(x) = \sum_{k=0}^N \frac{(D_q^k f)(a)}{[k]_q!} (x-a)_q^k.$$

Proof. The polynomials $(x-a)_q^k$, $0 \leq k \leq N$, are monic of distinct degrees and hence form a basis. Write $f = \sum c_k (x-a)_q^k$. Apply D_q^j and set $x = a$. By Theorem 11.3, every term with $k > j$ still contains the factor $x-a$, terms with $k < j$ vanish after too many derivatives, and the $k = j$ term equals $c_j [j]_q!$. Thus $c_j = (D_q^j f)(a) / [j]_q!$. □

Remark 11.5. When the nodes $a, aq, aq^2, \dots$ are distinct, this is a Newton interpolation formula on that geometric grid. In every case covered by the theorem, its basis zeros follow a multiplicative rather than an additive progression.

11.3 The quantum plane

Theorem 11.6 (Noncommutative q -binomial theorem). *Let X, Y belong to an associative algebra and satisfy*

$$YX = qXY.$$

Then

$$(X + Y)^n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q X^{n-k} Y^k.$$

Proof. Induct on n and multiply on the right by $X + Y$. Since $Y^k X = q^k X Y^k$, the coefficient of $X^{n+1-k} Y^k$ becomes

$$q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by the first q -Pascal recurrence. □

Corollary 11.7 (Factorization of q -exponentials). *Assume $|q| < 1$. Define*

$$e_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{[n]_q!} = \frac{1}{((1-q)x; q)_{\infty}}, \quad |(1-q)x| < 1,$$

and

$$E_q(x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} x^n}{[n]_q!} = (- (1-q)x; q)_{\infty}, \quad x \in \mathbb{C}.$$

Then $e_q(x)E_q(-x) = 1$ wherever the first series is defined. If $YX = qXY$, then, as an identity of formal series in total degree,

$$e_q(X + Y) = e_q(X)e_q(Y).$$

Proof. The product formulas and the stated convergence domains follow from Euler's identities after replacing z by $(1-q)x$. Their product is 1. For the noncommutative factorization, apply [Theorem 11.6](#) degree by degree and use

$$\frac{1}{[n]_q!} \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{[n-k]_q! [k]_q!}.$$

Reindex with $r = n - k$ to obtain the Cauchy product of $e_q(X)$ and $e_q(Y)$, with all X factors preceding all Y factors. Since every total degree receives only finitely many contributions, the formal rearrangement is valid. □

Proposition 11.8 (Eigenfunction properties). *Under the hypotheses and within the convergence domains of [Theorem 11.7](#), for scalar λ ,*

$$D_q e_q(\lambda x) = \lambda e_q(\lambda x), \quad D_q E_q(\lambda x) = \lambda E_q(q\lambda x).$$

Proof. Differentiate the series term by term and use $D_q x^n = [n]_q x^{n-1}$. □

11.4 Jackson's q -integral

Assume $0 < q < 1$. Define

$$\int_0^a f(t) d_q t = (1-q)a \sum_{n=0}^{\infty} q^n f(aq^n),$$

whenever the series converges.

Theorem 11.9 (Fundamental theorem of q -calculus). *Let*

$$F(x) = \int_0^x f(t) d_q t.$$

Whenever the defining series converges locally uniformly,

$$D_q F(x) = f(x).$$

Conversely, if $G(aq^n) \rightarrow G(0)$, then

$$\int_0^a D_q G(t) d_q t = G(a) - G(0).$$

Proof. From the definition,

$$F(x) = (1-q)x \sum_{n \geq 0} q^n f(xq^n),$$

while

$$F(qx) = (1-q)x \sum_{n \geq 1} q^n f(xq^n).$$

Their difference is $(1-q)xf(x)$, proving the first identity. For the second,

$$\begin{aligned} \int_0^a D_q G(t) d_q t &= (1-q)a \sum_{n \geq 0} q^n \frac{G(aq^n) - G(aq^{n+1})}{(1-q)aq^n} \\ &= \sum_{n \geq 0} (G(aq^n) - G(aq^{n+1})), \end{aligned}$$

which telescopes to $G(a) - G(0)$. □

Corollary 11.10 (Jackson integration by parts). *Under convergence assumptions,*

$$\int_0^a f(t) D_q g(t) d_q t = f(a)g(a) - f(0)g(0) - \int_0^a g(qt) D_q f(t) d_q t.$$

Proof. Use the product rule $D_q(fg) = fD_q g + g(q \cdot)D_q f$ and integrate with [Theorem 11.9](#). □

Example 11.11 (Monomials). For $\Re(s) > -1$,

$$\int_0^a t^s d_q t = (1-q)a^{s+1} \sum_{n \geq 0} q^{n(s+1)} = \frac{a^{s+1}}{[s+1]_q},$$

where $[u]_q = (1 - q^u)/(1 - q)$ for complex u .

Chapter 12

The q -gamma and q -beta functions

Throughout this chapter $0 < q < 1$ unless stated otherwise.

12.1 Definition and functional equation

Definition 12.1. For $x > 0$, define

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

With $q^z = \exp(z \log q)$, the same product quotient defines a meromorphic function of $z \in \mathbb{C}$; its possible poles occur where $q^{z+j} = 1$ for some $j \in \mathbb{N}_0$. Whenever complex arguments of Γ_q occur below, this meromorphic continuation is intended.

Theorem 12.2 (Basic properties). For $x > 0$,

$$\Gamma_q(1) = 1, \quad \Gamma_q(x+1) = [x]_q \Gamma_q(x).$$

Consequently, for $n \in \mathbb{N}$,

$$\Gamma_q(n) = [n-1]_q!.$$

The recurrence extends as an identity of meromorphic functions of a complex argument.

Proof. The value at 1 is immediate. The tail identity $(q^x; q)_\infty = (1 - q^x)(q^{x+1}; q)_\infty$ gives

$$\frac{\Gamma_q(x+1)}{\Gamma_q(x)} = (1 - q)^{-1} (1 - q^x) = [x]_q.$$

Iteration gives the integral values. The same tail identity is valid for complex x , and hence proves the meromorphic recurrence wherever both sides are finite; equality elsewhere follows in the sense of meromorphic continuation. $\square$

Proposition 12.3 (Logarithmic derivatives and convexity). For $x > 0$,

$$\begin{aligned} \frac{d}{dx} \log \Gamma_q(x) &= -\log(1 - q) + (\log q) \sum_{n=0}^{\infty} \frac{q^{n+x}}{1 - q^{n+x}}, \\ \frac{d^2}{dx^2} \log \Gamma_q(x) &= (\log q)^2 \sum_{n=0}^{\infty} \frac{q^{n+x}}{(1 - q^{n+x})^2} > 0. \end{aligned}$$

Thus Γ_q is strictly log-convex.

Proof. Differentiate the locally uniformly convergent logarithm of the defining infinite product. A second differentiation gives the positive series. $\square$

12.2 A q -Bohr–Mollerup theorem

Theorem 12.4 (q -Bohr–Mollerup characterization). *The function Γ_q is the unique function $f : (0, \infty) \rightarrow (0, \infty)$ satisfying*

$$f(1) = 1, \quad f(x+1) = [x]_q f(x),$$

and such that $\log f$ is convex.

Proof. Existence follows from [Theorems 12.2](#) and [12.3](#). For uniqueness let f have the stated properties and fix $0 < x \leq 1$. Convexity on the interval from n to $n+1$ gives

$$f(n+x) \leq f(n)^{1-x} f(n+1)^x.$$

A comparison of the adjacent secant slopes on $[n+x, n+1]$ and $[n+1, n+2]$ gives

$$f(n+x) \geq \frac{f(n+1)}{[n+1]_q^{1-x}}.$$

Using the recurrence,

$$f(n+x) = f(x) \prod_{j=0}^{n-1} [x+j]_q, \quad f(n) = [n-1]_q!, \quad f(n+1) = [n]_q!,$$

we obtain

$$\frac{[n]_q!}{[n+1]_q^{1-x} \prod_{j=0}^{n-1} [x+j]_q} \leq f(x) \leq \frac{[n-1]_q! [n]_q^x}{\prod_{j=0}^{n-1} [x+j]_q}.$$

The ratio of the upper bound to the lower bound is

$$\left(\frac{[n+1]_q}{[n]_q} \right)^{1-x} \rightarrow 1.$$

Hence the bounds determine $f(x)$ uniquely for $0 < x \leq 1$, and the recurrence determines it on all of $(0, \infty)$. Since Γ_q satisfies the same conditions, $f = \Gamma_q$. $\square$

12.3 Classical limit

Theorem 12.5 (Limit to Euler's gamma function). *For every $x > 0$,*

$$\lim_{q \rightarrow 1^-} \Gamma_q(x) = \Gamma(x).$$

The convergence is locally uniform on $(0, \infty)$.

Proof. It suffices by the functional equations to consider $0 < x \leq 1$. The bounds in the proof of [Theorem 12.4](#) hold for $\Gamma_q(x)$. Fix n and let $q \rightarrow 1^-$. Since $[u]_q \rightarrow u$, they become

$$\begin{aligned} \frac{n!}{(n+1)^{1-x} x(x+1) \cdots (x+n-1)} &\leq \liminf_{q \rightarrow 1^-} \Gamma_q(x) \\ &\leq \limsup_{q \rightarrow 1^-} \Gamma_q(x) \leq \frac{(n-1)! n^x}{x(x+1) \cdots (x+n-1)}. \end{aligned}$$

Both outer expressions tend to $\Gamma(x)$ as $n \rightarrow \infty$. To see this directly, use Euler's beta integral:

$$B(x, n) = \int_0^1 t^{x-1} (1-t)^{n-1} dt = \frac{(n-1)!}{x(x+1) \cdots (x+n-1)}.$$

After $t = u/n$, dominated convergence gives $n^x B(x, n) \rightarrow \int_0^\infty u^{x-1} e^{-u} du = \Gamma(x)$. The lower expression differs from the upper by a factor tending to 1. This proves pointwise convergence. We record the uniformity argument. On a compact interval $x \in [\delta, 1]$, the upper outer bound can be written as

$$n^x B(x, n) = \int_0^n u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} du.$$

For $n \geq 2$, its integrand is dominated uniformly in x by $u^{\delta-1}$ on $(0, 1]$ and by $e^{-u/2}$ on $[1, \infty)$. More explicitly, after extending the kernel by 0 for $u > n$,

$$\sup_{\delta \leq x \leq 1} \left| u^{x-1} \left(1 - \frac{u}{n}\right)_+^{n-1} - u^{x-1} e^{-u} \right| \rightarrow 0$$

for every $u > 0$, while this supremum is bounded by an integrable multiple of $u^{\delta-1} \mathbf{1}_{(0,1]}(u) + e^{-u/2} \mathbf{1}_{[1,\infty)}(u)$. The dominated convergence theorem therefore gives

$$\sup_{\delta \leq x \leq 1} |n^x B(x, n) - \Gamma(x)| \rightarrow 0.$$

The lower outer bound differs by the factor $(n/(n+1))^{1-x}$, which also tends uniformly to 1. For fixed n , the original q -bounds converge uniformly as $q \rightarrow 1^-$ because they contain only finitely many continuous factors. A uniform squeeze proves local uniform convergence on $(0, 1]$, and the recurrence extends it to arbitrary compact subsets of $(0, \infty)$. $\square$

12.4 The Jackson q -beta integral

Definition 12.6. For $x, y > 0$, define

$$B_q(x, y) = \int_0^1 t^{x-1} \frac{(qt; q)_\infty}{(q^y t; q)_\infty} d_q t.$$

Theorem 12.7 (q -beta evaluation). For $x, y > 0$,

$$B_q(x, y) = \frac{\Gamma_q(x) \Gamma_q(y)}{\Gamma_q(x+y)}.$$

Proof. Expand the Jackson integral:

$$\begin{aligned} B_q(x, y) &= (1-q) \sum_{n \geq 0} q^{nx} \frac{(q^{n+1}; q)_\infty}{(q^{n+y}; q)_\infty} \\ &= (1-q) \frac{(q; q)_\infty}{(q^y; q)_\infty} \sum_{n \geq 0} \frac{(q^y; q)_n}{(q; q)_n} q^{nx}. \end{aligned}$$

The infinite q -binomial theorem evaluates the sum as $(q^{x+y}; q)_\infty / (q^x; q)_\infty$. Thus

$$B_q(x, y) = (1-q) \frac{(q; q)_\infty (q^{x+y}; q)_\infty}{(q^x; q)_\infty (q^y; q)_\infty},$$

which is exactly the gamma quotient after substituting the definition of Γ_q . $\square$

Corollary 12.8 (Recurrences). *For $x, y > 0$,*

$$B_q(x+1, y) = \frac{[x]_q}{[x+y]_q} B_q(x, y), \quad B_q(x, y+1) = \frac{[y]_q}{[x+y]_q} B_q(x, y).$$

Proof. Apply the functional equation for Γ_q to the quotient in [Theorem 12.7](#). □

12.5 Multiplication and reflection-type formulas

Theorem 12.9 (q -multiplication formula). *For an integer $m \geq 1$ and $x > 0$,*

$$\Gamma_q(mx) \prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = [m]_q^{mx-1} \prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right).$$

Proof. Multiply the definitions on the right. By dissection,

$$\prod_{j=0}^{m-1} (q^{mx+j}; q^m)_\infty = (q^{mx}; q)_\infty.$$

Hence

$$\prod_{j=0}^{m-1} \Gamma_{q^m}\left(x + \frac{j}{m}\right) = (1 - q^m)^{(m+1)/2 - mx} \frac{(q^m; q^m)_\infty^m}{(q^{mx}; q)_\infty}.$$

Similarly,

$$\prod_{j=1}^{m-1} \Gamma_{q^m}\left(\frac{j}{m}\right) = (1 - q^m)^{(m-1)/2} \frac{(q^m; q^m)_\infty^m}{(q; q)_\infty}.$$

Taking the ratio and using $[m]_q = (1 - q^m)/(1 - q)$ produces the formula. □

Proposition 12.10 (Reflection-type product). *For $0 < x < 1$,*

$$\Gamma_q(x) \Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^2}{(q^x, q^{1-x}; q)_\infty}.$$

Proof. Multiply the two defining products; the powers of $1 - q$ combine to one. □

Remark 12.11. Unlike Euler's reflection formula, the right side is naturally a theta quotient rather than an elementary trigonometric function. Jacobi's triple product in the next part makes that theta structure explicit.

Part III

Theta products and partition identities

Chapter 13

Jacobi's triple product

13.1 A bilateral theta series

For $z \neq 0$ and $|q| < 1$, define

$$\Theta(z; q) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n.$$

The series converges locally uniformly on $\mathbb{C}^*$ because its terms decay quadratically in both directions.

Theorem 13.1 (Jacobi triple product). *For $z \neq 0$ and $|q| < 1$,*

$$\sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n = (-z, -q/z, q; q)_\infty, \quad (13.1)$$

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (z, q/z, q; q)_\infty. \quad (13.2)$$

Proof. It suffices to prove (13.1); replace z by $-z$ for (13.2). Put

$$F(z) = (-z; q)_\infty (-q/z; q)_\infty (q; q)_\infty.$$

The products converge locally uniformly on $\mathbb{C}^*$, so F has a Laurent expansion $F(z) = \sum_{n \in \mathbb{Z}} c_n z^n$. The shift identities give

$$F(qz) = \frac{1+z^{-1}}{1+z} F(z) = z^{-1} F(z).$$

Comparing Laurent coefficients yields

$$c_{n+1} = q^n c_n, \quad c_n = q^{\binom{n}{2}} c_0$$

for every integer n .

It remains to find c_0 . Euler's product expansions give

$$(-z; q)_\infty = \sum_{r \geq 0} \frac{q^{\binom{r}{2}} z^r}{(q; q)_r}, \quad (-q/z; q)_\infty = \sum_{s \geq 0} \frac{q^{\binom{s+1}{2}} z^{-s}}{(q; q)_s}.$$

Hence

$$c_0 = (q; q)_\infty \sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2}.$$

The sum is the generating function obtained by decomposing an arbitrary partition according to the side r of its Durfee square: the square contributes q^{r^2} , while the diagrams to its right and below are independent partitions with at most r parts, each having generating function $1/(q; q)_r$. Therefore

$$\sum_{r \geq 0} \frac{q^{r^2}}{(q; q)_r^2} = \frac{1}{(q; q)_\infty},$$

so $c_0 = 1$. Thus $F(z) = \sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n$. □

Corollary 13.2 (Quasi-periodicity and zeros). *The theta function satisfies*

$$\Theta(qz; q) = z^{-1} \Theta(z; q).$$

Its zeros are the simple points $z = -q^m$, $m \in \mathbb{Z}$.

Proof. The functional equation follows either from reindexing the bilateral series or from the proof above. The product in (13.1) vanishes exactly when $1 + zq^j = 0$ or $1 + q^{j+1}/z = 0$, which together give $z = -q^m$ for $m \in \mathbb{Z}$. Exactly one factor vanishes at each such point. □

13.2 Euler's pentagonal number theorem

Theorem 13.3 (Pentagonal number theorem). *For $|q| < 1$,*

$$(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2} = 1 + \sum_{n=1}^{\infty} (-1)^n \left(q^{n(3n-1)/2} + q^{n(3n+1)/2} \right).$$

Proof. In (13.2), replace the base q by q^3 and set $z = q$. The product side is

$$(q; q^3)_\infty (q^2; q^3)_\infty (q^3; q^3)_\infty = (q; q)_\infty$$

by dissection. The series exponent is

$$3 \binom{n}{2} + n = \frac{n(3n-1)}{2}.$$

Pair the terms n and $-n$ to obtain the second form. □

Corollary 13.4 (Euler's partition recurrence). *Let $p(0) = 1$ and $p(n) = 0$ for $n < 0$. Then for $n \geq 1$,*

$$p(n) = \sum_{j=1}^{\infty} (-1)^{j-1} \left(p\left(n - \frac{j(3j-1)}{2}\right) + p\left(n - \frac{j(3j+1)}{2}\right) \right).$$

Proof. The generating function for partitions is $\sum_{n \geq 0} p(n) q^n = 1/(q; q)_\infty$. Multiply this by the pentagonal expansion and compare the coefficient of q^n . □

13.3 Jacobi's cubic identity

Theorem 13.5 (Jacobi identity). *For $|q| < 1$,*

$$(q; q)_\infty^3 = \sum_{n=0}^{\infty} (-1)^n (2n+1) q^{n(n+1)/2}.$$

Proof. Differentiate (13.2) with respect to z and set $z = 1$. Since

$$(z; q)_\infty = (1 - z)(qz; q)_\infty,$$

the derivative of the product side at $z = 1$ is $-(q; q)_\infty^3$. On the series side it is

$$\sum_{m \in \mathbb{Z}} m(-1)^m q^{\binom{m}{2}}.$$

Pair $m = -n$ with $m = n + 1$ for $n \geq 0$. Their sum is

$$-(2n + 1)(-1)^n q^{n(n+1)/2}.$$

Cancelling the common minus sign proves the identity. □

13.4 A theta form of the q -gamma reflection product

Corollary 13.6. For $0 < x < 1$,

$$\Gamma_q(x)\Gamma_q(1-x) = (1-q) \frac{(q; q)_\infty^3}{\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn}}.$$

Proof. Set $z = q^x$ in (13.2):

$$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2} + xn} = (q^x, q^{1-x}, q; q)_\infty.$$

Substitute this in Theorem 12.10. □

Chapter 14

Partitions and product–sum identities

14.1 Unrestricted, distinct, and odd parts

Theorem 14.1 (Basic partition products). *Let $p(n)$ count unrestricted partitions, $d(n)$ partitions into distinct parts, and $o(n)$ partitions into odd parts. Then*

$$\begin{aligned}\sum_{n \geq 0} p(n)q^n &= \frac{1}{(q; q)_\infty}, \\ \sum_{n \geq 0} d(n)q^n &= (-q; q)_\infty, \\ \sum_{n \geq 0} o(n)q^n &= \frac{1}{(q; q^2)_\infty}.\end{aligned}$$

Moreover, $d(n) = o(n)$ for every n .

Proof. For unrestricted partitions, the part j may occur $0, 1, 2, \dots$ times and contributes the geometric factor $(1 - q^j)^{-1}$. For distinct partitions, the part j occurs zero or one time and contributes $1 + q^j$. For odd parts, retain only the odd geometric factors. Finally,

$$(-q; q)_\infty = \prod_{j \geq 1} \frac{1 - q^{2j}}{1 - q^j} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty} = \frac{1}{(q; q^2)_\infty},$$

so the two generating functions agree coefficientwise. □

Remark 14.2 (A bijection). The equality $d(n) = o(n)$ also has a direct proof. In a partition into distinct parts, write each part uniquely as $2^r u$ with u odd. Replace that part by 2^r copies of u . Distinct powers of two ensure that the process is reversible by binary expansion of the multiplicity of every odd part.

14.2 Bounded partitions

Proposition 14.3. *The following generating functions are equal:*

$$\frac{1}{(q; q)_r} = \sum_{n \geq 0} p_{\leq r}(n)q^n,$$

where $p_{\leq r}(n)$ counts partitions with largest part at most r , or equivalently partitions with at most r parts.

Proof. The product allows parts $1, \dots, r$ with arbitrary multiplicity. Conjugation of Ferrers diagrams exchanges the condition “largest part at most r ” with “at most r parts.” $\square$

Corollary 14.4 (Durfee decomposition of all partitions).

$$\frac{1}{(q; q)_{\infty}} = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q; q)_r^2}.$$

Proof. Decompose every partition by the side r of its Durfee square. The diagrams to the right and below are independent partitions with at most r parts. This is the identity used in the proof of Jacobi’s triple product. $\square$

14.3 Finite approximations and stabilization

Proposition 14.5. *For fixed N , the coefficient of q^N in $\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$ equals $p(N)$ whenever $m, k \geq N$.*

Proof. The Gaussian coefficient counts partitions inside a $k \times m$ rectangle. Any partition of N has at most N parts and largest part at most N , so for $m, k \geq N$ the rectangle restriction is vacuous. $\square$

Remark 14.6. Thus $\left[\begin{smallmatrix} m+k \\ k \end{smallmatrix} \right]_q$ converges coefficientwise to $1/(q; q)_{\infty}$ as both sides of the rectangle tend to infinity. This finite-to-infinite passage is a recurring bridge between Gaussian polynomials and partition identities.

Chapter 15

Bailey pairs and the Rogers–Ramanujan identities

15.1 Bailey pairs

For brevity, write $(a)_n = (a; q)_n$ in this chapter.

Definition 15.1. Sequences $(\alpha_n)_{n \geq 0}$ and $(\beta_n)_{n \geq 0}$ form a *Bailey pair relative to a* if

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q)_{n-r}(aq)_{n+r}} \quad (n \geq 0).$$

15.2 Bailey inversion

Lemma 15.2 (q -difference annihilation). *If $P(x)$ is a polynomial of degree less than N , then*

$$\sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s) = 0.$$

Proof. By linearity it suffices to take $P(x) = x^m$ with $0 \leq m < N$. Put $t = N - s$ and use the finite q -binomial theorem:

$$\begin{aligned} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q q^{ms} &= q^{mN} \sum_{t=0}^N (-1)^t q^{\binom{t}{2}} \begin{bmatrix} N \\ t \end{bmatrix}_q q^{-mt} \\ &= q^{mN} (q^{-m}; q)_N = 0, \end{aligned}$$

for the product contains the factor $1 - q^0$. □

Theorem 15.3 (Bailey inversion). *Assume $a \neq 1$ initially. The Bailey-pair relation is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \sum_{j=0}^n \frac{(a)_{n+j}}{(q)_{n-j}} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j.$$

At $a = 1$, the right side is interpreted by its removable limit.

Proof. Substitute the definition of β_j into the proposed inverse. The coefficient of α_r is

$$\frac{1 - aq^{2n}}{1 - a} \sum_{j=r}^n \frac{(a)_{n+j}}{(q)_{n-j}(q)_{j-r}(aq)_{j+r}} (-1)^{n-j} q^{\binom{n-j}{2}}.$$

If $n = r$, only $j = n$ occurs, and

$$\frac{1 - aq^{2n}}{1 - a} \frac{(a)_{2n}}{(aq)_{2n}} = 1.$$

Suppose $N = n - r > 0$ and put $j = r + s$. Then

$$\frac{(a)_{n+r+s}}{(aq)_{2r+s}} = (1 - a)(aq^{2r+s+1}; q)_{N-1}.$$

The coefficient becomes

$$\frac{1 - aq^{2n}}{(q)_N} \sum_{s=0}^N (-1)^{N-s} q^{\binom{N-s}{2}} \begin{bmatrix} N \\ s \end{bmatrix}_q P(q^s),$$

where $P(x) = (aq^{2r+1}x; q)_{N-1}$ has degree $N - 1$. It vanishes by [Theorem 15.2](#). Thus the two triangular matrices are inverse. Since every expression is rational in a and the singularity at $a = 1$ is removable, the limiting statement follows. $\square$

Corollary 15.4 (Two unit Bailey pairs). *Let $\beta_n = \delta_{n,0}$.*

(i) *Relative to $a = 1$,*

$$\alpha_0 = 1, \quad \alpha_n = (-1)^n q^{\binom{n}{2}} (1 + q^n) \quad (n > 0).$$

(ii) *Relative to $a = q$,*

$$\alpha_n = (-1)^n q^{\binom{n}{2}} [2n + 1]_q \quad (n \geq 0).$$

Proof. With $\beta_j = \delta_{j,0}$, Bailey inversion gives

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \frac{(a)_n}{(q)_n} (-1)^n q^{\binom{n}{2}}.$$

For $a \rightarrow 1$ and $n > 0$, $(a)_n/(1 - a) \rightarrow (q)_{n-1}$, so the prefactor reduces to $(1 - q^{2n})/(1 - q^n) = 1 + q^n$. For $a = q$, the ratio $(a)_n/(q)_n$ is 1 and the remaining factor is $[2n + 1]_q$. $\square$

15.3 The limiting Bailey lemma

Lemma 15.5 (Auxiliary finite identity). *For $N \geq 0$,*

$$\sum_{k=0}^N z^k q^{k(k-1)} \begin{bmatrix} N \\ k \end{bmatrix}_q (zq^k; q)_{N-k} = 1.$$

Proof. Call the left side $S_N(z)$. Using the second q -Pascal recurrence and splitting the last factor of $(zq^k; q)_{N+1-k}$ gives

$$S_{N+1}(z) = (1 - zq^N)S_N(z) + zq^N S_N(qz).$$

The identity is immediate for $N = 0$. If S_N is identically 1, the recurrence gives $S_{N+1} = 1 - zq^N + zq^N = 1$. Induction completes the proof. $\square$

Theorem 15.6 (Limiting Bailey lemma, finite form). *If (α_n, β_n) is a Bailey pair relative to a , define*

$$\alpha'_n = a^n q^{n^2} \alpha_n, \quad \beta'_n = \sum_{j=0}^n \frac{a^j q^{j^2}}{(q)_{n-j}} \beta_j.$$

Then (α'_n, β'_n) is also a Bailey pair relative to a .

Proof. Substitute the Bailey relation for β_j and interchange the finite sums. The coefficient of α_r in β'_n is

$$a^r q^{r^2} \sum_{s=0}^N \frac{a^s q^{s^2+2rs}}{(q)_{N-s}(q)_s(aq)_{2r+s}}, \quad N = n - r.$$

Let $A = aq^{2r+1}$. Since

$$a^s q^{s^2+2rs} = A^s q^{s(s-1)}, \quad (aq)_{2r+s} = (aq)_{2r}(A)_s,$$

the desired Bailey coefficient $a^r q^{r^2} / ((q)_N (aq)_{n+r})$ is equivalent, after clearing factors, to

$$\sum_{s=0}^N \begin{bmatrix} N \\ s \end{bmatrix}_q A^s q^{s(s-1)} (Aq^s)_{N-s} = 1.$$

This is [Theorem 15.5](#). □

Corollary 15.7 (Limiting Bailey lemma, infinite form). *If the series converge absolutely, then every Bailey pair relative to a satisfies*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_{\infty}} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n.$$

Proof. The transformed pair in [Theorem 15.6](#) satisfies

$$\beta'_N = \sum_{r=0}^N \frac{\alpha'_r}{(q)_{N-r}(aq)_{N+r}}.$$

Let $N \rightarrow \infty$. On the left,

$$\beta'_N \rightarrow \frac{1}{(q)_{\infty}} \sum_{j \geq 0} a^j q^{j^2} \beta_j.$$

On the right,

$$\beta'_N \rightarrow \frac{1}{(q)_{\infty}(aq)_{\infty}} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Absolute convergence permits dominated convergence. Cancel $(q)_{\infty}^{-1}$. □

15.4 The Rogers–Ramanujan identities

Theorem 15.8 (Rogers–Ramanujan). *For $|q| < 1$,*

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} = \frac{1}{(q; q^5)_{\infty} (q^4; q^5)_{\infty}}, \quad (15.1)$$

$$\sum_{n=0}^{\infty} \frac{q^{n(n+1)}}{(q; q)_n} = \frac{1}{(q^2; q^5)_{\infty} (q^3; q^5)_{\infty}}. \quad (15.2)$$

Proof. For the first identity, begin with the unit pair relative to $a = 1$ in [Theorem 15.4](#). Apply the finite limiting lemma once. The new pair is

$$\alpha'_n = q^{n^2} \alpha_n, \quad \beta'_n = \frac{1}{(q)_n}.$$

Applying the infinite limiting lemma to this new pair gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2}}{(q)_n} &= \frac{1}{(q)_\infty} \left(1 + \sum_{r \geq 1} (-1)^r q^{2r^2 + \binom{r}{2}} (1 + q^r) \right) \\ &= \frac{1}{(q)_\infty} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2}. \end{aligned}$$

By Jacobi's triple product with base q^5 and $z = q^2$,

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 - r)/2} = (q^2, q^3, q^5; q^5)_\infty.$$

Dissection gives

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty,$$

which proves (15.1).

For the second identity use the unit pair relative to $a = q$. After one finite limiting step,

$$\alpha'_n = (-1)^n q^{n^2 + n + \binom{n}{2}} [2n + 1]_q, \quad \beta'_n = \frac{1}{(q)_n}.$$

The infinite limiting lemma gives

$$\begin{aligned} \sum_{n \geq 0} \frac{q^{n^2 + n}}{(q)_n} &= \frac{1}{(q^2; q)_\infty} \sum_{r \geq 0} q^{r^2 + r} \alpha'_r \\ &= \frac{1}{(q; q)_\infty} \sum_{r \geq 0} (-1)^r q^{(5r^2 + 3r)/2} (1 - q^{2r+1}) \\ &= \frac{1}{(q; q)_\infty} \sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2}. \end{aligned}$$

In the last step, the subtracted term indexed by $r \geq 0$ is the term with bilateral index $-r - 1$. Jacobi's triple product with base q^5 and $z = q^4$ gives

$$\sum_{r \in \mathbb{Z}} (-1)^r q^{(5r^2 + 3r)/2} = (q, q^4, q^5; q^5)_\infty.$$

Divide by the fivefold dissection of $(q; q)_\infty$ to obtain (15.2). □

Remark 15.9 (Partition interpretation). The product side of (15.1) counts partitions into parts congruent to 1 or 4 modulo 5; the product side of (15.2) counts parts congruent to 2 or 3 modulo 5. The sum sides count partitions with successive parts differing by at least 2, with an additional smallest-part condition in the second identity. Establishing the latter combinatorial interpretations requires a separate bijective or finite-polynomial argument; the proof above instead reveals the hypergeometric engine.

Part IV

Cyclotomic arithmetic and extended indices

Chapter 16

Cyclotomic factorization and congruences

The arithmetic of Gaussian coefficients is governed by cyclotomic polynomials. This is the precise sense in which a primitive root of unity plays the role of a prime. The point of this chapter is that the essential congruences do not require mysterious cancellation: they follow from product factorization, coefficient extraction, and a careful accounting of blocks.

16.1 Cyclotomic valuations and carries

Let $\Phi_d(q)$ denote the d th cyclotomic polynomial, normalized by

$$q^n - 1 = \prod_{d|n} \Phi_d(q).$$

For a nonzero polynomial $F \in \mathbb{Z}[q]$, write $\nu_d(F)$ for the exponent of $\Phi_d(q)$ in its factorization.

Theorem 16.1 (Cyclotomic factorization of a shifted factorial). *For every $n \geq 0$,*

$$(q; q)_n = (-1)^n \prod_{d=1}^n \Phi_d(q)^{\lfloor n/d \rfloor}.$$

Consequently,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{d=1}^n \Phi_d(q)^{e_d(n,k)}, \quad e_d(n,k) = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor. \quad (16.1)$$

Moreover $e_d(n,k)$ is always either 0 or 1.

Proof. Since $1 - q^j = -(q^j - 1)$ and $q^j - 1 = \prod_{d|j} \Phi_d(q)$,

$$(q; q)_n = \prod_{j=1}^n (1 - q^j) = (-1)^n \prod_{j=1}^n \prod_{d|j} \Phi_d(q).$$

For fixed d , the multiples of d among $1, \dots, n$ are $d, 2d, \dots, \lfloor n/d \rfloor d$, so the exponent of Φ_d is $\lfloor n/d \rfloor$. Dividing the factorizations of $(q; q)_n$, $(q; q)_k$, and $(q; q)_{n-k}$ gives (16.1); the signs cancel.

Write $k = ad + r$ and $n - k = bd + s$ with $0 \leq r, s < d$. Then

$$e_d(n, k) = \lfloor (r + s)/d \rfloor,$$

which is 0 or 1. □

Corollary 16.2 (Square-free cyclotomic structure). *Every Gaussian polynomial is a product of distinct cyclotomic polynomials. More precisely,*

$$\Phi_d(q) \mid \left[\begin{matrix} n \\ k \end{matrix} \right]_q \iff (k \bmod d) > (n \bmod d). \quad (16.2)$$

Equivalently, Φ_d occurs exactly when adding k and $n - k$ produces a carry in the units place in base d .

Proof. The first assertion follows because every exponent in (16.1) belongs to $\{0, 1\}$. If $n = cd + t$ and $k = ad + r$ with $0 \leq r, t < d$, then the residue of $n - k$ is $t - r$ when $r \leq t$ and $t - r + d$ when $r > t$. Thus the two residues add to t in the first case and to $t + d$ in the second. The latter is precisely the carry case, and by the proof of Theorem 16.1 it is equivalent to $e_d(n, k) = 1$. $\square$

Example 16.3. For $n = 10$ and $k = 4$, the values of d for which $4 \bmod d > 10 \bmod d$ are $d = 5, 7, 8, 9$. Hence

$$\left[\begin{matrix} 10 \\ 4 \end{matrix} \right]_q = \Phi_5(q)\Phi_7(q)\Phi_8(q)\Phi_9(q).$$

The degrees add to $4 + 6 + 4 + 6 = 20 = 4(10 - 4)$, as required by reciprocity.

16.2 The q -Lucas congruence

The following theorem is often called the q -Lucas theorem. It is naturally proved by grouping a finite product into complete root-of-unity blocks.

Lemma 16.4 (A complete root-of-unity block). *If ζ is a primitive d th root of unity, then*

$$\prod_{j=0}^{d-1} (1 + x\zeta^j) = 1 - (-x)^d. \quad (16.3)$$

Proof. The roots in x of the left side are $-\zeta^{-j}$, which are exactly the roots of $1 - (-x)^d$. Both polynomials have constant term 1, so they are equal. $\square$

Theorem 16.5 (q -Lucas). *Let $a, r \geq 0$, let $d \geq 1$, and let $0 \leq b, s < d$. Then*

$$\left[\begin{matrix} ad + b \\ rd + s \end{matrix} \right]_q \equiv \binom{a}{r} \left[\begin{matrix} b \\ s \end{matrix} \right]_q \pmod{\Phi_d(q)}. \quad (16.4)$$

Here the right side is understood as 0 when $r > a$ or $s > b$.

Proof. It suffices to prove equality after substituting an arbitrary primitive d th root ζ for q , because Φ_d is the minimal polynomial of ζ over $\mathbb{Q}$.

The finite q -binomial theorem gives

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = \sum_{h=0}^{ad+b} \zeta^{\binom{h}{2}} \left[\begin{matrix} ad + b \\ h \end{matrix} \right]_{\zeta} x^h. \quad (16.5)$$

Group the product into a complete blocks and one incomplete block. By Theorem 16.4,

$$\prod_{j=0}^{ad+b-1} (1 + x\zeta^j) = (1 - (-x)^d)^a \prod_{j=0}^{b-1} (1 + x\zeta^j).$$

The coefficient of x^{rd+s} on the right is

$$\binom{a}{r} (-1)^{r(d+1)} \zeta^{\binom{s}{2}} \left[\begin{matrix} b \\ s \end{matrix} \right]_{\zeta}. \quad (16.6)$$

Comparing (16.5) and (16.6) leaves only the phase. We claim that

$$(-1)^{r(d+1)} \zeta^{\binom{s}{2} - \binom{rd+s}{2}} = 1. \quad (16.7)$$

Indeed,

$$2 \left(\binom{rd+s}{2} - \binom{s}{2} \right) = rd(rd+2s-1).$$

If d is odd, $r(rd+2s-1)$ is even, so the exponent difference is a multiple of d , while $r(d+1)$ is even. If d is even, then $\zeta^{d/2} = -1$ and $rd+2s-1$ is odd; hence the second factor in (16.7) equals $(-1)^r$, as does the first. Their product is 1. The coefficient comparison therefore gives (16.4) at $q = \zeta$, completing the proof. $\square$

Corollary 16.6 (Evaluation at $q = -1$). For $0 \leq k \leq n$,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q \Big|_{q=-1} = \begin{cases} \binom{n/2}{k/2}, & n \text{ even and } k \text{ even,} \\ 0, & n \text{ even and } k \text{ odd,} \\ \binom{(n-1)/2}{\lfloor k/2 \rfloor}, & n \text{ odd.} \end{cases}$$

Proof. Apply Theorem 16.5 with $d = 2$. Write $n = 2a + b$ and $k = 2r + s$, where $b, s \in \{0, 1\}$. Since $\left[\begin{matrix} b \\ s \end{matrix} \right]_{-1}$ is 1 when $s \leq b$ and 0 otherwise, the three cases follow. $\square$

16.3 Cyclic sieving for subsets

A triple $(X, C, F(q))$, where a finite cyclic group $C = \langle c \rangle$ acts on a finite set X and $F(q) \in \mathbb{Z}[q]$, exhibits the *cyclic sieving phenomenon* when

$$\# \text{Fix}(c^j) = F(\omega^j)$$

for every j , where ω is a primitive $|C|$ th root of unity.

Theorem 16.7 (Cyclic sieving for k -subsets). Let X be the set of k -element subsets of $\mathbb{Z}/n\mathbb{Z}$, and let C_n act by translation. Then

$$\left(X, C_n, \left[\begin{matrix} n \\ k \end{matrix} \right]_q \right)$$

exhibits the cyclic sieving phenomenon.

Proof. Fix j , and let d be the order of translation by j . Its orbits on $\mathbb{Z}/n\mathbb{Z}$ all have size d , and there are n/d such orbits. A subset is fixed precisely when it is a union of these orbits. Thus

$$\# \text{Fix}(c^j) = \begin{cases} \binom{n/d}{k/d}, & d \mid k, \\ 0, & d \nmid k. \end{cases} \quad (16.8)$$

The number ω^j is a primitive d th root. Write $n = (n/d)d$ and $k = rd + s$ with $0 \leq s < d$. By q -Lucas,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q \Big|_{q=\omega^j} = \binom{n/d}{r} \left[\begin{matrix} 0 \\ s \end{matrix} \right]_{\omega^j}.$$

This is 0 if $s > 0$, and it is $\binom{n/d}{k/d}$ if $s = 0$, exactly as in (16.8). $\square$

16.4 A q -Babbage congruence modulo a cyclotomic square

The modulus $\Phi_n(q)^2$ records not only the value at a primitive n th root, but also first-order contact there. The next congruence is a particularly transparent example. Its block proof is also a useful template for stronger q -congruences; see Andrews [3] and Straub [30].

Theorem 16.8 (q -Babbage congruence). *For integers $a \geq b \geq 0$ and $n \geq 1$,*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}} \pmod{\Phi_n(q)^2}. \quad (16.9)$$

Proof. Split the finite product into a consecutive blocks of length n :

$$\prod_{j=0}^{an-1} (1 + xq^j) = \prod_{i=0}^{a-1} \prod_{j=0}^{n-1} (1 + xq^{in+j}).$$

Apply the finite q -binomial theorem to each block. Comparing the coefficient of x^{bn} gives

$$q^{\binom{bn}{2}} \begin{bmatrix} an \\ bn \end{bmatrix}_q = \sum_{\substack{0 \leq k_0, \dots, k_{a-1} \leq n \\ k_0 + \dots + k_{a-1} = bn}} q^{\sum_i \binom{k_i}{2} + n \sum_i i k_i} \prod_{i=0}^{a-1} \begin{bmatrix} n \\ k_i \end{bmatrix}_q. \quad (16.10)$$

If $0 < k_i < n$, then $\Phi_n(q)$ divides $\begin{bmatrix} n \\ k_i \end{bmatrix}_q$ by Theorem 16.2. A tuple contributing to (16.10) cannot have exactly one such intermediate entry: all other entries would be 0 or n , making the sum of the k_i nonzero modulo n . Therefore every tuple containing an intermediate entry contributes a multiple of $\Phi_n(q)^2$.

Modulo $\Phi_n(q)^2$, only tuples with each $k_i \in \{0, n\}$ survive. Such a tuple is specified by a b -element subset $S \subseteq \{0, 1, \dots, a-1\}$. After division by $q^{\binom{bn}{2}}$, its exponent is

$$n^2 \sum_{i \in S} i + b \binom{n}{2} - \binom{bn}{2} = n^2 \left(\sum_{i \in S} i - \binom{b}{2} \right).$$

Hence

$$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \sum_{\substack{S \subseteq \{0, \dots, a-1\} \\ |S|=b}} (q^{n^2})^{\sum_{i \in S} i - \binom{b}{2}} \pmod{\Phi_n(q)^2}.$$

The subset form of the finite q -binomial theorem identifies the sum as $\begin{bmatrix} a \\ b \end{bmatrix}_{q^{n^2}}$. $\square$

Remark 16.9 (Why the square appears). The decisive observation in the proof is combinatorial: because the target degree is a multiple of the block length, nontrivial block choices occur in pairs. Each nontrivial block supplies one factor $\Phi_n(q)$, so two of them supply the square. Congruences modulo Φ_n^3 require genuinely finer first- and second-order information; the q -Ljunggren congruences of Straub and subsequent refinements belong to that deeper layer.

Corollary 16.10 (Primitive-root value and derivative). *Let ζ be a primitive n th root of unity. Then*

$$\begin{bmatrix} an \\ bn \end{bmatrix}_{\zeta} = \begin{bmatrix} a \\ b \end{bmatrix},$$

and the derivatives at $q = \zeta$ of the two sides of (16.9) are equal.

Proof. The first assertion follows either from q -Lucas or from (16.9), because $\zeta^{n^2} = 1$. If a polynomial is divisible by Φ_n^2 , it has a zero of order at least two at every primitive n th root, so both its value and first derivative vanish there. $\square$

Chapter 17

Negative upper indices and geometric Newton interpolation

Ordinary binomial coefficients acquire a useful extension to negative upper arguments. Gaussian coefficients do as well, but the powers of q are essential: omitting them destroys the Pascal recurrences. The same algebra leads naturally to interpolation on a geometric grid.

17.1 Generalized Gaussian coefficients

Definition 17.1. For an integer N and $k \geq 0$, define

$$\begin{bmatrix} N \\ k \end{bmatrix}_q := \frac{(q^{N-k+1}; q)_k}{(q; q)_k}. \quad (17.1)$$

For $N \geq 0$ this agrees with the usual Gaussian coefficient, including the value 0 when $k > N$.

Proposition 17.2 (Negative-index reflection). *For $m \geq 1$ and $k \geq 0$,*

$$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q. \quad (17.2)$$

Proof. From the definition,

$$\begin{aligned} (q^{-m-k+1}; q)_k &= \prod_{j=0}^{k-1} (1 - q^{-(m+k-1-j)}) \\ &= (-1)^k q^{-\sum_{j=0}^{k-1} (m+k-1-j)} \prod_{j=0}^{k-1} (1 - q^{m+k-1-j}) \\ &= (-1)^k q^{-mk - \binom{k}{2}} (q^m; q)_k. \end{aligned}$$

After division by $(q; q)_k$, the final ratio is $\begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$. □

Theorem 17.3 (Pascal recurrences for all integral upper indices). *For every integer N and every $k \geq 1$,*

$$\begin{bmatrix} N \\ k \end{bmatrix}_q = \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} N-1 \\ k \end{bmatrix}_q, \quad (17.3)$$

$$= q^{N-k} \begin{bmatrix} N-1 \\ k-1 \end{bmatrix}_q + \begin{bmatrix} N-1 \\ k \end{bmatrix}_q. \quad (17.4)$$

Both formulas are identities in $\mathbb{Q}(q)$.

Proof. Put the two terms on the right of (17.3) over the common denominator $(q; q)_k$. Since

$$\left[\begin{matrix} N-1 \\ k-1 \end{matrix} \right]_q = \frac{(q^{N-k+1}; q)_{k-1} (1 - q^k)}{(q; q)_k}$$

the numerator of the right side is

$$\begin{aligned} & (q^{N-k+1}; q)_{k-1} [(1 - q^k) + q^k (1 - q^{N-k})] \\ &= (q^{N-k+1}; q)_{k-1} (1 - q^N) = (q^{N-k+1}; q)_k, \end{aligned}$$

which proves (17.3). For (17.4), the corresponding bracket is

$$q^{N-k} (1 - q^k) + (1 - q^{N-k}) = 1 - q^N.$$

The same common-denominator calculation applies. $\square$

Corollary 17.4 (Reciprocal product series). *For $m \geq 1$ and $|z| < 1$,*

$$\frac{1}{(z; q)_m} = \sum_{k=0}^{\infty} \left[\begin{matrix} m+k-1 \\ k \end{matrix} \right]_q z^k = \sum_{k=0}^{\infty} (-1)^k q^{mk + \binom{k}{2}} \left[\begin{matrix} -m \\ k \end{matrix} \right]_q z^k. \quad (17.5)$$

The first equality also holds in the formal power-series ring $\mathbb{Z}[q][[z]]$.

Proof. The reciprocal finite q -binomial theorem, already proved in Theorem 3.4, gives the first expansion. Substitute (17.2) to obtain the second. Formal validity follows because each coefficient of z^k is a polynomial and multiplication by the finite product $(z; q)_m$ can be checked coefficientwise. $\square$

17.2 An arbitrary upper parameter

When a branch of q^α has been fixed, it is often convenient to define

$$\left[\begin{matrix} \alpha \\ k \end{matrix} \right]_q = \frac{(q^{\alpha-k+1}; q)_k}{(q; q)_k}. \quad (17.6)$$

The proofs of the Pascal recurrences are purely algebraic in q^α and therefore remain valid. A related, and usually more analytic, expansion is the following.

Theorem 17.5 (Generalized upper-parameter series). *For $|q| < 1$, $|z| < 1$, and any parameter α for which the factors are defined,*

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k=0}^{\infty} \left[\begin{matrix} \alpha+k \\ k \end{matrix} \right]_q z^k. \quad (17.7)$$

Proof. Apply the infinite q -binomial theorem with $a = q^{\alpha+1}$:

$$\frac{(q^{\alpha+1}z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^{\alpha+1}; q)_k}{(q; q)_k} z^k.$$

The coefficient is exactly $\left[\begin{matrix} \alpha+k \\ k \end{matrix} \right]_q$ by (17.6). $\square$

17.3 Two complex arguments and generalized binomial series

The extension in [Theorem 1.11](#) also permits both arguments of a Gaussian coefficient to be complex. Throughout this section take $0 < q < 1$ and set $q^z = \exp(z \log q)$, using the real logarithm of q .

Definition 17.6. For $\alpha, \beta \in \mathbb{C}$, define

$$\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q := \frac{(q^{\beta+1}, q^{\alpha-\beta+1}; q)_\infty}{(q, q^{\alpha+1}; q)_\infty}, \quad (17.8)$$

whenever the denominator is nonzero.

Theorem 17.7 (Gamma representation and Pascal laws). *Wherever the expressions are defined,*

$$\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q = \frac{\Gamma_q(\alpha + 1)}{\Gamma_q(\beta + 1)\Gamma_q(\alpha - \beta + 1)}, \quad (17.9)$$

$$\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q = \left[\begin{matrix} \alpha \\ \alpha - \beta \end{matrix} \right]_q, \quad (17.10)$$

$$\left[\begin{matrix} \alpha + 1 \\ \beta \end{matrix} \right]_q = q^\beta \left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q + \left[\begin{matrix} \alpha \\ \beta - 1 \end{matrix} \right]_q, \quad (17.11)$$

$$= \left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q + q^{\alpha+1-\beta} \left[\begin{matrix} \alpha \\ \beta - 1 \end{matrix} \right]_q. \quad (17.12)$$

If $\beta = k \in \mathbb{N}_0$, this definition agrees with (17.6).

Proof. Substitution of the product definition of Γ_q proves (17.9); all powers of $1 - q$ cancel. Symmetry is immediate either from that quotient or directly from (17.8). For $\beta = k \in \mathbb{N}_0$, tail cancellation gives

$$\frac{(q^{k+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(q; q)_k}, \quad \frac{(q^{\alpha-k+1}; q)_\infty}{(q^{\alpha+1}; q)_\infty} = (q^{\alpha-k+1}; q)_k,$$

which is exactly (17.6).

For the first Pascal law, divide every term by $\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q$. The gamma functional equation gives

$$\frac{\left[\begin{matrix} \alpha+1 \\ \beta \end{matrix} \right]_q}{\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q} = \frac{[\alpha + 1]_q}{[\alpha - \beta + 1]_q}, \quad \frac{\left[\begin{matrix} \alpha \\ \beta-1 \end{matrix} \right]_q}{\left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q} = \frac{[\beta]_q}{[\alpha - \beta + 1]_q}.$$

The identity $[\alpha + 1]_q = q^\beta [\alpha - \beta + 1]_q + [\beta]_q$ proves (17.11). The alternative identity $[\alpha + 1]_q = [\alpha - \beta + 1]_q + q^{\alpha+1-\beta} [\beta]_q$ proves (17.12). $\square$

Corollary 17.8 (Classical generalized binomial). *If $\alpha > \beta > -1$ are real, then*

$$\lim_{q \rightarrow 1^-} \left[\begin{matrix} \alpha \\ \beta \end{matrix} \right]_q = \frac{\Gamma(\alpha + 1)}{\Gamma(\beta + 1)\Gamma(\alpha - \beta + 1)}.$$

Proof. Apply the locally uniform q -gamma limit in [Theorem 12.5](#) to (17.9). $\square$

Theorem 17.9 (Generalized finite and reciprocal q -binomial theorems). *Let $0 < q < 1$. If $|zq^\alpha| < 1$, then*

$$(z; q)_\alpha = \sum_{k=0}^{\infty} (-1)^k q^{\binom{k}{2}} \left[\begin{matrix} \alpha \\ k \end{matrix} \right]_q z^k. \quad (17.13)$$

If $|z| < 1$, then

$$\frac{1}{(z; q)_\alpha} = \sum_{k=0}^{\infty} \begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q z^k. \quad (17.14)$$

For a nonnegative integral α , the first series terminates and becomes Euler's finite theorem; in that case the restriction $|zq^\alpha| < 1$ can be removed because both sides are polynomials. For a positive integral α , the second becomes the reciprocal finite theorem.

Proof. Apply the infinite q -binomial theorem with $a = q^{-\alpha}$ and argument zq^α :

$$\frac{(z; q)_\infty}{(zq^\alpha; q)_\infty} = \sum_{k \geq 0} \frac{(q^{-\alpha}; q)_k}{(q; q)_k} (zq^\alpha)^k.$$

The left side is $(z; q)_\alpha$. Reversing the finite numerator product gives

$$\frac{(q^{-\alpha}; q)_k}{(q; q)_k} q^{\alpha k} = (-1)^k q^{\binom{k}{2}} \begin{bmatrix} \alpha \\ k \end{bmatrix}_q,$$

which proves (17.13).

For the reciprocal formula, apply the infinite theorem with $a = q^\alpha$ and argument z :

$$\frac{(q^\alpha z; q)_\infty}{(z; q)_\infty} = \sum_{k \geq 0} \frac{(q^\alpha; q)_k}{(q; q)_k} z^k.$$

The left side is $(z; q)_\alpha^{-1}$ and the coefficient is $\begin{bmatrix} \alpha + k - 1 \\ k \end{bmatrix}_q$. The integral specializations follow from the consistency results already proved. $\square$

17.4 Newton interpolation on the grid $1, q, q^2, \dots$

The usual Newton basis is adapted to an arbitrary sequence of distinct nodes. At geometric nodes its basis polynomials are q -Pochhammer symbols.

Theorem 17.10 (Geometric Newton interpolation). *Assume $q^i \neq q^j$ whenever $0 \leq i < j \leq n$. For arbitrary values $y_0, \dots, y_n$ there is a unique polynomial P of degree at most n satisfying*

$$P(q^j) = y_j \quad (0 \leq j \leq n).$$

It has the Newton expansion

$$P(x) = \sum_{k=0}^n c_k \prod_{r=0}^{k-1} (x - q^r) = \sum_{k=0}^n c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k, \quad (17.15)$$

where

$$c_k = \sum_{j=0}^k \frac{y_j}{\prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r)} \quad (17.16)$$

$$= \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} y_j. \quad (17.17)$$

Proof. For distinct nodes $x_0, \dots, x_n$, Lagrange interpolation gives existence and uniqueness. The coefficient of the Newton basis polynomial $\prod_{r=0}^{k-1} (x - x_r)$ is the divided difference

$$y[x_0, \dots, x_k] = \sum_{j=0}^k \frac{y_j}{\prod_{r \neq j} (x_j - x_r)};$$

this follows by comparing the coefficient of x^k in the Lagrange interpolant through the first $k + 1$ nodes. Taking $x_j = q^j$ proves (17.16).

It remains to simplify the denominator. Splitting the factors below and above j gives

$$\begin{aligned} \prod_{\substack{0 \leq r \leq k \\ r \neq j}} (q^j - q^r) &= \left[\prod_{r=0}^{j-1} -q^r (1 - q^{j-r}) \right] \left[\prod_{r=j+1}^k q^j (1 - q^{r-j}) \right] \\ &= (-1)^j q^{\binom{j}{2} + j(k-j)} (q; q)_j (q; q)_{k-j} \\ &= (-1)^j q^{jk - \binom{j+1}{2}} (q; q)_j (q; q)_{k-j}. \end{aligned}$$

Taking reciprocals yields (17.17). Finally,

$$\prod_{r=0}^{k-1} (x - q^r) = (-1)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1} (1 - xq^{-r}) = (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k,$$

which proves the second form of (17.15). $\square$

Corollary 17.11 (Triangular reconstruction). *The coefficients in (17.15) can be recovered successively by*

$$c_0 = y_0, \quad c_k = \frac{y_k - \sum_{r=0}^{k-1} c_r \prod_{j=0}^{r-1} (q^k - q^j)}{\prod_{j=0}^{k-1} (q^k - q^j)}. \quad (17.18)$$

Proof. Evaluate (17.15) at $x = q^k$. Every term with index greater than k vanishes, while the coefficient multiplying c_k is nonzero by the distinct-node hypothesis. Solving for c_k gives the formula. $\square$

Remark 17.12. Kupersmidt's q -Newton viewpoint emphasizes that the classical binomial theorem, Newton interpolation, and q -difference calculus are not separate constructions: each is controlled by a basis whose successive factors form a geometric progression. The finite formula above is algebraic and needs no convergence hypothesis; infinite Newton series require additional growth conditions on the data.

Part V

Geometric grids and the dyadic half-base theory

Chapter 18

Denominator-free Gaussian algebra and geometric filters

The quotient

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}$$

is convenient over a field of rational functions, but it conceals two facts that matter at roots of unity, in positive characteristic, and in formal verification. First, the Gaussian coefficient is a polynomial in q before any division is performed. Second, the identities that make geometric interpolation work are identities over arbitrary commutative rings. We now develop that division-free form and then identify the exact residual of every moment filter supported on a geometric grid.

18.1 Recursive Gaussian polynomials over an arbitrary ring

Definition 18.1 (Recursive Gaussian polynomial). For $n, k \in \mathbb{N}_0$, define $G_{n,k}(q) \in \mathbb{Z}[q]$ by

$$G_{0,0}(q) = 1, \quad G_{n,k}(q) = 0 \quad (k < 0 \text{ or } k > n),$$

and

$$G_{n+1,k}(q) = G_{n,k}(q) + q^{n+1-k} G_{n,k-1}(q). \quad (18.1)$$

For a commutative ring R and $q \in R$, the notation $\begin{bmatrix} n \\ k \end{bmatrix}_q$ means the evaluation of $G_{n,k}$ at that element of R .

This definition makes sense when $q = 0$, when q is a root of unity, when the coefficient ring has zero divisors, and even in the zero ring.

Theorem 18.2 (Denominator-free finite q -binomial theorem). In $\mathbb{Z}[q, z]$,

$$\prod_{j=0}^{n-1} (1 - zq^j) = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} G_{n,k}(q) z^k. \quad (18.2)$$

Consequently the same identity holds after substituting arbitrary elements q, z of any commutative ring.

Proof. The assertion is $1 = 1$ for $n = 0$. Assume it for n and multiply by $1 - zq^n$. The coefficient of z^k becomes

$$\begin{aligned} & (-1)^k q^{\binom{k}{2}} G_{n,k}(q) - q^n (-1)^{k-1} q^{\binom{k-1}{2}} G_{n,k-1}(q) \\ &= (-1)^k q^{\binom{k}{2}} (G_{n,k}(q) + q^{n+1-k} G_{n,k-1}(q)), \end{aligned}$$

because $n + \binom{k-1}{2} = \binom{k}{2} + n + 1 - k$. The bracket is $G_{n+1,k}(q)$ by (18.1). Induction proves the polynomial identity, and every evaluation homomorphism preserves it. $\square$

Corollary 18.3 (Polynomiality, symmetry, degree, and reciprocity). *For $0 \leq k \leq n$, the polynomial $G_{n,k}(q)$ has nonnegative integral coefficients, degree $k(n-k)$, constant term 1, and satisfies*

$$G_{n,k}(q) = G_{n,n-k}(q), \quad q^{k(n-k)} G_{n,k}(q^{-1}) = G_{n,k}(q). \quad (18.3)$$

Proof. Nonnegativity and the constant term follow from the recursion. In $\mathbb{Q}(q)$, the quotient

$$H_{n,k}(q) = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}$$

has the same initial conditions and Pascal recursion as $G_{n,k}$; hence $G_{n,k} = H_{n,k}$. Symmetry is immediate. Replacing q by q^{-1} and extracting powers from $1 - q^{-j} = -q^{-j}(1 - q^j)$ gives reciprocity. Since the constant term is 1, reciprocity shows that the top exponent is $k(n-k)$. All equalities proved in $\mathbb{Q}(q)$ are equalities of the polynomials already constructed in $\mathbb{Z}[q]$. $\square$

Theorem 18.4 (Denominator-free factorial identities). *For $0 \leq k \leq n$, the following identities hold in $\mathbb{Z}[q]$:*

$$(q; q)_k G_{n,k}(q) = (q^{n-k+1}; q)_k, \quad (18.4)$$

$$(q; q)_n = (q; q)_k (q; q)_{n-k} G_{n,k}(q). \quad (18.5)$$

They therefore remain valid after evaluating q in any commutative ring.

Proof. In $\mathbb{Q}(q)$ both formulas follow by inserting the quotient expression for $G_{n,k}$. Every term in the resulting equations belongs to $\mathbb{Z}[q]$; since $\mathbb{Z}[q] \hookrightarrow \mathbb{Q}(q)$ is injective, the identities already hold in $\mathbb{Z}[q]$. Evaluation preserves polynomial identities. Notice that this does not cancel a possibly vanishing factor after specialization. $\square$

18.2 Complete homogeneous functions on a geometric alphabet

For variables $x_0, \dots, x_p$, let $h_r(x_0, \dots, x_p)$ be the complete homogeneous symmetric polynomial characterized by

$$\prod_{i=0}^p \frac{1}{1 - x_i t} = \sum_{r=0}^{\infty} h_r(x_0, \dots, x_p) t^r. \quad (18.6)$$

Here the identity is formal, $h_0 = 1$, and $h_r = 0$ for $r < 0$.

Theorem 18.5 (Principal specialization). *For all $p, r \in \mathbb{N}_0$,*

$$h_r(1, q, \dots, q^p) = G_{p+r,p}(q), \quad (18.7)$$

$$h_r(c, cq, \dots, cq^p) = c^r G_{p+r,p}(q). \quad (18.8)$$

These are polynomial identities over $\mathbb{Z}[q, c]$ and hence survive every commutative-semiring specialization.

Proof. Put

$$H_p(t) = \prod_{j=0}^p (1 - q^j t)^{-1} = \sum_{r \geq 0} h_{p,r} t^r.$$

The relation $(1 - q^p t) H_p(t) = H_{p-1}(t)$ gives

$$h_{p,r} = h_{p-1,r} + q^p h_{p,r-1}.$$

The alternative Pascal recurrence

$$G_{N,K} = G_{N-1,K-1} + q^K G_{N-1,K}$$

follows either from the quotient in $\mathbb{Q}(q)$ or from symmetry and (18.1). With $N = p + r$ and $K = p$, it is the same recurrence with the same boundary values. Thus $h_{p,r} = G_{p+r,p}(q)$. Scaling every variable by c multiplies a homogeneous polynomial of degree r by c^r . $\square$

Equivalently,

$$\frac{1}{(t; q)_{p+1}} = \sum_{r=0}^{\infty} \begin{bmatrix} p+r \\ p \end{bmatrix}_q t^r. \quad (18.9)$$

No analytic hypothesis is present in this formal identity.

18.3 A universal moment-residual theorem

Theorem 18.6 (All-degree residual from the first surviving moment). *Let R be a commutative ring, let $x_0, \dots, x_p, w_0, \dots, w_p \in R$, and put $M_m = \sum_{i=0}^p w_i x_i^m$. If*

$$M_0 = M_1 = \dots = M_{p-1} = 0, \quad M_p = K, \quad (18.10)$$

then, for every $r \geq 0$,

$$M_{p+r} = K h_r(x_0, \dots, x_p). \quad (18.11)$$

Proof. In $R[[t]]$,

$$M(t) := \sum_{m \geq 0} M_m t^m = \sum_{i=0}^p \frac{w_i}{1 - x_i t}.$$

Let $E(t) = \prod_{i=0}^p (1 - x_i t)$. Then $E(t)M(t)$ is a polynomial of degree at most p . Its coefficient of t^m , for $m \leq p$, depends only on $M_0, \dots, M_m$; the hypotheses make it zero for $m < p$ and K for $m = p$. Therefore $E(t)M(t) = K t^p$. Divide by $E(t)$ and use (18.6); coefficient comparison gives (18.11). $\square$

Corollary 18.7 (Geometric Gaussian residual). *Under the preceding hypotheses, if $x_i = cq^i$, then*

$$\sum_{i=0}^p w_i (cq^i)^{p+r} = K c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q \quad (r \geq 0). \quad (18.12)$$

Proof. Combine (18.11) with (18.8). $\square$

Corollary 18.8 (Formal-series filter). *If $f(z) = \sum_{m \geq 0} a_m z^m \in R[[z]]$, then*

$$\sum_{i=0}^p w_i f(x_i z) = K \sum_{r=0}^{\infty} a_{p+r} h_r(x_0, \dots, x_p) z^{p+r}. \quad (18.13)$$

On the geometric grid $x_i = cq^i$, the multiplier of a_{p+r} is $K c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q$.

Proof. The coefficient of z^m on the left is $a_m M_m$. Apply the preceding theorem coefficientwise; the outer sum is finite. $\square$

18.4 Explicit geometric interpolation and Richardson weights

Theorem 18.9 (Leading-coefficient filter on a geometric grid). *Let K be a field, let $c, q \in K^\times$, and suppose $q^d \neq 1$ for $1 \leq d \leq p$. Set $x_i = cq^i$ and*

$$\lambda_i = \frac{1}{\prod_{j \neq i} (x_i - x_j)}.$$

Then

$$\lambda_i = \frac{(-1)^i c^{-p} q^{\binom{i+1}{2} - pi}}{(q; q)_i (q; q)_{p-i}} = \frac{(-1)^i c^{-p} q^{\binom{i+1}{2} - pi}}{(q; q)_p} \begin{bmatrix} p \\ i \end{bmatrix}_q. \quad (18.14)$$

For every polynomial P of degree at most p ,

$$[t^p]P(t) = \sum_{i=0}^p \lambda_i P(x_i), \quad (18.15)$$

and, for every $r \geq 0$,

$$\sum_{i=0}^p \lambda_i (cq^i)^{p+r} = c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q. \quad (18.16)$$

Proof. For $j < i$, $q^i - q^j = -q^j(1 - q^{i-j})$, while for $j > i$, $q^i - q^j = q^i(1 - q^{j-i})$. Hence

$$\prod_{j \neq i} (x_i - x_j) = c^p (-1)^i q^{pi - \binom{i+1}{2}} (q; q)_i (q; q)_{p-i},$$

which gives the weight. Lagrange interpolation and extraction of the leading coefficient prove (18.15). Applied to monomials, this gives moments 0 below degree p and 1 at degree p ; now use Theorem 18.7. $\square$

Theorem 18.10 (Exact geometric Richardson extrapolation). *Under the same assumptions, define*

$$\mu_i = \prod_{j \neq i} \frac{-x_j}{x_i - x_j} = \frac{(-1)^{p-i} q^{\binom{p-i+1}{2}}}{(q; q)_i (q; q)_{p-i}}. \quad (18.17)$$

Then $\sum_i \mu_i P(x_i) = P(0)$ for every polynomial of degree at most p . If $M_m = \sum_i \mu_i x_i^m$, then

$$M_0 = 1, \quad M_m = 0 \quad (1 \leq m \leq p), \quad M_{p+1+r} = (-1)^p e_{p+1}(x_0, \dots, x_p) h_r(x_0, \dots, x_p). \quad (18.18)$$

Consequently, for $f(z) = \sum_{m \geq 0} a_m z^m$,

$$\sum_{i=0}^p \mu_i f(cq^i z) = a_0 + (-1)^p c^{p+1} q^{\binom{p+1}{2}} \sum_{r \geq 0} a_{p+1+r} c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q z^{p+1+r}. \quad (18.19)$$

Proof. The first identity is Lagrange interpolation evaluated at 0; simplifying the products gives (18.17). Thus the first $p+1$ moments have the stated values. Let $E(t) = \prod_i (1 - x_i t)$. The polynomial $E(t) \sum_{m \geq 0} M_m t^m$ has degree at most p and agrees through degree p with $E(t)$; hence it equals $E(t) - (-1)^{p+1} e_{p+1} t^{p+1}$. Division by $E(t)$ proves (18.18). On geometric nodes, $e_{p+1} = c^{p+1} q^{\binom{p+1}{2}}$ and $h_r = c^r \begin{bmatrix} p+r \\ p \end{bmatrix}_q$. Apply the identity coefficientwise to f . $\square$

The last formula is an exact all-order error expansion, not merely an asymptotic estimate: the first omitted Taylor coefficient is multiplied by the product of all nodes, and every later coefficient receives a Gaussian residual.

Chapter 19

The Fabius distribution, moments, and dyadic Appell calculus

The special base $q = \frac{1}{2}$ occurring below is not a deformation parameter tending to 1; it is forced by binary dilation. We derive the relevant formulas from a probabilistic model, so the connection does not depend on the formalized repository development as a black box.

19.1 The random series and its self-similarity

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and define

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j, \quad F(x) = \mathbb{P}(X \leq x). \quad (19.1)$$

The series converges absolutely, $0 \leq X \leq 1$, and F is the bounded Fabius function [35, 33].

Proposition 19.1 (Affine distributional recursion and smoothness). *If X' is an independent copy of X , then*

$$X \stackrel{d}{=} \frac{U_1 + X'}{2}. \quad (19.2)$$

After extending F by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$,

$$F(x) = \int_0^1 F(2x - u) \, du, \quad (19.3)$$

$$F'(x) = 2F(2x) - 2F(2x - 1), \quad (19.4)$$

$$F(1 - x) = 1 - F(x), \quad (19.5)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (19.6)$$

Moreover $F \in C^\infty(\mathbb{R})$ and $0 \leq F'(x) \leq 2$, so F is 2-Lipschitz.

Proof. Separating the first summand in the random series and reindexing the tail gives (19.2). Conditioning on $U_1 = u$ gives (19.3), equivalently $F(x) = \int_{2x-1}^{2x} F(v) \, dv$. This first makes F continuous and then, by differentiating the moving integral, proves (19.4). The right side is a linear combination of affine rescalings of F ; induction therefore yields $F \in C^m$ for every m . The left-half formula follows because $F(2x - 1) = 0$ there.

Replacing all U_j by $1 - U_j$ preserves their joint law and changes X to $1 - X$. The distribution is atomless because it is convolved with a uniform law, so this proves symmetry. The derivative formula gives $0 \leq F' \leq 2$ on the left half; differentiated symmetry transfers the bound to the right half. $\square$

Define Rvachev's compactly supported even function by

$$\text{up}(t) = \begin{cases} F(1 - |t|), & |t| \leq 1, \\ 0, & |t| > 1. \end{cases} \quad (19.7)$$

19.2 Moments, the entire product, and inverse-dyadic values

Set

$$d_n = \mathbb{E}X^n, \quad a_n = \frac{d_n}{n!}, \quad d_0 = a_0 = 1, \quad (19.8)$$

and let $\Phi(z) = (e^z - 1)/z$, with $\Phi(0) = 1$.

Proposition 19.2 (Half-moment interpretation). *For $n \geq 1$,*

$$d_n = n \int_0^1 t^{n-1} \text{up}(t) \, dt. \quad (19.9)$$

Proof. The layer-cake identity $x^n = \int_0^x nt^{n-1} \, dt$ and Tonelli's theorem give $d_n = n \int_0^1 t^{n-1} \mathbb{P}(X > t) \, dt$. Atomlessness and symmetry give $\mathbb{P}(X > t) = 1 - F(t) = F(1 - t) = \text{up}(t)$. $\square$

Theorem 19.3 (Dyadic moment product). *The moment generating function*

$$A(z) = \mathbb{E}e^{zX} = \sum_{n=0}^{\infty} a_n z^n \quad (19.10)$$

is entire and satisfies

$$A(z) = \prod_{j=1}^{\infty} \Phi\left(\frac{z}{2^j}\right), \quad (19.11)$$

$$A(2z) = \Phi(z)A(z), \quad (19.12)$$

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n - j + 1)!} \quad (n \geq 1). \quad (19.13)$$

Proof. For $X_N = \sum_{j=1}^N 2^{-j} U_j$, independence gives

$$\mathbb{E}e^{zX_N} = \prod_{j=1}^N \int_0^1 e^{zu/2^j} \, du = \prod_{j=1}^N \Phi(z/2^j).$$

The convergence is locally uniform because $\Phi(w) = 1 + O(w)$ and $\sum_j 2^{-j} < \infty$; it also follows from dominated convergence on compact sets. This proves the entire product. Replace z by $2z$ and separate the first factor to obtain the functional equation. Since $\Phi(z) = \sum_{m \geq 0} z^m / (m + 1)!$, coefficient comparison gives the recurrence. $\square$

Theorem 19.4 (Moment formula for inverse powers of two). *For every $n \geq 0$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (19.14)$$

Proof. The case $n = 0$ is $F(1) = 1$. Iterating (19.6) on $0 \leq x \leq 2^{-n}$ gives $F^{(n)}(x) = 2^{\binom{n+1}{2}} F(2^n x)$. All derivatives of lower order vanish at 0 because F is smooth and identically zero on the negative half-line. Repeated integration therefore gives

$$\begin{aligned} F(2^{-n}) &= \frac{1}{(n-1)!} \int_0^{2^{-n}} (2^{-n} - t)^{n-1} F^{(n)}(t) dt \\ &= \frac{2^{-\binom{n}{2}}}{(n-1)!} \int_0^1 (1-u)^{n-1} F(u) du. \end{aligned}$$

After $s = 1 - u$, the integral is $\int_0^1 s^{n-1} \text{up}(s) ds = d_n/n$. $\square$

19.3 Appell polynomials and a box-spline formula

Definition 19.5 (Fabius moment Appell polynomials). For $n \geq 0$ and $\theta \in \mathbb{C}$, put

$$A_n(\theta) = \mathbb{E}(X + \theta)^n = \sum_{j=0}^n \binom{n}{j} d_{n-j} \theta^j. \quad (19.15)$$

Equivalently,

$$\sum_{n=0}^{\infty} A_n(\theta) \frac{z^n}{n!} = e^{\theta z} A(z). \quad (19.16)$$

Proposition 19.6 (Appell and reflection laws). For $n \geq 1$,

$$A'_n(\theta) = n A_{n-1}(\theta), \quad (19.17)$$

$$A_n(\theta + c) = \sum_{j=0}^n \binom{n}{j} A_{n-j}(\theta) c^j, \quad (19.18)$$

$$A_n(-1 - \theta) = (-1)^n A_n(\theta). \quad (19.19)$$

Proof. Differentiate the generating function or multiply it by e^{cz} . For the reflection, use $1 - X \stackrel{d}{=} X$. $\square$

For $y \in \mathbb{R}$, write $y_+ = \max(y, 0)$.

Lemma 19.7 (Distribution of a sum of unequal uniforms). Let V_j be independent and uniform on $[0, c_j]$, where $c_j > 0$, and set $S = V_1 + \cdots + V_n$. Then

$$\mathbb{P}(S \leq t) = \frac{1}{n! \prod_{j=1}^n c_j} \sum_{J \subseteq \{1, \dots, n\}} (-1)^{|J|} \left(t - \sum_{j \in J} c_j \right)_+^n. \quad (19.20)$$

Proof. The probability is the volume of $\{v_j \in [0, c_j] : \sum v_j \leq t\}$ divided by $\prod c_j$. In the positive orthant, the simplex $\{v_j \geq 0 : \sum v_j \leq y\}$ has volume $y_+^n/n!$. Impose the upper bounds by inclusion–exclusion; translating v_j by c_j for every violated index $j \in J$ gives the displayed shifted simplex. $\square$

Let $\varepsilon_h = (-1)^{\text{wt}_2(h)}$, where $\text{wt}_2(h)$ is the binary Hamming weight.

Theorem 19.8 (Exact formula on every dyadic cell). Let $n \geq 1$, $0 \leq m \leq 2^n$, and $0 \leq \theta < 1$, with $m + \theta \leq 2^n$. Then

$$F\left(\frac{m + \theta}{2^n}\right) = \varepsilon_m F\left(\frac{\theta}{2^n}\right) + \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h A_n(m - h - 1 + \theta). \quad (19.21)$$

The sum is empty when $m = 0$.

Proof. Split the random series after n terms:

$$2^n X = \sum_{j=1}^n 2^{n-j} U_j + X^{(n)}, \quad X^{(n)} \stackrel{d}{=} X,$$

with independent tail. Conditional on $X^{(n)} = x$, apply [Theorem 19.7](#) to the side lengths $2^{n-1}, \dots, 1$, whose product is $2^{\binom{n}{2}}$. Subsets of these lengths have sums $h = 0, \dots, 2^n - 1$, with sign ε_h , so

$$F(t/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{2^n-1} \varepsilon_h \mathbb{E}(t - h - X)_+^n. \quad (19.22)$$

Take $t = m + \theta$. Terms with $h > m$ vanish. For $h < m$, symmetry gives $m + \theta - h - X \stackrel{d}{=} X + m - h - 1 + \theta$, which is nonnegative. The term $h = m$ is $\varepsilon_m \mathbb{E}(\theta - X)_+^n$; applying (19.22) at $t = \theta < 1$ identifies its normalization with $\varepsilon_m F(\theta/2^n)$. $\square$

Corollary 19.9 (One-shot Appell formula for dyadic rationals). *For $n \geq 1$ and $0 \leq m \leq 2^n$,*

$$F(m/2^n) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{h=0}^{m-1} \varepsilon_h A_n(m - h - 1). \quad (19.23)$$

Proof. Set $\theta = 0$ in (19.21) and use $F(0) = 0$. $\square$

Corollary 19.10 (Local reduction identity). *For $0 \leq \theta < 1$ and $n \geq 1$,*

$$F\left(\frac{1+\theta}{2^n}\right) + F\left(\frac{\theta}{2^n}\right) = \frac{2^{-\binom{n}{2}}}{n!} A_n(\theta). \quad (19.24)$$

Proof. Use [Theorem 19.8](#) with $m = 1$ and $\varepsilon_1 = -1$. $\square$

19.4 Binary digits and a sparse global expansion

Let $x \in [0, 1)$ have the canonical binary expansion $x = 0.b_1 b_2 \dots$ that is not eventually all ones. Put

$$m_j = \lfloor 2^j x \rfloor, \quad \theta_j = 2^j x - m_j, \quad m_0 = 0. \quad (19.25)$$

Then $m_j = 2m_{j-1} + b_j$ and $\theta_{j-1} = (b_j + \theta_j)/2$.

Theorem 19.11 (Digit-sparse Appell expansion). *For every $x \in [0, 1)$,*

$$F(x) = \sum_{j=1}^{\infty} b_j \varepsilon_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j). \quad (19.26)$$

If $S_N(x)$ is the first N terms, then

$$F(x) - S_N(x) = \varepsilon_{m_N} F(\theta_N/2^N), \quad |F(x) - S_N(x)| \leq \frac{2^{-\binom{N}{2}}}{N!}. \quad (19.27)$$

The series converges absolutely and uniformly and terminates for a dyadic rational in its canonical terminating expansion.

Proof. Set $P_N = F(x) - \varepsilon_{m_N} F(\theta_N/2^N)$. If $b_j = 0$, then the residual term is unchanged. If $b_j = 1$, then $\varepsilon_{m_j} = -\varepsilon_{m_{j-1}}$ and $\theta_{j-1}/2^{j-1} = (1 + \theta_j)/2^j$; hence [Theorem 19.10](#) gives

$$P_j - P_{j-1} = \varepsilon_{m_{j-1}} \frac{2^{-\binom{j}{2}}}{j!} A_j(\theta_j).$$

Telescoping proves the finite identity and exact residual. Since $0 \leq \theta_N < 1$, monotonicity and [\(19.14\)](#) give $F(\theta_N/2^N) \leq 2^{-\binom{N}{2}} d_N/N! \leq 2^{-\binom{N}{2}}/N!$. Finally $|A_j(\theta_j)| \leq 2^j$, so the terms are uniformly dominated by the summable sequence $2^{j-\binom{j}{2}}/j!$. $\square$

Chapter 20

Thue–Morse cancellation and exact half-base Gaussian formulas

Set

$$\rho = \frac{1}{2}, \quad P_n = (\rho; \rho)_n = \prod_{j=1}^n (1 - 2^{-j}). \quad (20.1)$$

This fixed base encodes binary dilation rather than a limit process.

20.1 Mersenne arithmetic and binary subspaces

Proposition 20.1 (Two meanings of the half-base normalization). *For $n \geq 0$ and $0 \leq k \leq n$,*

$$P_n = 2^{-\binom{n+1}{2}} \prod_{j=1}^n (2^j - 1), \quad (20.2)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = 2^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2. \quad (20.3)$$

Thus the half-base coefficient is a power-of-two normalization of the number of k -dimensional subspaces of $\mathbb{F}_2^n$.

Proof. Extract 2^{-j} from each factor $1 - 2^{-j}$ to obtain the first formula. The second is Gaussian reciprocity $q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = \begin{bmatrix} n \\ k \end{bmatrix}_q$ at $q = 2$. The finite-field interpretation was proved in [Chapter 6](#). $\square$

20.2 The dyadic nodal annihilator

For $0 \leq k \leq n$, define

$$w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \quad x_k = -2^k, \quad C_n = 2^{\binom{n+1}{2}} P_n. \quad (20.4)$$

Theorem 20.2 (Geometric moment annihilation). *For every $d \geq 0$,*

$$\sum_{k=0}^n w_{n,k} x_k^d = (-1)^d (2^d; 1/2)_n. \quad (20.5)$$

Consequently

$$\sum_{k=0}^n w_{n,k} x_k^d = \begin{cases} 0, & 0 \leq d < n, \\ C_n, & d = n. \end{cases} \quad (20.6)$$

Proof. Since $x_k^d = (-1)^d 2^{kd}$, the sum is

$$(-1)^d \sum_{k=0}^n (-1)^k (1/2)^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^d)^k,$$

which is $(-1)^d (2^d; 1/2)_n$ by the finite q -binomial theorem. If $d < n$, the factor indexed by $j = d$ vanishes. For $d = n$,

$$(-1)^n (2^n; 1/2)_n = \prod_{r=1}^n (2^r - 1) = 2^{\binom{n+1}{2}} P_n.$$

□

20.3 Thue–Morse blocks and their forced zero

Theorem 20.3 (Thue–Morse exponential product and Prouhet cancellation). *For every $k \geq 0$,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r e^{rz} = \prod_{j=0}^{k-1} (1 - e^{2^j z}). \quad (20.7)$$

Consequently, for every translation τ ,

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r + \tau)^d = 0 \quad (0 \leq d < k), \quad (20.8)$$

$$\sum_{r=0}^{2^k-1} \varepsilon_r (r + \tau)^k = (-1)^k 2^{\binom{k}{2}} k!. \quad (20.9)$$

Proof. Write $r = \sum_{j=0}^{k-1} b_j 2^j$. Then $\varepsilon_r = \prod_j (-1)^{b_j}$, and independent summation over the bits gives the product. Its zero at $z = 0$ has exact order k and leading term $(-1)^k 2^{\binom{k}{2}} z^k$. Multiplication by $e^{\tau z}$ does not change that order or coefficient. Comparing exponential-series coefficients proves the power-sum identities. □

Define

$$\Psi_k(z) = e^{-2^k z} \prod_{j=0}^{k-1} \Phi(2^j z) = e^{-z} \prod_{j=0}^{k-1} \Phi(-2^j z). \quad (20.10)$$

The equality follows from $\Phi(-u) = e^{-u} \Phi(u)$.

Proposition 20.4 (Normalized block coefficients). *For $\theta \in \mathbb{C}$, write*

$$e^{-2^k \theta z} \Psi_k(z) = \sum_{m=0}^{\infty} c_{k,m}(\theta) z^m. \quad (20.11)$$

Then

$$c_{k,m}(\theta) = (-1)^k 2^{-\binom{k}{2}} \frac{1}{(m+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k(1+\theta))^{m+k}, \quad (20.12)$$

and

$$e^{-2^k \theta z} \Psi_k(z) A(-z) = e^{-z} (e^{\theta u} A(u))_{u=-2^k z}. \quad (20.13)$$

Proof. Using $1 - e^u = -u\Phi(u)$ in (20.7),

$$\sum_{r < 2^k} \varepsilon_r e^{(r-2^k(1+\theta))z} = (-1)^k 2^{\binom{k}{2}} z^k e^{-2^k \theta z} \Psi_k(z).$$

Compare the coefficient of z^{m+k} . Iterating $A(2z) = \Phi(z)A(z)$ at $-z, -2z, \dots, -2^{k-1}z$ gives $A(-2^k z) = A(-z) \prod_{j < k} \Phi(-2^j z)$; multiplication by the remaining exponentials gives (20.13). $\square$

One factor $2^{-\binom{k}{2}}$ comes from the Gaussian weights and another from removing the forced Thue–Morse zero. Their product is $4^{-\binom{k}{2}}$.

20.4 A coefficient-extraction principle

Theorem 20.5 (Dyadic Gaussian coefficient extraction). *Let $B_k(z) = \sum_{m \geq 0} b_{k,m} z^m$ and $H(z) = \sum_{m \geq 0} h_m z^m$, with $A(0) = H(0) = 1$. Suppose*

$$B_k(z) A(-z) = E(z) H(-2^k z), \quad (20.14)$$

where $E(z)$ is independent of k and $E(0) = 1$. Then

$$\sum_{k=0}^n w_{n,k} b_{k,d} = 0 \quad (0 \leq d < n), \quad (20.15)$$

$$\sum_{k=0}^n w_{n,k} b_{k,n} = C_n h_n. \quad (20.16)$$

Proof. The coefficient of z^d on the left of (20.14) is $\sum_{i=0}^d b_{k,i} (-1)^{d-i} a_{d-i}$. If $E(z) = \sum e_j z^j$, the coefficient on the right is $\sum_{j+\ell=d} e_j h_\ell (-2^k)^\ell$. Multiply by $w_{n,k}$ and sum over k . For $d < n$, every exponent $\ell \leq d$ is annihilated by (20.6); induction on d then annihilates all terms $i < d$ on the left, leaving the first formula. For $d = n$, all terms with $\ell < n$ vanish and the sole survivor is $e_0 h_n \sum_k w_{n,k} (-2^k)^n = C_n h_n$. The lower-index terms on the left have already vanished, proving the second formula. $\square$

The common factor $E(z)$ may contain an arbitrary translation parameter. Its nonconstant coefficients multiply nodal powers below degree n and are invisible to the extractor.

20.5 Finite Gaussian formulas for the Appell polynomials

Theorem 20.6 (Centered half-base representation). *For every $n \geq 0$ and $\theta \in \mathbb{C}$,*

$$\frac{A_n(\theta)}{n!} = \frac{2^{-(n+1)}}{P_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k(1+\theta))^{n+k}. \quad (20.17)$$

Proof. In Theorem 20.5, take $B_k(z) = e^{-2^k \theta z} \Psi_k(z)$, $E(z) = e^{-z}$, and $H(u) = e^{\theta u} A(u)$. By (19.16), $h_n = A_n(\theta)/n!$. Insert (20.12) into (20.16). The two signs cancel and the two triangular powers combine into $4^{-\binom{k}{2}}$. Divide by $C_n = 2^{\binom{n+1}{2}} P_n$. $\square$

Theorem 20.7 (Raw translated representation). *For every $n \geq 0$ and every $\theta, \eta \in \mathbb{C}$,*

$$\frac{A_n(\theta)}{n!} = \frac{(-1)^n 2^{-(n+1)}}{P_n} \sum_{k=0}^n \frac{[k]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r + \eta + 2^k \theta)^{n+k}. \quad (20.18)$$

The complete finite expression is independent of η .

Proof. After removing the forced zero, the exponential generating function of the raw power sum is $B_k(z) = e^{(\eta+2^k(\theta+1))z} \Psi_k(z)$. Thus

$$B_k(z) A(-z) = e^{(\eta-1)z} (e^{(-1-\theta)u} A(u))_{u=-2^k z}.$$

Apply [Theorem 20.5](#). The n th coefficient of the function of u is $A_n(-1-\theta)/n! = (-1)^n A_n(\theta)/n!$. The same normalization as in the preceding proof yields the formula. Since η occurs only in the common factor $E(z)$, the extracted value is independent of it. $\square$

Setting $\theta = 0$ and multiplying by $2^{-\binom{n}{2}}$ gives

$$F(2^{-n}) = \frac{1}{2^{n^2} P_n} \sum_{k=0}^n \frac{[k]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^{n+k}. \quad (20.19)$$

20.6 Arbitrary dyadic numerators in one shot

Theorem 20.8 (Translated one-shot dyadic formula). *Let $n \geq 1$, $0 \leq m \leq 2^n$, and $\tau \in \mathbb{C}$. Then*

$$F(m/2^n) = \frac{1}{2^{n^2} P_n} \sum_{k=0}^n \frac{[k]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{s=0}^{m2^k-1} \varepsilon_s (s - m2^k + \tau)^{n+k}. \quad (20.20)$$

For fixed m, n , the right side is independent of τ .

Proof. Write $s = h2^k + r$, with $0 \leq h < m$ and $0 \leq r < 2^k$. There is no binary carry, so $\varepsilon_s = \varepsilon_h \varepsilon_r$. After division by the forced factor $(-1)^k 2^{\binom{k}{2}} z^k$, the exponential generating function of the long block, call it $B_{m,k,\tau}(z)$, satisfies

$$B_{m,k,\tau}(z) A(-z) = e^{(\tau-1)z} \sum_{h=0}^{m-1} \varepsilon_h (e^{(m-h-1)u} A(u))_{u=-2^k z}. \quad (20.21)$$

The coefficient of u^n in the right-hand function is $n!^{-1} \sum_{h < m} \varepsilon_h A_n(m-h-1)$, which equals $2^{\binom{n}{2}} F(m/2^n)$ by [\(19.23\)](#). Apply the extraction theorem and insert the normalized power sum. The resulting factor is $C_n 2^{\binom{n}{2}} = 2^{n^2} P_n$. The common exponential proves translation invariance. $\square$

20.7 The arbitrary-argument nested q -series

Theorem 20.9 (Digit-sparse half-base q -series). *With the binary-prefix notation of [\(19.25\)](#), every $x \in [0, 1)$ satisfies*

$$F(x) = \sum_{j=1}^{\infty} \frac{b_j \varepsilon_{m_{j-1}}}{2^{j^2} P_j} \sum_{k=0}^j \frac{[k]_{1/2}}{4^{\binom{k}{2}} (j+k)!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k(1 + \theta_j))^{j+k}. \quad (20.22)$$

The j th nested summand is exactly the j th summand in (19.26); hence the series is absolutely and uniformly convergent, has the exact residual (19.27), and terminates at every dyadic rational with its canonical terminating expansion.

Proof. Insert (20.17), with $n = j$ and $\theta = \theta_j$, into (19.26). The powers multiply as $2^{-\binom{j}{2}} 2^{-\binom{j+1}{2}} = 2^{-j^2}$. This is a termwise replacement by an exact finite identity, so all convergence and residual assertions are unchanged. $\square$

Chapter 21

Bernoulli–Bell formulas, sinc products, and a q -Toeplitz limit

The moment product (19.11) admits three complementary expansions. Expanding its factors directly gives a sum over compositions; taking its logarithm gives Bernoulli cumulants and complete Bell polynomials; and centering the underlying random series gives Rvachev’s infinite sinc product. The same half-base Gaussian row that extracted exact dyadic values also acts as a regular Toeplitz filter on finite uniform-spline approximants. We develop all four descriptions and prove that the Toeplitz scheme converges uniformly on the whole real line.

21.1 A composition formula for the reciprocal dyadic values

Write a composition of $n \geq 1$ as $\mathbf{r} = (r_1, \dots, r_\ell) \models n$, where every $r_i \geq 1$ and $r_1 + \dots + r_\ell = n$. Its tail sums are

$$t_h = r_h + r_{h+1} + \dots + r_\ell \quad (1 \leq h \leq \ell).$$

Lemma 21.1 (Geometric simplex sum). *For positive integers $r_1, \dots, r_\ell$,*

$$\sum_{1 \leq j_1 < \dots < j_\ell} 2^{-\sum_{i=1}^\ell j_i r_i} = \prod_{h=1}^\ell \frac{1}{2^{t_h} - 1}. \quad (21.1)$$

Proof. Put $d_1 = j_1$ and $d_h = j_h - j_{h-1}$ for $h \geq 2$. This is a bijection between strictly increasing positive integer tuples $(j_1, \dots, j_\ell)$ and positive integer tuples $(d_1, \dots, d_\ell)$. Since $j_i = d_1 + \dots + d_i$,

$$\sum_{i=1}^\ell j_i r_i = \sum_{h=1}^\ell d_h (r_h + \dots + r_\ell) = \sum_{h=1}^\ell d_h t_h.$$

The multiple sum therefore factors into independent geometric series:

$$\prod_{h=1}^\ell \sum_{d_h=1}^\infty 2^{-d_h t_h} = \prod_{h=1}^\ell \frac{2^{-t_h}}{1 - 2^{-t_h}} = \prod_{h=1}^\ell \frac{1}{2^{t_h} - 1}.$$

□

Theorem 21.2 (Composition formula). *For $n \geq 1$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{(r_1, \dots, r_\ell) \models n} \prod_{h=1}^{\ell} \frac{1}{(r_h + 1)! (2^{r_h} - 1)!}. \quad (21.2)$$

The empty composition gives the value $F(1) = 1$ when $n = 0$.

Proof. By (19.11),

$$A(z) = \prod_{j=1}^{\infty} \Phi(z/2^j), \quad \Phi(w) = \frac{e^w - 1}{w} = 1 + \sum_{r=1}^{\infty} \frac{w^r}{(r+1)!}.$$

To obtain z^n , choose a nonconstant term from finitely many factors, whose indices we write as $1 \leq j_1 < \dots < j_\ell$, and choose positive degrees $r_1 + \dots + r_\ell = n$. Thus

$$a_n = [z^n]A(z) = \sum_{r \models n} \frac{1}{\prod_h (r_h + 1)!} \sum_{1 \leq j_1 < \dots < j_\ell} 2^{-\sum_h j_h r_h}.$$

The interchange is harmless because, for a fixed coefficient, the resulting nonnegative series is absolutely convergent; it also follows formally by first truncating the product and then passing monotonically to the limit. Apply Theorem 21.1 and finally use $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$ from Theorem 19.4. $\square$

21.2 A lower-Hessenberg determinant

The triangular recurrence (19.13) may be solved by a continuant-like determinant. Put

$$\beta_{n,k} = \frac{1}{(2^n - 1)(n - k + 1)!} \quad (0 \leq k < n). \quad (21.3)$$

For $n \geq 1$, let H_n be the lower-Hessenberg matrix

$$H_n = \begin{pmatrix} \beta_{1,0} & -1 & 0 & \cdots & 0 \\ \beta_{2,0} & \beta_{2,1} & -1 & \ddots & \vdots \\ \beta_{3,0} & \beta_{3,1} & \beta_{3,2} & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ \beta_{n,0} & \beta_{n,1} & \beta_{n,2} & \cdots & \beta_{n,n-1} \end{pmatrix}. \quad (21.4)$$

Theorem 21.3 (Hessenberg determinant). *For $n \geq 1$,*

$$a_n = \det H_n, \quad F(2^{-n}) = 2^{-\binom{n}{2}} \det H_n. \quad (21.5)$$

Proof. Let $D_0 = 1$ and $D_n = \det H_n$. Expand D_n along its last row. For the entry in column $k + 1$, the part of the minor to its upper right is triangular with -1 on its diagonal, while the lower-left part is exactly H_k . The cofactor sign cancels the product of those -1 entries, leaving

$$D_n = \sum_{k=0}^{n-1} \beta_{n,k} D_k.$$

This is the recurrence (19.13), with $D_0 = a_0 = 1$. Induction gives $D_n = a_n$. The second identity follows from (19.14). $\square$

21.3 Bernoulli cumulants and complete Bell polynomials

Let the Bernoulli numbers be defined by

$$\frac{z}{e^z - 1} = \sum_{m=0}^{\infty} B_m \frac{z^m}{m!}, \quad B_1 = -\frac{1}{2}. \quad (21.6)$$

Let $\mathcal{B}_n(x_1, \dots, x_n)$ denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{m=1}^{\infty} x_m \frac{z^m}{m!}\right) = \sum_{n=0}^{\infty} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (21.7)$$

Lemma 21.4 (Logarithm of the elementary moment factor). *Near $z = 0$,*

$$\log \Phi(z) = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m} \frac{z^{2m}}{(2m)!}. \quad (21.8)$$

Proof. Logarithmic differentiation gives

$$\frac{d}{dz} \log \Phi(z) = \frac{e^z}{e^z - 1} - \frac{1}{z} = 1 + \frac{1}{e^z - 1} - \frac{1}{z}.$$

From (21.6),

$$\frac{1}{e^z - 1} = \frac{1}{z} - \frac{1}{2} + \sum_{m=1}^{\infty} B_{2m} \frac{z^{2m-1}}{(2m)!},$$

because $B_{2m+1} = 0$ for $m \geq 1$. Therefore

$$(\log \Phi)'(z) = \frac{1}{2} + \sum_{m=1}^{\infty} B_{2m} \frac{z^{2m-1}}{(2m)!}.$$

Integrate termwise and use $\Phi(0) = 1$, hence $\log \Phi(0) = 0$. □

Theorem 21.5 (Cumulants of the Fabius distribution). *The logarithm of the moment generating function is*

$$\log A(z) = \frac{z}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2^{2m} - 1)} \frac{z^{2m}}{(2m)!}. \quad (21.9)$$

Consequently, if $\log A(z) = \sum_{r \geq 1} \kappa_r z^r / r!$, then

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2m} = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0 \quad (m \geq 1). \quad (21.10)$$

Moreover

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{n!} \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (21.11)$$

Proof. Take logarithms in (19.11) and use Theorem 21.4. Local uniform convergence of the product permits termwise summation near the origin. The linear terms sum to

$$\frac{z}{2} \sum_{j=1}^{\infty} 2^{-j} = \frac{z}{2},$$

while, for $m \geq 1$,

$$\sum_{j=1}^{\infty} 2^{-2mj} = \frac{1}{2^{2m} - 1}.$$

This proves (21.9) and the stated cumulants. Exponentiating and comparing with (21.7) gives $d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n)$ and $a_n = d_n/n!$. Now apply (19.14). □

21.4 The Rvachev density and its infinite sinc product

Let $V_j = 2U_j - 1$, so each V_j is uniform on $[-1, 1]$, and put

$$Z = 2X - 1 = \sum_{j=1}^{\infty} 2^{-j} V_j. \quad (21.12)$$

The density of Z is Rvachev's even compactly supported function

$$\text{up}(t) = \begin{cases} F(1+t), & -1 \leq t \leq 0, \\ F(1-t), & 0 < t \leq 1, \\ 0, & |t| > 1. \end{cases} \quad (21.13)$$

This is the normalization used in the Fabius literature [35, 33, 34]. We use $\text{sinc } w = \sin(w)/w$, with $\text{sinc } 0 = 1$.

Theorem 21.6 (Sinc and hyperbolic products). *For every real t and every complex z ,*

$$\int_{-1}^1 \text{up}(x) e^{itx} dx = \prod_{j=1}^{\infty} \text{sinc}(t/2^j), \quad (21.14)$$

$$e^{-z} A(2z) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (21.15)$$

Both products converge locally uniformly in their complex argument.

Proof. The characteristic function of $2^{-j}V_j$ is

$$\frac{1}{2} \int_{-1}^1 e^{it2^{-j}v} dv = \text{sinc}(t/2^j).$$

Independence gives the finite product for the partial sums of (21.12). Those partial sums converge almost surely and uniformly boundedly to Z , so dominated convergence gives the limit characteristic function. Since up is the density of Z , this is (21.14).

Similarly,

$$\mathbb{E} e^{zZ} = e^{-z} \mathbb{E} e^{2zX} = e^{-z} A(2z).$$

The moment generating factor of $2^{-j}V_j$ is $\sinh(z/2^j)/(z/2^j)$, so independence gives (21.15). On each compact set the j th factor is $1 + O(4^{-j})$, uniformly; the Weierstrass criterion therefore proves local uniform convergence of both products. $\square$

21.5 Finite centered uniform splines

Define the midpoint-tail approximation

$$X_p = \sum_{j=1}^p 2^{-j} U_j + 2^{-p-1}, \quad S_p(x) = \mathbb{P}(X_p \leq x). \quad (21.16)$$

For $r \in \mathbb{N}_0$, retain $\varepsilon_r = (-1)^{\text{wt}_2(r)}$ and write $y_+ = \max(y, 0)$.

Theorem 21.7 (Finite Thue–Morse spline). *For $p \geq 1$ and every real x ,*

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r=0}^{2^p-1} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p. \quad (21.17)$$

Moreover

$$\sup_{x \in \mathbb{R}} |S_p(x) - F(x)| \leq 2^{-p}. \quad (21.18)$$

Proof. For positive weights $a_1, \dots, a_p$, inclusion–exclusion on the box $[0, 1]^p$ gives

$$\mathbb{P}\left(\sum_{j=1}^p a_j U_j \leq y\right) = \frac{1}{p! \prod_j a_j} \sum_{J \subseteq \{1, \dots, p\}} (-1)^{|J|} \left(y - \sum_{j \in J} a_j\right)_+^p. \quad (21.19)$$

Indeed, after the change of variables $v_j = a_j u_j$, the desired probability is $(\prod a_j)^{-1}$ times the volume of $\{0 \leq v_j \leq a_j : \sum v_j \leq y\}$. Without the upper bounds its volume is $y_+^p/p!$; inclusion–exclusion over the violated inequalities $v_j > a_j$, followed by translation in those coordinates, proves (21.19).

Take $a_j = 2^{-j}$ and $y = x - 2^{-p-1}$. The subset sums $\sum_{j \in J} 2^{p-j}$ run bijectively through 0, $\dots, 2^p - 1$, and their subset parity is the binary digit sum parity. Since $\prod_j a_j = 2^{-p(p+1)/2}$ and extracting 2^{-p} from the positive part contributes 2^{-p^2} , the total prefactor is $2^{-\binom{p}{2}}/p!$. This proves (21.17).

Couple X_p and X with the same U_j . Their difference is the tail minus its midpoint:

$$|X - X_p| = \left| \sum_{j > p} 2^{-j} U_j - 2^{-p-1} \right| \leq 2^{-p-1}.$$

The law of X has a density bounded by 2: condition on $R = \sum_{j \geq 2} 2^{-j} U_j$ and average the density $2\mathbf{1}_{[R, R+1/2]}$ of $R + U_1/2$. Hence F is 2-Lipschitz. The coupling implies $F(x - \delta) \leq S_p(x) \leq F(x + \delta)$ with $\delta = 2^{-p-1}$, and the Lipschitz bound gives (21.18). $\square$

21.6 The half-base Gaussian Toeplitz filter

For $p \geq 1$ define the *clamped midpoint cutoff*

$$N_p(x) = \min\left(2^p, \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right)\right). \quad (21.20)$$

The upper clamp is essential: it is what makes the formulas below describe the bounded distribution F also for $x \geq 1$, rather than a signed global extension.

For $n \geq 1$ define

$$D_n(x) = \frac{1}{2^{n^2} P_n} \sum_{k=0}^n \frac{\left[n\right]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{N_{n+k}(x)-1} \varepsilon_r \left(r - 2^{n+k} x + \frac{1}{2}\right)^{n+k}, \quad (21.21)$$

$$\omega_{n,j} = \frac{(-1)^j 2^{-\binom{j+1}{2}} \left[n\right]_{1/2}}{P_n} \quad (0 \leq j \leq n). \quad (21.22)$$

As usual, a sum with upper endpoint -1 is empty.

Theorem 21.8 (Exact Toeplitz decomposition). *For every $n \geq 1$ and every $x \in \mathbb{R}$,*

$$D_n(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}(x). \quad (21.23)$$

The row-generating polynomial is

$$W_n(z) = \sum_{j=0}^n \omega_{n,j} z^j = \frac{(z/2; 1/2)_n}{(1/2; 1/2)_n}. \quad (21.24)$$

Consequently

$$\sum_{j=0}^n \omega_{n,j} = 1, \quad (21.25)$$

$$V_n := \sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n}, \quad (21.26)$$

and (V_n) is bounded.

Proof. The cutoff (21.20) is exactly the set of indices for which the positive part in (21.17) is active. Thus, for every $p \geq 1$ and every real x ,

$$\frac{1}{p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)^p = (-1)^p 2^{\binom{p}{2}} S_p(x). \quad (21.27)$$

Insert $p = n + k$ into (21.21). Then put $j = n - k$ and use half-base symmetry $\left[\begin{smallmatrix} n \\ n-j \end{smallmatrix}\right]_{1/2} = \left[\begin{smallmatrix} n \\ j \end{smallmatrix}\right]_{1/2}$. The sign becomes $(-1)^{n+k} = (-1)^j$, and the power of 2 simplifies by

$$\binom{n+k}{2} - n^2 - 2\binom{k}{2} = -\binom{j+1}{2}.$$

This gives (21.23).

The finite q -binomial theorem at $q = 1/2$ gives

$$(z/2; 1/2)_n = \sum_{j=0}^n (-1)^j 2^{-\binom{j}{2}} (z/2)^j \left[\begin{smallmatrix} n \\ j \end{smallmatrix}\right]_{1/2},$$

which is exactly $P_n W_n(z)$. Setting $z = 1$ proves the mass identity. All Gaussian coefficients are positive at $q = 1/2$, so replacing each sign by its absolute value is the same as setting $z = -1$, proving (21.26). Finally,

$$V_n = \prod_{h=1}^n \frac{1 + 2^{-h}}{1 - 2^{-h}}$$

converges to a finite positive limit because the logarithm of its h th factor is $O(2^{-h})$. Hence the variations are uniformly bounded. $\square$

Theorem 21.9 (Uniform half-base q -Toeplitz limit). *For every $n \geq 1$,*

$$\sup_{x \in \mathbb{R}} |D_n(x) - F(x)| \leq V_n 2^{-n}. \quad (21.28)$$

In particular, $D_n \rightarrow F$ uniformly on $\mathbb{R}$.

Proof. By (21.25) and (21.23),

$$D_n(x) - F(x) = \sum_{j=0}^n \omega_{n,j} (S_{2n-j}(x) - F(x)).$$

Apply (21.18). Since $2n - j \geq n$,

$$|D_n(x) - F(x)| \leq \sum_{j=0}^n |\omega_{n,j}| 2^{-(2n-j)} \leq 2^{-n} V_n.$$

The bound is independent of x , and V_n is bounded by Theorem 21.8; therefore it tends to zero. $\square$

21.7 Synthesis of the half-base theory

The preceding chapters exhibit four mathematically distinct roles of the same finite Gaussian row:

1. it is the coefficient row of the nodal polynomial $(z/2; 1/2)_n$ and hence annihilates the first n geometric moments;
2. it extracts one Appell coefficient from normalized Thue–Morse blocks, producing exact values of F at dyadic arguments;
3. it gives the complete residual multiplier of a formal or analytic series sampled on a geometric grid;
4. after normalization by P_n , it becomes a regular Toeplitz row that combines finite box splines and converges uniformly to the Fabius CDF.

These are not analogies. They are all direct consequences of the same finite q -binomial factorization, specialized at the dyadic base $q = 1/2$. The denominator-free treatment in Chapter 18 shows which parts survive at singular specializations; the probabilistic and analytic arguments in Chapters 19 and 21 explain why this particular specialization governs the Fabius–Rvachev theory.

Part VI

Central sums, limits, and effective methods

Chapter 22

Central q -binomial sums and Lambert series

Central Gaussian coefficients frequently become tractable only after a seemingly extraneous factor is inserted. The elementary identity in the next lemma shows why the factor $(-q; q)_{2k}$ is natural: it converts a central coefficient at base q^2 into a single quotient of shifted factorials. This chapter develops both a regular q -beta evaluation and a deleted-singularity identity proved recently by Chern, Dilcher, and Jiu [7].

22.1 The central-factor reduction

Lemma 22.1. *For every $k \geq 0$,*

$$\left[\begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{1}{(-q; q)_{2k}} = \frac{(q; q^2)_k}{(q^2; q^2)_k}. \quad (18.1)$$

Proof. The elementary product identity

$$(q; q)_{2k}(-q; q)_{2k} = (q^2; q^2)_{2k}$$

gives

$$\begin{aligned} \left[\begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{1}{(-q; q)_{2k}} &= \frac{(q^2; q^2)_{2k}}{(q^2; q^2)_k^2} \frac{(q; q)_{2k}}{(q^2; q^2)_{2k}} \\ &= \frac{(q; q^2)_k (q^2; q^2)_k}{(q^2; q^2)_k^2}, \end{aligned}$$

which is the claimed quotient. □

For a complex parameter z and $0 < q < 1$, we use the analytic extension

$$[z]_q = \frac{1 - q^z}{1 - q},$$

with the real logarithm in $q^z = \exp(z \log q)$.

22.2 A q -beta evaluation

Theorem 22.2 (A regular central q -binomial sum). *Let $0 < q < 1$, and let $\alpha \in \mathbb{C}$ be such that none of the denominators below vanishes. Then*

$$\begin{aligned} \sum_{k=0}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [2k+1+\alpha]_q} \\ = \frac{\Gamma_{q^2}(3/2) \Gamma_{q^2}((\alpha+1)/2)}{\Gamma_{q^2}((\alpha+2)/2)}. \end{aligned} \quad (18.2)$$

In particular, for real α it is enough to exclude $\alpha = -1, -3, -5, \dots$

Proof. By Theorem 22.1 and the telescoping quotient

$$\frac{1}{[2k+1+\alpha]_q} = \frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+1}; q^2)_k}{(q^{\alpha+3}; q^2)_k},$$

the left side is

$$\frac{1}{[\alpha+1]_q} {}_2\phi_1 \left[\begin{matrix} q, q^{\alpha+1} \\ q^{\alpha+3} \end{matrix}; q^2, q \right].$$

The q -Gauss sum from Theorem 9.4, with base q^2 , gives

$$\frac{1}{[\alpha+1]_q} \frac{(q^{\alpha+2}, q^2; q^2)_{\infty}}{(q^{\alpha+3}, q; q^2)_{\infty}}. \quad (18.3)$$

Now

$$(q^{\alpha+1}; q^2)_{\infty} = (1 - q^{\alpha+1})(q^{\alpha+3}; q^2)_{\infty}, \quad (q; q^2)_{\infty} = (1 - q)(q^3; q^2)_{\infty}.$$

Using these two relations to simplify (18.3), we obtain

$$\frac{(q^2; q^2)_{\infty} (q^{\alpha+2}; q^2)_{\infty}}{(q^3; q^2)_{\infty} (q^{\alpha+1}; q^2)_{\infty}}. \quad (18.4)$$

Finally insert

$$\Gamma_{q^2}(z) = (1 - q^2)^{1-z} \frac{(q^2; q^2)_{\infty}}{(q^{2z}; q^2)_{\infty}}.$$

In the quotient on the right of (18.2), all powers of $1 - q^2$ cancel, and the remaining products are exactly (18.4). $\square$

Corollary 22.3 (Classical limit of the regular sum). *For $\alpha \notin \{-1, -3, -5, \dots\}$,*

$$\sum_{k=0}^{\infty} \binom{2k}{k} \frac{1}{4^k (2k+1+\alpha)} = \frac{\Gamma(3/2) \Gamma((\alpha+1)/2)}{\Gamma((\alpha+2)/2)}. \quad (18.5)$$

Proof. The series is

$$\frac{1}{\alpha+1} {}_2F_1 \left(\begin{matrix} 1/2, (\alpha+1)/2 \\ (\alpha+3)/2 \end{matrix}; 1 \right).$$

Gauss's classical summation gives

$$\frac{1}{\alpha+1} \frac{\Gamma((\alpha+3)/2) \Gamma(1/2)}{\Gamma((\alpha+2)/2)}.$$

Since $\Gamma((\alpha+3)/2) = ((\alpha+1)/2) \Gamma((\alpha+1)/2)$ and $\Gamma(1/2)/2 = \Gamma(3/2)$, this is (18.5). The same identity is also the $q \rightarrow 1^-$ limit of Theorem 22.2 by the q -gamma limit in Theorem 12.5. $\square$

22.3 Deleting a singular term

The regular formula becomes singular when $\alpha = -2m - 1$. Removing the offending $k = m$ term does not merely produce a finite part of the same gamma quotient; it yields an alternating Lambert-series tail.

Theorem 22.4 (Deleted-singularity central identity). *Let $m \geq 0$ and $0 < q < 1$. Then*

$$\begin{aligned} & \sum_{\substack{k \geq 0 \\ k \neq m}} \left[\begin{matrix} 2k \\ k \end{matrix} \right]_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ &= \left[\begin{matrix} 2m \\ m \end{matrix} \right]_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q}. \end{aligned} \quad (18.6)$$

Proof. By Theorem 22.1, the left side is

$$(1 - q^2) \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - q^{2k-2m})}. \quad (18.7)$$

Introduce an auxiliary parameter a and define

$$\begin{aligned} F(a, q) := & \frac{1}{1 - aq^{-2m}} \left({}_2\phi_1 \left[\begin{matrix} q, aq^{-2m} \\ aq^{-2m+2}; q^2, q \end{matrix} \right] \right. \\ & \left. - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right). \end{aligned} \quad (18.8)$$

The elementary cancellation

$$\frac{(aq^{-2m}; q^2)_k}{(aq^{-2m+2}; q^2)_k} = \frac{1 - aq^{-2m}}{1 - aq^{2k-2m}}$$

shows that

$$F(a, q) = \sum_{\substack{k \geq 0 \\ k \neq m}} \frac{(q; q^2)_k q^k}{(q^2; q^2)_k (1 - aq^{2k-2m})}. \quad (18.9)$$

Thus (18.7) equals $(1 - q^2) \lim_{a \rightarrow 1} F(a, q)$.

Apply q -Gauss with base q^2 to the ${}_2\phi_1$ in (18.8). We obtain

$$\begin{aligned} F(a, q) &= \frac{1}{1 - aq^{-2m}} \left(\frac{(q^2, aq^{-2m+1}; q^2)_{\infty}}{(q, aq^{-2m+2}; q^2)_{\infty}} - \frac{(q, aq^{-2m}; q^2)_m q^m}{(q^2, aq^{-2m+2}; q^2)_m} \right) \\ &= \frac{H(a) - C_m}{1 - a}, \end{aligned} \quad (18.10)$$

where

$$\begin{aligned} H(a) &= \frac{(q^2, aq; q^2)_{\infty} (aq^{-2m+1}; q^2)_m}{(q, aq^2; q^2)_{\infty} (aq^{-2m}; q^2)_m}, \\ C_m &= \frac{(q; q^2)_m q^m}{(q^2; q^2)_m}. \end{aligned}$$

The reversal identity for finite shifted factorials gives

$$\frac{(q^{-2m+1}; q^2)_m}{(q^{-2m}; q^2)_m} = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m},$$

so $H(1) = C_m$. Hence L'Hospital's rule yields

$$\lim_{a \rightarrow 1} F(a, q) = -H'(1) = -H(1) \left. \frac{\partial}{\partial a} \log H(a) \right|_{a=1}. \quad (18.11)$$

Differentiating the logarithms of the product factors gives

$$\lim_{a \rightarrow 1} F(a, q) = C_m \left(\sum_{r=1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r} + \sum_{r=1}^{2m} \frac{(-1)^{r-1} q^{-r}}{1 - q^{-r}} \right). \quad (18.12)$$

For $r \geq 1$,

$$\frac{q^{-r}}{1 - q^{-r}} = -\frac{1}{1 - q^r}.$$

Consequently, the two summands in (18.12) combine, for $1 \leq r \leq 2m$, to $(-1)^r$. Their sum is 0 because $2m$ is even. Therefore

$$\lim_{a \rightarrow 1} F(a, q) = C_m \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{1 - q^r}. \quad (18.13)$$

Multiply by $1 - q^2 = (1 + q)(1 - q)$, use $(1 - q)/(1 - q^r) = 1/[r]_q$, and finally use (18.1) at $k = m$ to rewrite

$$C_m = q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} = q^m \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{1}{(-q; q)_{2m}}.$$

The result is exactly (18.6). $\square$

Remark 22.5 (The mechanism). The deleted term is encoded by the auxiliary parameter a . The q -Gauss product has a removable zero-over-zero singularity at $a = 1$, and its logarithmic derivative turns products into Lambert series. This parameter-differentiation mechanism recurs throughout the theory of harmonic-number and Lambert-series refinements of hypergeometric summations.

22.4 Separate finite and infinite components

The deleted sum in Theorem 22.4 naturally splits at the missing index. Each component has a simpler one-sided Lambert-series evaluation.

Theorem 22.6 (Infinite and finite components). *Let $m \geq 0$ and $0 < q < 1$, and put*

$$P_m(q) = \begin{bmatrix} 2m \\ m \end{bmatrix}_{q^2} \frac{(1 + q)q^m}{(-q; q)_{2m}}.$$

Then

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} \\ &= P_m(q) \left(\sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} \right), \end{aligned} \quad (18.14)$$

and

$$\sum_{k=0}^{m-1} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k - m]_{q^2}} = -P_m(q) \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \quad (18.15)$$

For $m = 0$, the finite sum and the sum on its right are both empty.

Proof. By [Theorem 22.1](#), shifting $k \mapsto k + m$ in the infinite component gives

$$\begin{aligned} & \sum_{k=m+1}^{\infty} \begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} \frac{q^k}{(-q; q)_{2k} [k-m]_{q^2}} \\ &= (1 - q^2) q^m \frac{(q; q^2)_m}{(q^2; q^2)_m} \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1 - q^{2k})}. \end{aligned}$$

Introduce

$$U(a, q) = \frac{1}{1-a} \left({}_2\phi_1 \left[\begin{matrix} a, q^{2m+1} \\ q^{2m+2} \end{matrix}; q^2, a^{-1}q \right] - 1 \right).$$

For a near 1, absolute convergence permits termwise passage to the limit. Since

$$\frac{(a; q^2)_k}{1-a} = (aq^2; q^2)_{k-1} \quad (k \geq 1),$$

we obtain

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{k=1}^{\infty} \frac{(q^{2m+1}; q^2)_k q^k}{(q^{2m+2}; q^2)_k (1 - q^{2k})}.$$

The q -Gauss summation, at base q^2 , evaluates the auxiliary series as

$$U(a, q) = \frac{1}{1-a} \left(\frac{(q, a^{-1}q^{2m+2}; q^2)_{\infty}}{(a^{-1}q, q^{2m+2}; q^2)_{\infty}} - 1 \right).$$

The product quotient tends to 1 as $a \rightarrow 1$. L'Hospital's rule and termwise logarithmic differentiation, justified locally uniformly, give

$$\lim_{a \rightarrow 1} U(a, q) = \sum_{r=1}^{\infty} \frac{q^{2r-1}}{1 - q^{2r-1}} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{1 - q^{2r+2m}}.$$

Multiplication by the prefactor above, followed by $(1 - q)/(1 - q^j) = 1/[j]_q$ and [Theorem 22.1](#) at $k = m$, proves (18.14).

To obtain the finite component, subtract (18.14) from the full deleted sum (18.6). Indeed,

$$\begin{aligned} \sum_{j=2m+1}^{\infty} \frac{(-1)^{j-1} q^j}{[j]_q} &= \sum_{r=m+1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=m+1}^{\infty} \frac{q^{2r}}{[2r]_q} \\ &= \sum_{r=1}^{\infty} \frac{q^{2r-1}}{[2r-1]_q} - \sum_{r=1}^{\infty} \frac{q^{2r+2m}}{[2r+2m]_q} - \sum_{r=1}^m \frac{q^{2r-1}}{[2r-1]_q}. \end{aligned}$$

The first two sums constitute the infinite component, leaving exactly (18.15). □

22.5 The classical finite part

Define the alternating harmonic number

$$\overline{H}_N = \sum_{r=1}^N \frac{(-1)^{r-1}}{r}.$$

Theorem 22.7 (Classical deleted-singularity identity). *For every $m \geq 0$,*

$$\sum_{\substack{k \geq 0 \\ k \neq m}} \binom{2k}{k} \frac{1}{4^k (k-m)} = \binom{2m}{m} \frac{\log 2 - \bar{H}_{2m}}{2^{2m-1}}. \quad (18.16)$$

Proof. Put $c_k = (1/2)_k / k! = \binom{2k}{k} / 4^k$. For a away from the nonpositive integers, the series

$$T(a) = \sum_{k=0}^{\infty} \frac{c_k}{k+a}$$

converges locally uniformly and defines a meromorphic function. For $\Re a > 0$, termwise integration of $(1-x)^{-1/2} = \sum_{k \geq 0} c_k x^k$ gives

$$T(a) = \int_0^1 x^{a-1} (1-x)^{-1/2} dx = B(a, 1/2) = \frac{\Gamma(a)\Gamma(1/2)}{\Gamma(a+1/2)}. \quad (18.17)$$

Analytic continuation makes (18.17) valid wherever both sides are defined. The desired deleted sum is the finite part at $a = -m$:

$$\lim_{a \rightarrow -m} \left(T(a) - \frac{c_m}{a+m} \right). \quad (18.18)$$

Write $a = -m + \varepsilon$. The standard expansions of the gamma function at a negative integer and at a regular point are

$$\begin{aligned} \Gamma(-m + \varepsilon) &= \frac{(-1)^m}{m! \varepsilon} \left(1 + (H_m - \gamma)\varepsilon + O(\varepsilon^2) \right), \\ \Gamma(1/2 - m + \varepsilon) &= \Gamma(1/2 - m) \left(1 + \psi(1/2 - m)\varepsilon + O(\varepsilon^2) \right). \end{aligned}$$

Moreover,

$$\frac{(-1)^m \Gamma(1/2)}{m! \Gamma(1/2 - m)} = c_m.$$

Thus the finite part in (18.18) is

$$c_m (H_m - \gamma - \psi(1/2 - m)). \quad (18.19)$$

Using $\psi(1/2) = -\gamma - 2 \log 2$ and the recurrence $\psi(z+1) = \psi(z) + 1/z$, we find

$$\psi(1/2 - m) = -\gamma - 2 \log 2 + 2 \sum_{r=1}^m \frac{1}{2r-1}.$$

Since

$$\bar{H}_{2m} = \sum_{r=1}^m \frac{1}{2r-1} - \frac{1}{2} H_m,$$

the bracket in (18.19) is $2(\log 2 - \bar{H}_{2m})$. Hence the finite part equals

$$2 \frac{\binom{2m}{m}}{4^m} (\log 2 - \bar{H}_{2m}),$$

which is (18.16). □

Proposition 22.8 (Limit of the Lambert tail). *For fixed $m \geq 0$,*

$$\lim_{q \rightarrow 1^-} \sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1} q^r}{[r]_q} = \log 2 - \bar{H}_{2m}. \quad (18.20)$$

Proof. For $0 < q < 1$, set $a_r(q) = q^r / [r]_q$. These numbers are positive and decrease with r , since

$$\frac{a_{r+1}(q)}{a_r(q)} = q \frac{[r]_q}{[r+1]_q} < 1.$$

Moreover,

$$a_r(q) = \frac{q^r}{1 + q + \cdots + q^{r-1}} \leq \frac{1}{r},$$

because every summand in the denominator is at least q^{r-1} . The alternating-series remainder after N terms therefore has absolute value at most $a_{N+1}(q) \leq 1/(N+1)$, uniformly in q . For each fixed r , $a_r(q) \rightarrow 1/r$ as $q \rightarrow 1^-$. First pass to the limit in a finite partial sum and then let $N \rightarrow \infty$. The result is the tail of the alternating harmonic series,

$$\sum_{r=2m+1}^{\infty} \frac{(-1)^{r-1}}{r} = \log 2 - \bar{H}_{2m}.$$

□

Chapter 23

Asymptotic regimes and Bernoulli expansions

A q -analogue should explain both the deformation near $q = 1$ and behavior far from it. Gaussian coefficients have especially clean asymptotic descriptions because their logarithms are finite sums of a single elementary function.

23.1 A complete expansion as $q \rightarrow 1$

Let the Bernoulli numbers be defined by

$$\frac{x}{e^x - 1} = \sum_{r=0}^{\infty} B_r \frac{x^r}{r!}.$$

Lemma 23.1 (The basic logarithmic expansion). *As $x \rightarrow 0$,*

$$\log \frac{1 - e^{-x}}{x} = -\frac{x}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!} x^{2r}. \quad (19.1)$$

The series converges for $|x| < 2\pi$.

Proof. Differentiate the left side:

$$\frac{d}{dx} \log \frac{1 - e^{-x}}{x} = \frac{1}{e^x - 1} - \frac{1}{x}.$$

By the Bernoulli generating function,

$$\frac{1}{e^x - 1} - \frac{1}{x} = -\frac{1}{2} + \sum_{r=1}^{\infty} B_{2r} \frac{x^{2r-1}}{(2r)!}.$$

Integrate term by term and use the value 0 at $x = 0$. The nearest nonzero singularities are at $x = \pm 2\pi i$, giving radius 2π . $\square$

Theorem 23.2 (Bernoulli expansion of a Gaussian coefficient). *Let $0 \leq k \leq n$, put $D = k(n - k)$, and set $q = e^{-t}$. As $t \rightarrow 0$,*

$$\log \frac{\begin{bmatrix} n \\ k \end{bmatrix} e^{-t}}{\binom{n}{k}} = -\frac{D}{2}t + \sum_{r=1}^{\infty} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^k \left((n - k + j)^{2r} - j^{2r} \right). \quad (19.2)$$

For fixed n, k , the displayed series converges whenever $|t| < 2\pi/n$.

Proof. From the product formula,

$$\begin{aligned} \left[\begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} &= \prod_{j=1}^k \frac{1 - e^{-(n-k+j)t}}{1 - e^{-jt}} \\ &= \binom{n}{k} \prod_{j=1}^k \frac{(1 - e^{-(n-k+j)t}) / ((n-k+j)t)}{(1 - e^{-jt}) / (jt)}. \end{aligned}$$

Take logarithms and apply [Theorem 23.1](#) to every factor. The linear term is

$$-\frac{t}{2} \sum_{j=1}^k (n-k) = -\frac{1}{2} k(n-k)t = -\frac{D}{2}t,$$

and the even terms give the sum in (19.2). Every argument of the basic series has modulus below 2π when $|t| < 2\pi/n$, so the finite sum converges there. $\square$

Corollary 23.3 (First terms). *With $D = k(n-k)$,*

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} = \binom{n}{k} \exp \left(-\frac{D}{2}t + \frac{D(n+1)}{24}t^2 + O(t^4) \right), \quad (19.3)$$

$$= \binom{n}{k} \left(1 - \frac{D}{2}t + \left(\frac{D^2}{8} + \frac{D(n+1)}{24} \right) t^2 + O(t^3) \right). \quad (19.4)$$

Proof. For $r = 1$ in (19.2),

$$\begin{aligned} \sum_{j=1}^k \left((n-k+j)^2 - j^2 \right) &= k(n-k)^2 + (n-k)k(k+1) \\ &= k(n-k)(n+1) = D(n+1). \end{aligned}$$

Since $B_2/(2 \cdot 2!) = 1/24$, the logarithmic formula follows. Expanding the exponential gives the second formula. $\square$

Remark 23.4 (Reciprocity and centering). The factor $e^{-Dt/2}$ is forced by reciprocity:

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}} = e^{-Dt} \left[\begin{matrix} n \\ k \end{matrix} \right]_{e^t}.$$

Therefore the centered quantity $e^{Dt/2} \left[\begin{matrix} n \\ k \end{matrix} \right]_{e^{-t}}$ is an even analytic function of t . This explains, before any calculation, why all odd powers beyond the linear term vanish from the logarithm.

23.2 Fixed q and growing parameters

Theorem 23.5 (Fixed-column limit). *If $0 < |q| < 1$ and k is fixed, then*

$$\lim_{n \rightarrow \infty} \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_k}. \quad (19.5)$$

More precisely,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{1}{(q; q)_k} \left(1 + O(|q|^{n-k+1}) \right). \quad (19.6)$$

Proof. The product formula gives

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{(q^{n-k+1}; q)_k}{(q; q)_k}.$$

The numerator tends to 1. Since the number of factors is fixed,

$$\prod_{j=0}^{k-1} (1 - q^{n-k+1+j}) = 1 + O(|q|^{n-k+1}),$$

which proves both assertions. $\square$

Theorem 23.6 (Central fixed- q limit). *For $0 < |q| < 1$,*

$$\lim_{m \rightarrow \infty} \left[\begin{matrix} 2m \\ m \end{matrix} \right]_q = \frac{1}{(q; q)_\infty}. \quad (19.7)$$

Proof. By the factorial form,

$$\left[\begin{matrix} 2m \\ m \end{matrix} \right]_q = \frac{(q; q)_{2m}}{(q; q)_m^2}.$$

Every finite shifted factorial tends to $(q; q)_\infty$ as its index tends to infinity. The quotient therefore tends to $(q; q)_\infty / (q; q)_\infty^2$. $\square$

Proposition 23.7 (Uniform elementary bound). *For $0 < q < 1$ and $0 \leq k \leq n$,*

$$1 \leq \left[\begin{matrix} n \\ k \end{matrix} \right]_q \leq \frac{1}{(q; q)_\infty}. \quad (19.8)$$

Proof. The lower bound follows from the partition interpretation: the constant term is 1 and every coefficient is nonnegative. The same interpretation says that $\left[\begin{matrix} n \\ k \end{matrix} \right]_q$ is the generating function of partitions fitting in a k by $(n - k)$ rectangle. Dropping the rectangle restriction embeds this set into all partitions, whose generating function is $(q; q)_\infty^{-1}$. Evaluating at a positive $q < 1$ gives the upper bound. $\square$

23.3 The regime $q > 1$

Corollary 23.8 (Transfer through reciprocity). *Let $q > 1$ and $D = k(n - k)$. Then*

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = q^D \left[\begin{matrix} n \\ k \end{matrix} \right]_{q^{-1}}. \quad (19.9)$$

Consequently, for fixed k ,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{q^{k(n-k)}}{(q^{-1}; q^{-1})_k} \left(1 + O(q^{-n+k-1}) \right), \quad (19.10)$$

and

$$\left[\begin{matrix} 2m \\ m \end{matrix} \right]_q \sim \frac{q^{m^2}}{(q^{-1}; q^{-1})_\infty}. \quad (19.11)$$

Proof. The reciprocity identity was proved in [Theorem 2.5](#). Apply [Theorems 23.5](#) and [23.6](#) with q^{-1} in place of q . $\square$

23.4 Extreme specializations

Proposition 23.9. For $0 \leq k \leq n$,

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_q \Big|_{q=0} = 1, \quad \lim_{q \rightarrow 1} \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \binom{n}{k},$$

and the leading term as $q \rightarrow \infty$ is $q^{k(n-k)}$ with coefficient 1.

Proof. The partition generating polynomial has constant term 1, proving the first assertion. The second follows by replacing every ratio $(1 - q^a)/(1 - q^b)$ in the product formula by its limit a/b . Palindromicity and degree $k(n - k)$ show that the leading coefficient equals the constant coefficient, namely 1. $\square$

Chapter 24

Computation, certification, and formal proof architecture

The identities in this book are well suited to exact computation because their natural objects are polynomials, rational functions, and formal power series. They are also well suited to proof assistants, provided algebraic identities are separated cleanly from analytic convergence. Lau, Lee, and Ono’s 2026 formalization project [20] demonstrates this architecture for q -Pochhammer symbols, Bailey pairs, Jacobi’s triple product, and the Rogers–Ramanujan identities.

24.1 Dynamic programming for Gaussian polynomials

A polynomial is represented here by its coefficient vector. Multiplication by q^h is therefore a shift by h places.

Algorithm 24.1 (Pascal dynamic program). Given $n \geq 0$ and $0 \leq k \leq n$:

1. Replace k by $\min(k, n - k)$, using symmetry.
2. Initialize $G_0(q) = 1$ and $G_s(q) = 0$ for $1 \leq s \leq k$.
3. For $r = 1, 2, \dots, n$, update in descending order

$$G_s(q) \leftarrow G_s(q) + q^{r-s} G_{s-1}(q) \quad (s = \min(r, k), \dots, 1).$$

4. Return $G_k(q)$.

Theorem 24.2 (Correctness and complexity). *Theorem 24.1* returns $\begin{bmatrix} n \\ k \end{bmatrix}_q$. If $D = k(n - k)$ after the symmetry reduction, its coefficient-level arithmetic cost is $O(nkD)$ and its storage cost is $O(kD)$.

Proof. Let $G_s^{(r)}$ denote the value after row r . We prove by induction on r that

$$G_s^{(r)} = \begin{bmatrix} r \\ s \end{bmatrix}_q \quad (0 \leq s \leq \min(r, k)).$$

The assertion is true at $r = 0$. Descending updates ensure that both G_s and G_{s-1} on the right are still values from row $r - 1$. Hence the update is

$$G_s^{(r)} = \begin{bmatrix} r-1 \\ s \end{bmatrix}_q + q^{r-s} \begin{bmatrix} r-1 \\ s-1 \end{bmatrix}_q = \begin{bmatrix} r \\ s \end{bmatrix}_q$$

by the second q -Pascal recurrence. At $r = n$ this gives the desired output.

After replacing k by $\min(k, n - k)$, we have $k \leq n/2$. Every intermediate polynomial $\left[\begin{smallmatrix} r \\ s \end{smallmatrix} \right]_q$ with $s \leq k$ has degree $s(r - s) \leq k(n - k) = D$. There are at most nk shift-and-add updates, each touching at most $D + 1$ coefficients. Storing $k + 1$ such vectors uses $O(kD)$ coefficients. $\square$

Remark 24.3 (Numerical specialization). When only a numerical value is required, one should not first expand the polynomial. For positive q near 1, the stable formula is

$$\log \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q = \sum_{j=1}^k \left[\log(-\expm1((n - k + j) \log q)) - \log(-\expm1(j \log q)) \right],$$

where $\expm1$ avoids cancellation in $e^x - 1$. For exact symbolic work, the Pascal algorithm or the cyclotomic factorization is preferable.

24.2 A factorization algorithm

Algorithm 24.4 (Cyclotomic factor list). For each $1 \leq d \leq n$, compute

$$e_d = \lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n - k)/d \rfloor.$$

Return the list of d for which $e_d = 1$.

Theorem 24.5 (Correctness of the factor list). *The output of Theorem 24.4 is exactly the set of cyclotomic factors of $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$, each with multiplicity one.*

Proof. This is Theorem 16.1: the exponent of $\Phi_d(q)$ is e_d , and the floor calculation in that theorem proves $e_d \in \{0, 1\}$. $\square$

Remark 24.6. The factor-list algorithm takes only $O(n)$ integer floor operations. It is therefore far cheaper than expanding $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$ when the purpose is root-of-unity evaluation, divisibility testing, or congruence work. Polynomial expansion may be postponed until the selected cyclotomic factors are actually multiplied.

24.3 Telescoping certificates for finite sums

A term $T(n, k)$ is called q -hypergeometric if the ratios

$$\frac{T(n, k + 1)}{T(n, k)}, \quad \frac{T(n + 1, k)}{T(n, k)}$$

are rational functions of q^n and q^k (and of any fixed parameters). Shifted-factorial terms have this property because adjacent factors cancel.

Theorem 24.7 (Finite telescoping certificate). *Let*

$$S_n = \sum_{k=L(n)}^{U(n)} T(n, k).$$

Suppose there are functions $A_0(n), \dots, A_r(n)$ and a certificate $G(n, k)$ such that

$$\sum_{j=0}^r A_j(n) T(n + j, k) = G(n, k + 1) - G(n, k) \tag{20.1}$$

for all relevant k , after extending T by zero outside its natural finite range. If $G(n, k)$ vanishes at both ends of the resulting finite summation interval, then

$$\sum_{j=0}^r A_j(n) S_{n+j} = 0. \quad (20.2)$$

Proof. Sum (20.1) over an interval containing the support of every $T(n + j, k)$. The left side becomes $\sum_j A_j(n) S_{n+j}$. The right side telescopes to the terminal value of G minus its initial value, which is 0 by hypothesis. $\square$

Corollary 24.8 (Identity certification). *If two finite sums satisfy the same recurrence of order r and agree for r consecutive initial values at every nonsingular parameter choice, then they are equal wherever the recurrence determines subsequent values.*

Proof. Subtract the two sequences. The difference satisfies the homogeneous recurrence and has r consecutive zero initial values. Induction through the recurrence gives zero thereafter. $\square$

Remark 24.9 (Creative telescoping). The q -Zeilberger method searches algorithmically for a rational multiple $G(n, k) = R(q^n, q^k)T(n, k)$ satisfying (20.1). Once found, the certificate is independently checkable by rational-function arithmetic. The search may be sophisticated, but verification is elementary: divide by $T(n, k)$ and clear denominators.

24.4 Denominator clearing and specialization

Lemma 24.10 (Polynomial identity principle). *Let $R(q), S(q) \in \mathbb{Q}(q)$. If $R(q) = S(q)$ at infinitely many complex values of q where both sides are defined, then $R = S$ in $\mathbb{Q}(q)$.*

Proof. Write $R - S = A/B$ with $A, B \in \mathbb{Q}[q]$ and $B \neq 0$. The hypothesis says that A has infinitely many zeros. A nonzero polynomial has only finitely many zeros, so $A = 0$. $\square$

Corollary 24.11 (Safe generic specialization). *A rational q -identity proved under generic nonvanishing assumptions remains valid after any specialization at which the denominator-cleared identity is defined. Apparent excluded values may therefore be recovered by cancellation or by taking limits.*

Proof. Clear all denominators to obtain a polynomial identity. Polynomial evaluation is valid at every specialization. If the original rational expressions have removable singularities, divide out common factors or take their uniquely determined limits. $\square$

This principle is particularly important at roots of unity. One should first rewrite a Gaussian coefficient as its polynomial, not substitute a root into the factorial quotient and accept an indeterminate $0/0$.

24.5 Formal series versus analytic series

Theorem 24.12 (Formal-to-analytic transfer). *Suppose*

$$\sum_{n \geq 0} a_n z^n = \sum_{n \geq 0} b_n z^n$$

coefficientwise in $\mathbb{C}[[z]]$. If both series converge for $|z| < R$, then they define the same analytic function on that disk. Conversely, if two power series converge in a disk and define the same function on a subset with an accumulation point in the disk, then their coefficients are equal.

Proof. The forward implication is immediate because the two partial sums agree term by term. For the converse, the identity theorem makes the analytic functions equal throughout the disk; differentiating n times at 0 gives $n!a_n = n!b_n$. $\square$

Principle 24.13 (A scalable proof architecture). *For a mixed algebraic-analytic q -identity, the following order minimizes hidden assumptions:*

1. *define finite shifted factorials and prove their algebraic laws;*
2. *prove terminating identities after denominator clearing;*
3. *define formal infinite products by coefficient stabilization when possible;*
4. *separately establish analytic convergence for $|q| < 1$;*
5. *transfer the formal coefficient identity to analytic functions;*
6. *only then take limits, differentiate parameters, or specialize at roots of unity.*

Proof. Each step is justified by results already proved: finite product algebra in [Chapter 1](#); polynomial identity and specialization by [Theorems 24.10](#) and [24.11](#); formal-to-analytic transfer by [Theorem 24.12](#); and the convergence theorems in [Chapters 8](#) to [10](#). The ordering prevents an analytic operation from being used before its convergence hypothesis has been supplied. $\square$

Historical note 24.14. Formal verification forces distinctions that handwritten mathematics often suppresses: a finite product indexed by a natural number is not the same object as a bilateral product; a polynomial Gaussian coefficient is not definitionally the same as a rational factorial quotient; and a formal power-series identity does not automatically contain a proof of analytic convergence. The Lean development of Lau, Lee, and Ono [20] turns these distinctions into reusable interfaces rather than obstacles.

Appendix A

Formula atlas

This appendix is a navigation table, not a substitute for hypotheses and proofs. Each formula points back to the theorem where its exact domain and derivation are given.

Name	Formula	Location
Concatenation	$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n$	Theorem 1.2
Dissection	$(a; q)_{rn} = \prod_{j=0}^{r-1} (aq^j; q^r)_n$	Theorem 1.3
Base reversal	$(a; q)_n = (-a)^n q^{\binom{n}{2}} (a^{-1} q^{1-n}; q)_n$	Theorem 1.4
Negative index	$(a; q)_{-n} = 1 / (aq^{-n}; q)_n$	Theorem 1.6
Infinite logarithm	$\log(a; q)_\infty = -\sum_{r \geq 1} a^r / (r(1 - q^r))$	Theorem 1.14
Gaussian product	$\begin{bmatrix} n \\ k \end{bmatrix}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$	Theorem 2.2
Pascal I	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Pascal II	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$	Theorem 2.3
Reciprocity	$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^{k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}}$	Theorem 2.5
Finite theorem	$(z; q)_n = \sum_k (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k$	Theorem 3.1
Reciprocal theorem	$(z; q)_{n+1}^{-1} = \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q z^k$	Theorem 3.4
q -Vandermonde	$\begin{bmatrix} m+n \\ r \end{bmatrix}_q = \sum_k q^{\binom{m-k}{2} + \binom{r-k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q \begin{bmatrix} n \\ r-k \end{bmatrix}_q$	Theorem 3.5
Infinite theorem	$\sum_{n \geq 0} (a; q)_n z^n / (q; q)_n = (az; q)_\infty / (z; q)_\infty$	Theorem 8.1
q -Gauss	${}_2\phi_1(a, b; c; q, c/(ab)) = (c/a, c/b; q)_\infty / (c, c/(ab); q)_\infty$	Theorem 9.4
q -Pfaff–Saalschütz	Terminating balanced ${}_3\phi_2$ product evaluation	Theorem 9.6
q -Chu–Vandermonde	Two terminating ${}_2\phi_1$ evaluations	Theorem 9.7
Ramanujan ${}_1\psi_1$	${}_1\psi_1(a; b; q, z) = (q, b/a, az, q/(az); q)_\infty / (b, q/a, z, b/(az); q)_\infty$	Theorem 10.2
Jacobi triple product	$\sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n = (q, z, q/z; q)_\infty$	Theorem 13.1
Pentagonal theorem	$(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2}$	Theorem 13.3
q -derivative	$D_q f(x) = (f(x) - f(qx)) / ((1 - q)x)$	Theorem 11.1
Jackson integral	$\int_0^a f(x) d_q x = (1 - q)a \sum_{n \geq 0} q^n f(aq^n)$	Theorem 11.9
q -gamma	$\Gamma_q(x) = (1 - q)^{1-x} (q; q)_\infty / (q^x; q)_\infty$	Theorem 12.2
Cyclotomic factorization	$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_d \Phi_d(q)^{\lfloor n/d \rfloor - \lfloor k/d \rfloor - \lfloor (n-k)/d \rfloor}$	Theorem 16.1
q -Lucas	$\begin{bmatrix} ad+bs \\ rd+rs \end{bmatrix}_q \equiv \begin{bmatrix} a \\ r \end{bmatrix}_q \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}$	Theorem 16.5
q -Babbage	$\begin{bmatrix} an \\ bn \end{bmatrix}_q \equiv \begin{bmatrix} a \\ b \end{bmatrix}_q n^2 \pmod{\Phi_n(q)^2}$	Theorem 16.8
Negative upper index	$\begin{bmatrix} -m \\ k \end{bmatrix}_q = (-1)^k q^{-mk - \binom{k}{2}} \begin{bmatrix} m+k-1 \\ k \end{bmatrix}_q$	Theorem 17.2
Geometric Newton basis	$P(x) = \sum_k c_k (-1)^k q^{\binom{k}{2}} (x; q^{-1})_k$	Theorem 17.10
Central reduction	$\begin{bmatrix} 2k \\ k \end{bmatrix}_{q^2} / (-q; q)_{2k} = (q; q^2)_k / (q^2; q^2)_k$	Theorem 22.1
Bernoulli expansion	All-orders expansion of $\log(\begin{bmatrix} n \\ k \end{bmatrix}_{e^{-t}} / \binom{n}{k})$	Theorem 23.2

Appendix B

Limit and specialization dictionary

B.1 Elementary limits

For fixed nonnegative integers, the following transitions are direct consequences of the product formulas:

$$\begin{aligned}\lim_{q \rightarrow 1} [n]_q &= n, & \lim_{q \rightarrow 1} (1-q)^{-n} (q^a; q)_n &= (a)_n, \\ \lim_{q \rightarrow 1} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \binom{n}{k}, & \lim_{n \rightarrow \infty} \begin{bmatrix} n \\ k \end{bmatrix}_q &= \frac{1}{(q; q)_k} \quad (|q| < 1), \\ \lim_{q \rightarrow 1^-} \Gamma_q(x) &= \Gamma(x), & \lim_{q \rightarrow 0} \begin{bmatrix} n \\ k \end{bmatrix}_q &= 1 \quad (0 \leq k \leq n).\end{aligned}$$

Each has been proved in the main text; the relevant references are [Theorems 1.17, 2.6, 12.5, 23.5](#) and [23.9](#).

B.2 Classical shadows of the main summations

Under the standard scaling $a = q^A$, $b = q^B$, and $q \rightarrow 1^-$, terminating basic hypergeometric sums tend to ordinary hypergeometric sums. The mechanism is

$$\frac{(q^A; q)_n}{(1-q)^n} \longrightarrow (A)_n, \quad \frac{(q; q)_n}{(1-q)^n} \longrightarrow n!.$$

Thus:

1. the finite q -binomial theorem tends to Newton's binomial theorem;
2. q -Chu–Vandermonde tends to Chu–Vandermonde;
3. q -Pfaff–Saalschütz tends to Pfaff–Saalschütz;
4. the Jackson q -beta integral tends to Euler's beta integral;
5. [Theorem 22.2](#) tends to [Theorem 22.3](#);
6. [Theorem 22.4](#) tends, at the level of its Lambert tail and prefactor, to [Theorem 22.7](#).

B.3 Roots of unity

At a root of unity, factorial quotients should be replaced by polynomial formulas before evaluation. The basic dictionary is:

$$\begin{aligned} \Phi_d(q) \mid \begin{bmatrix} n \\ k \end{bmatrix}_q &\iff k \bmod d > n \bmod d, \\ \begin{bmatrix} ad + b \\ rd + s \end{bmatrix}_q &\equiv \begin{pmatrix} a \\ r \end{pmatrix} \begin{bmatrix} b \\ s \end{bmatrix}_q \pmod{\Phi_d(q)}, \\ \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=-1} &\text{ is given by [Theorem 16.6](#).} \end{aligned}$$

These statements turn apparently singular substitutions into finite enumerations and are the algebraic basis of the subset cyclic-sieving theorem.

Appendix C

Proof-dependency guide

The monograph can be read linearly, but several shorter routes are useful.

Route to Gaussian combinatorics

finite products $\longrightarrow$ q -Pascal $\longrightarrow$ finite q -binomial theorem $\longrightarrow$ partitions, paths, subspaces.

Read [Chapters 1 to 3, 5 and 6](#).

Route to product–sum identities

infinite q -binomial theorem $\longrightarrow$ Heine transformation $\longrightarrow$ q -Gauss,
negative indices and bilateral convergence $\longrightarrow$ ${}_1\psi_1$,
Euler expansions $\longrightarrow$ Jacobi triple product $\longrightarrow$ Bailey pairs.

Read [Chapters 8 to 10, 13 and 15](#).

Route to arithmetic

$1 - q^n = \prod_{d|n} \Phi_d(q) \longrightarrow$ carry criterion $\longrightarrow$ q -Lucas $\longrightarrow$ cyclic sieving and q -Babbage.

Read [Chapter 16](#).

Route to analytic deformation

$\log(a; q)_\infty \longrightarrow$ Lambert series and dilogarithms $\longrightarrow$ q -gamma
 $\longrightarrow$ central sums and Bernoulli asymptotics.

Read [Chapters 1, 12, 22 and 23](#).

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Index

- $\mathfrak{sl}_2$, 26
- q -Pochhammer symbol, *see* q -shifted factorial, 104
 - base one half, 76
- q -beta function, 42
- q -binomial coefficient, *see* Gaussian coefficient, *see* Gaussian coefficient
 - base one half, 76
- q -binomial theorem
 - infinite, 29
- q -derivative, 38
- q -gamma function, 42, 104
- q -shifted factorial, 2
- Appell polynomial
 - Fabius moments, 71
- Bailey pair, 52, 104
- basic hypergeometric series, 32, 104
- Bell polynomial
 - Fabius moments, 81
- bilateral basic hypergeometric series, 36
- binary word, 20
- cyclic sieving, 104
- cyclotomic polynomial, 104
- dyadic rational, 71
- Fabius function, 71
 - composition formula, 81
- finite field, 23
- Gaussian coefficient, 9, 104
 - denominator-free form, 67
- geometric grid, 67
- Grassmannian, 23
- interpolation
 - geometric nodes, 67
- inversion statistic, 20
- Jackson integral, 38, 104
- Jacobi triple product, 47, 104
- Lambert series, 104
- Newton interpolation, 104
- partition
 - inside a rectangle, 20
- partition generating function, 50
- Prouhet cancellation, 76
- quantum plane, 38
- Ramanujan ${}_1\psi_1$ summation, 36
- Rogers–Ramanujan identities, 52, 104
- Rvachev function, 71
- Rvachev up-function, 81
- Schubert cell, 23
- sinc product, 81
- symmetric function, 17
- theta function, 47
- Thue–Morse sequence, 76
- Toeplitz filter, 81
- unimodality, 26
- weighted binomial coefficient, 17